

Bio 330 Evolution, Sex & The Birth Control Pill

Numbered Transcript

BIO 330 • Dr. Sarah Hill

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Lecture 1: Evolutionary Psychology

At a glance — Hill introduces evolutionary psychology and how it frames sex, hormones, and the birth control pill.

- 1.1 Welcome, everybody. I am Dr. Sarah Hill, and today I'm going to be talking to you about evolution, sex, and the birth control pill. Before we get started, I just am going to give you a little background on me. I am a professor, and I'm a professor of evolutionary psychology. I earned my PhD at the University of Texas, working with Dr. David Buss, who's one of the founding field members in the area of evolutionary psychology; he's one of the originals.
- 1.2 During my time working with him, I learned how to use and apply evolutionary principles to try to understand human behavior. I was specifically interested in trying to understand women's sexual psychology and women's behavior because much of the research that had been done in the field up until that point had been done from a male perspective and had been done really through the lens of looking at men.
- 1.3 So most of my early career research looked at things like mate choice, attraction, and women's intersexual competition, the context in which women engage in risky sexual behavior, and so on. It was only more recently in my career, within the last 10 years, that I really became interested in trying to unpack the impact that hormones have in creating the experience of being female.
- 1.4 I've written a couple of books now—This Is Your Brain on Birth Control was my first book. I'm talking about the way that women's hormones and the hormonal changes that they have when they're on the pill influence women in the way that they think, feel, and experience the world. Then later on, my most recent book is called The Period Brain, and it really unpacks the way that women's cycling hormones impact their experiences in the world.
- 1.5 Today, and throughout this lecture series, we're going to be learning about three parts of information. The first thing that we're going to do is I'm going to give you a really short crash course on the theory of evolution by natural selection. I'm going to do this because it's a theory that a lot of people don't really understand. There's also a lot of misunderstandings that surround the idea of what this theory is and what it means. So we're going to do a quick little crash in what that theory is.
- 1.6 Then we're going to talk about why this theory is so useful in terms of understanding sex and why it's so useful in terms of understanding sex differences. We're going to talk about the ways that we can use an evolutionary approach to understand reliably occurring differences between men and women.
- 1.7 Then we're going to learn about hormones. We're going to learn not only about the why between the different sexes, which the evolutionary approach provides us, but we're also going to understand how those differences are created. That's a question of

hormones. So we're going to learn a little bit about men's and women's primary sex hormones and the way that they create the experiences of being female and being male. Then lastly, we're going to be transitioning into talking about how changing women's hormones changes women. We're going to talk about the birth control pill.

- 1.8** Today, we're going to be starting off by talking about evolution in sex. We're going to do this by first giving you a primer on evolutionary biology. I like to start this class—and I start every evolutionary psychology class that I teach—asking people to imagine that they encounter a Martian who's landed here from Mars. It gets out of his spaceship. He's looking around at the world around him, and he says, I've been noticing that people tend to like sweet and fatty foods more than they tend to like things like vegetables. Why is that?
- 1.9** You reply to that Martian, as many people do; you come up with explanations for this. Some of the explanations that you might give are, "It tastes better," because we know that sweet and fatty foods taste better than broccoli and cauliflower and Brussels sprouts and all those bitter types of foods that we have to learn to love. One answer might be that it tastes better.
- 1.10** If you're a neuroscientist or you're a neuroscience major, you might try to impress the Martian with your understanding of dopamine release. You'd say, "Oh, the reason that people like to eat those kinds of things instead of these kinds of things is because when you eat things like fat and sugar and chocolate, it's going to release dopamine, which activates the reward centers of the brain, so that makes us feel good."
- 1.11** Or if you're somebody who has any background in childhood development, you might say, "Well, we tend to get rewarded with these things as children. It creates these positive associations where we think of sweet and fatty foods, and we think of being good and pleasantness, and we think of things like vegetables, and we think of things like not good, and it doesn't have those positive associations."
- 1.12** Each one of these answers is right. We know from research that dopamine gets released when people eat sweet and fatty food. We know from research that people tend to think that sweet and fatty foods taste better, especially initially, when they're first trying it, than they do things that are bitter like vegetables.
- 1.13** And we also know that kids tend to get rewarded with these types of foods. We tend to give our children things like a treat after they've done something that we asked them to, and that treat is not usually broccoli. It's usually something that's sweet and gives that dopamine release. All of these things are correct.
- 1.14** But this is only one layer of answer to this question of why people eat more sweet and fatty things than they do bitter tasting things. This is because there's nothing inherently delicious about sweet and fatty foods. While it's true that, yes, dopamine gets released when we eat sweet and fatty foods, and we prefer the flavor of sweet and fatty foods, there's nothing inherently better about them than there is about bitter-tasting foods.

- 1.15** So there's another type of question that we can ask when it comes to our food preferences, because there's nothing inherently tasty about fat and sugar. Instead, it's all about the fact that we possess a brain that tends to reward us when we eat sweet and fatty foods.
- 1.16** Our preference for sweet and fatty foods also has a second type of answer. This is an answer that's about the human brain. What is it about the human brain that provides a reward when we have sugar molecules, but then doesn't give that same reward when we have molecules that are related to things like a bitter taste in vegetables? Because there's nothing about these types of foods that necessarily is better than the other.
- 1.17** We know from much cross-species work that finding sweet and fatty things pleasurable isn't something that we find across species because the types of things that tend to activate the dopamine release that we get from sweet and fatty food is not the same as it is for other organisms.
- 1.18** For example, this over here is a gray lemur, and gray lemurs like to eat papyrus. Papyrus is something that's very dangerous to humans; it's poisonous to us. If we took a bite of papyrus, it would make us gag. We would have a terrible response to it because it's very bitter, and most people would inherently spit it out. But to a gray lemur, that's what's activating dopamine release. So lemur brains find papyrus molecules more pleasurable than a chocolate cake molecule.
- 1.19** A dung beetle finds dung more appealing than a chocolate cake molecule. Their brains get lit up when they're around and when they consume and when they build things out of dung. With humans, we have to ask ourselves: Why is it that we have a brain that is wired to find fat and sugar molecules more pleasurable than we find vegetable molecules? In other words, we need to ask ourselves: What is it about our brain that's rewarding us for the preference for sweet and fatty things compared to things that have a bitter flavor?
- 1.20** Another way of thinking about this is we're asking ourselves, is might this happen on purpose? Might we have inherited this tendency to find sweet and fatty foods more desirable, initially, especially, than we find bitter foods because it provided an advantage to our ancestors that allowed us to inherit that tendency.
- 1.21** To think about this, I want to do a little thought experiment. I want you to imagine that you're a human being living millions of years ago in the wild African savanna. We know that humans have been around as a species for a couple of million years—most estimates are about 5 million years. We spent most of our time living as hunter-gatherers, living on the savannas of Africa, living in small communal groups where hunting and gathering food was the way that we got access to the calories that we need. I want you to imagine that you're living in that context. I want you to imagine that there's a certain set of genes that play a role in determining which foods you find most immediately pleasurable.

- 1.22** You're either born with the A gene, the B gene, or the C gene. The A gene causes you to experience immense pleasure in response to all flavor profiles equally. You eat something bitter: dopamine release. You eat something sweet: dopamine release. Your brain is just rewarding you for whatever you eat. The B gene causes you to experience most pleasure in response to sweet flavors and a weariness and caution toward bitter flavors. We're getting dopamine release from the sweet and fatty things and caution and weariness toward things that are bitter. Or you have the C gene, which causes you to experience the most pleasure in response to bitterness, and then a weariness toward sweet and fatty foods.
- 1.23** I want you to imagine that there are three couples, and each couple has a baby: one has the A gene, one has the B gene, and one has the C gene. Knowing nothing else, which one of these babies would we expect to have the most children and the most grandchildren? When we walk through this little thought experiment, it seems pretty clear that, first of all, the C gene, that one's probably not going to get inherited very frequently, because we know that bitter foods in nature, oftentimes, are poisonous.
- 1.24** If you fall immediately in love with the flavor of a bitter food, that could mean trouble. In most of our evolutionary history, we didn't have grocery stores telling us that arugula is something that's safe to eat. We didn't know that this bitter flavor wasn't something that was going to make us sick. So we tend to have this inherent caution toward bitter foods because many times, bitterness is a cue to danger. We can probably guess that the C gene might not have been inherited very frequently because oftentimes it would end up leading you to bite the evolutionary dust when you end up accidentally eating some papyrus.
- 1.25** The A gene, which causes you to experience immense pleasure in response to all profiles, this one might do okay for a while. But again, if you're experiencing pleasure and having no weariness toward bitter foods, sometimes your genes are going to bite the evolutionary dust. You're going to end up going down a bad path.
- 1.26** So we can imagine that the B gene, which causes you to experience pleasure in response to sweet and fatty foods and having an inherent caution toward bitter foods, that those individuals— because things that taste sweet to us in nature are not poisonous, so there's not anything in nature that we can eat that it tastes sweet to us that will kill us, which is an interesting fun fact for you to be able to walk home with today. Our brain falls immediately in love with this flavor profile. But with bitter foods, we have an inherent caution toward them that we have to learn ourselves out of, to learn to love broccoli and learn to love Brussels sprouts and learn to love cauliflower.
- 1.27** So we can imagine how we could, over time, get the change in gene frequencies related to the types of foods that we most prefer because of the survival costs and survival advantages that could be afforded by different tendencies when it comes to food flavors. We can understand how, over the course of evolutionary time, we might expect to see that a greater number of those genes that are related to a preference for sweet

and fatty foods would be passed down, and that these other genes eventually would be weeded out of the population.

- 1.28** When we get change over time, these are evolutionary changes. These are called—when we're talking about traits and trying to understand, ultimately: why is it that we have this tendency, ultimately, why is it that we have a preference for sweet and fatty foods and caution toward bitter foods? That is an ultimate level explanation. It's called an ultimate level explanation, not because we think that it's necessarily better than any other explanation; all of these explanations are true. We know that dopamine gets released when we eat sweet and fatty foods, so that's a correct answer.
- 1.29** But ultimately, the reason that dopamine gets released when we eat sweet and fatty foods is because our brain is rewarding us from doing something that's going to get us access to sufficient calories to be able to survive and not going to poison us. We have a brain, we have inherited a tendency to prefer certain types of foods compared to other types of foods because of the evolutionary advantage that it provided. We're asking ourselves when we're using an evolutionary approach to try to understand human behavior, ultimately: Why does this trait persist?
- 1.30** If we're trying to understand the nature of human nature, that is, what are the qualities that most humans tend to have, we can understand these qualities in terms of the fact that people should be expected to have inherited tendencies that allowed our ancestors to be able to successfully survive and successfully reproduce. Because traits and qualities that are functional—and in an evolutionary situation, we consider something functional when it promotes survival or promotes reproduction.
- 1.31** That's how we define functional, is that it gets passed down. It's a trait that's going to get inherited. Traits and behaviors that promote successful survival and traits and behaviors that promote successful reproduction; those traits tend to get passed down from one generation to the next because they're doing exactly what they need to do to get passed down: they're promoting survival, they're promoting reproduction. Traits that don't, don't. If you have traits that cause a survival problem, they're not going to get inherited. If you don't survive long enough to be able to reproduce, you're not passing those genes that are code for those problematic tendencies onto anybody.
- 1.32** Same is true if you have genes related to not reproducing. If you have genes that make you sterile, those aren't going to be inherited by anyone. Sterility does not get inherited. We expect that over time, populations are going to change in response to the fact that individuals are going to have a tendency to inherit genes that help promote survival and reproduction, and that traits that don't promote survival and reproduction tend to get eliminated from the population over time. This process is called natural selection. When it changes populations, we call it evolution.
- 1.33** So evolution by natural selection is just changes in gene frequencies over time due to differential survival and reproductive success. It is how we can understand many of the

qualities that tend to define a species, that species' nature, because many of the qualities that we have have an impact on survival and have an impact on reproduction, and those that work get inherited and those that do not, do not.

- 1.34** Darwin, when he first developed his theory of evolution by natural selection—again, this tendency that we should expect to see that gene frequencies will change over time based on those genes that provide a survival and reproductive advantage will be inherited, and those genes that provide a survival and reproductive disadvantages will tend to get removed from the population— this was a really parsimonious theory because it helped to explain a lot of things that up until the time of Darwin's theory, nobody was really able to provide an explanation for.
- 1.35** One of those things was simply this idea of change over time. People in archaeology had been digging up fossils for a long time that looked similar to existing species, but not exactly the same. Before the theory of evolution by selection was put out there, nobody knew what to make of these things. It's like, "What is this skull? Is this a human? Is this a nonhuman?" Because it's not quite human, but it looks human-like. It explains that, "Oh, human skull— the shape of the human skull has changed over time due to differential survival and reproductive success."
- 1.36** What we tend to find with the human skull is that ancient human skulls tended to be more robust because we had to have these big, huge muscles to be able to chew our food. Then once we started cooking, that took selection pressure off the need to have these big, expensive jaw muscles. What we see is that more modern human skulls are more gracile. They're prettier. They're smoother because they don't need to have all these muscles attaching to all these different structures, which explains some of these ridges that we see on some of these older human skulls.
- 1.37** So, having Darwin's theory helped explain change over time. It helped explain the fossil record, which was something that nobody was really able to explain before. And it also explained the apparent purposive quality of so many traits that organisms have. When you look at the natural world, there are so many adaptations, there are so many traits that you see that organisms have that seem improbably well-suited to the problem that they solve.
- 1.38** For example, the porcupine quill is a really great tool for a porcupine to avoid predation. Or if we look at something like the prehensile tail on New World monkeys, this allows them to hang from trees—prehensile means grasping. So having this grasping tail allows them to be able to swing from the trees. It allows them to be able to eat, and it allows them to be able to carry their offspring.
- 1.39** This is really nice; we're able to explain these traits in terms of the function that they serve, and that so many organisms have so many traits that seem to serve these survival functions. These were things that were never really appreciated before Darwin's theory: the idea that we should inherit traits that work because they allow us

to survive, and they allow us to reproduce. So, we should expect to see that many of the qualities that organisms have are going to be functional and are going to serve some purpose.

- 1.40** In Darwin's theory, also, once he was able to elaborate on it by developing the theory of sexual selection, it was also able to explain some of the things that his original theory wasn't able to explain, and that was things like sex differentiation and costly traits. When Darwin first developed his theory of evolution by selection, one of the things that really, really bothered him was the sight of the peacock tail. Because the male peacock tail is big, elaborate, metabolically costly, and it makes the males at a greater risk of predation. Darwin would see this, and he couldn't understand why it is that male peacocks are inheriting the genes related to these tails, given how expensive they are.
- 1.41** Well, it turns out that the female peacocks really like the tail. The bigger the tail, the bigger the mating success for the males. The sexual selection theory tells us that even having traits that are really good at survival are only going to get inherited in as much as they also allow you to be able to reproduce. What this means is that we'll also expect to see, in addition to the fact that many organisms are going to have qualities that make them really good survivors, is that many organisms are also going to have a lot of traits that just make them really good at being able to have sex.
- 1.42** For a male peacock, having a big elaborate tail increases the probability of being chosen as a mate by the female peahen. That's why we tend to see the evolution, or we tend to see that so many male peacocks have these big elaborate tails—because it's been inherited, because it provides those males with a mating advantage.
- 1.43** The other thing that Darwin's theory of sexual selection was also able to be able to explain was sex differentiation. That is: Why is it that across so many species, the males and the females don't look similar to one another? It was when he had this aha moment that he had when he was able to explain things like the peacock's tail, he was also able to explain sex differentiation.
- 1.44** That is that sometimes you're going to have a male path and a female path of evolutionary change because of differences in mate preferences. Whereas, for example, with a female lioness, if she prefers to mate with males who have that big elaborate mane, then we're going to tend to get the evolution and selection of genes related to largemanes in males that get inherited because of the female preference. If the male preference isn't the same, it's going to lead to sex differentiated traits.
- 1.45** Darwin's theory was a real landmark in terms of being able to understand the behavior and the appearance and the traits that are possessed by animals across the animal kingdom. It's only more recently that it's begun to be applied to understanding human behavior. The field of evolutionary psychology uses these principles to try to understand the way that humans think, feel, and experience the world.

- 1.46** Understanding this idea that not only should we expect to see that things like the shape of the human skull has been something that has been acted on by the process of evolution by selection, where you see change over time based on differential reproductive success, but that we also can use these principles to understand the nature of human nature. We can try to use these theories, or we can apply these principles to understand what it is that characterizes humans as a species.
- 1.47** One of the things that this theory is most remarkably good at is being able to understand and provide a theoretical framework for understanding differences between the sexes and understanding why it is that even in a species like humans, where we have all these amazing, wonderful cultural influences on us, why at the core, there tends to be sex differentiation, not only in things like physical appearance, with, of course, males having larger body size than females, but also psychological traits.
- 1.48** That's one of the next things that we're going to be talking about: we're going to be talking about biological sex. We're going to be talking about what sex is from an evolutionary perspective, and why it is that something really small, really small differences between males and females, can ultimately lead men and women to go on two different but similar evolutionary paths.
- 1.49** To start with, let's talk about some definitional things. When you're a biologist, when you're talking about the differences between the sexes, when we're defining what sex is, it has everything to do with the size of your sex cells. If you have large, relatively immobile, expensive gametes or sex cells—gamete is just a fancy word for a sex cell—if you have large, immobile gametes, we call you a female. Females have eggs; that is how we define females.
- 1.50** Males have smaller, more mobile gametes that they're able to produce into perpetuity. They're relatively inexpensive to produce. There's a lot of them. They tend to move around. We call you—if these describe your gametes—we call you a male. So females are the ones who have eggs. Males are the ones who have sperm. This small difference, this little tiny difference in the size of our sex cells, actually ends up setting the stage for sex differentiation and provides a foundation for all reliably occurring sex differences between males and females.
- 1.51** One of the things that we're going to be doing now is walking through how it is that this tiny little difference creates such a difference in terms of the evolutionary path that has been followed by males and by females. Why does gamete size matter so much? Well, one of the reasons it matters so much is if you imagine a couple, before a man and a woman ever even meet, if they are going to have children, the female has already invested more in their children than the man has. They haven't even met; they haven't even had a first date, and the woman has already invested more in their future offspring than the man.

- 1.52** The reason for this is that eggs are metabolically expensive. They're in limited supply, and they're more difficult to get right than sperm. All of these things create a context in which women invest more initially before even meeting a man in the production of offspring. But beyond that, it also lays the groundwork for continuing differences in parental investment. We consider investing in the size of your sex cells. We consider that a form of parental investment. But the differences in levels of parental investment continue.
- 1.53** The reason that we tend to see that the size of your sex cells lays the groundwork for continuing differences in obligate parental investment is the following. Females produce large gametes; we know that, and that's something that requires greater energetic investment. They're more metabolically costly to produce. What this means is that females are going to have more to lose if their offspring is not successful. This means that evolution is going to reinforce whatever additional investment is needed to ensure that the offspring is going to succeed.
- 1.54** Think about it like a game of poker. If you're playing a game of poker and you put in \$5, you've got a lot less to lose than somebody who puts in \$100. If you put in \$100, you're going to keep on putting in additional money to make sure that you get that \$100 back. So it goes in poker, so it goes in evolutionary biology. We tend to see that whichever sex makes the larger initial investment also continues to make the larger continued investment.
- 1.55** There are some exceptions to this that we'll get to that are actually super interesting, but for the most part, in most species, what we tend to see is that not only do females have the larger initial investment, they also have the larger continued investment. This lays the groundwork for differences in what's known as obligatory parental investment.
- 1.56** First, some definitional things: What is obligatory parental investment? Obligatory parental investment is the minimum amount of investment that needs to be made by a male or a female in reproduction. In human beings, the absolute bare minimum investment that a woman must make is nine months: pregnancy. That's not even counting the fact that throughout most of our evolutionary history, lactation wasn't something that was up for debate. You had to breastfeed. So that minimum investment in an ancestral environment was at least a year, a year and a half—a very large minimum investment.
- 1.57** The minimum investment that a human male must make in reproduction is 5 minutes, 10 minutes, whatever amount of time it takes to be able to convince a woman to have sex, have sex and then leave. That's the bare minimum investment that a man must make in reproduction.
- 1.58** Now, this isn't the typical male investment in reproduction. Human males are actually really special across the different species of mammals in how much males typically invest in provisioning and caring for offspring. But these minimum differences matter.

The bare minimum investment differences between males and females in reproduction are very different, with the females having a much larger minimum investment than what is required of men.

- 1.59** It turns out that the optimal course for getting genes into the next generation differs depending on whether you invest less or whether you invest more. We know from this theory—this is Trivers' 1972 parental investment theory—one of the most important theories in evolutionary biology since Darwin's original theory. According to Trivers' theory, we should expect to see that whichever sex it is that must invest more obligatorily should be choosier about mates.
- 1.60** The reason for this, of course, being that they have more to lose if they make a bad decision, because that minimum investment is nine months plus. If you mess that up, you're on the hook. For a male, if you make a bad mating decision, you change your phone number, and you get out of town. That's your minimum investment. So we should expect to see that the sex that invests more should be more choosy about mates, while the sex that invests less, because they're having to compete for access to the choosier sex, and being choosy by definition, means saying no more frequently than being less choosy, they're going to need to be more competitive for access to the greater investing sex.
- 1.61** Already, just based on the minimum levels of investment necessary for reproduction, we get males and females going off on two separate evolutionary paths. This means that we're going to get sex differentiation in mating strategies and competitive strategies. To give you an example to work through of this, I want you to consider the costs and benefits of short-term mating. In short-term mating in the world of evolutionary psychology, we're talking about short-term sex, casual sexual encounters.
- 1.62** Let's talk about the benefits and costs related to short-term sexual encounters, depending on whether you have a male or female body. Some of the benefits—just to talk about the benefits of short-term sex—these are sex differentiated. For the lesser investing sex, and in human beings, again, that's men, short-term sex can create a low-cost opportunity to get your genes passed into the next generation.
- 1.63** Consider the benefits and costs related to short-term mating. For the lesser investing sex, having a short-term casual sexual encounter can provide a low-cost opportunity to get genes into the next generation. We know this because of something that's known as the Bateman principle. This was a principle that was first discovered in fruit flies. What the researcher noted was that males, male fruit flies, were able to increase their reproductive success—which we're going to define as their number of offspring that they produce— simply by gaining access to additional partners.
- 1.64** As you can see from this blue line here, this is looking at reproductive success based on mating success. Mating success is the number of different partners you get access to.

What you can see is that for the male fruit fly, with each additional female that he's able to mate with, he's able to increase his reproductive success.

- 1.65** The same is not true for females, because females with fruit flies, just like it is in humans, females are the greater investing sex. The females have a long incubation period before they're able to reproduce when they're having one set of babies. What they found in the female fruit flies is that females' reproductive success hits a plateau after they get access to one partner. If you have one partner, it's no different than having two partners or three partners or four partners in terms of sexual access because the number of offspring that they're able to produce are the same.
- 1.66** The same is true for humans. If a human male—let's imagine a set of identical twins, and they're 25 years old, so nice peak reproductive years, and they're going to go and, over the next six months, have as much unprotected sex as they possibly can. Let's say that they each end up having had sex with 500 different partners. The male in the twin pair could end up having 500 potential offspring as a result of those 500 different sexual encounters.
- 1.67** For the female, it's one. That's a lot of wasted energy and a lot of costs for very little benefit. But for the male, that's a potential huge evolutionary win in terms of getting your genes passed down to the next generation. So with short-term or opportunistic sexual behavior, you get a greater benefit of those types of behaviors if you are in a male body than you do if you're in a female body. Not only that, but for the greater investing sex, having casual sex can be something that's incredibly costly. There are social costs related to casual sex for both males and females, but the costs to females are greater than they are for males because they are the ones who get stuck raising the offspring.
- 1.68** This here is some data looking at the survivorship of children among people living in the Ache tribe. This is a contemporary hunter-gatherer group that lives in the rainforest of Paraguay. They live in an environmental context that's very similar to the one that we think that humans spent most of history in. What this research did, they were looking at the survivorship of the children of women based on whether or not they had an investing father around to help care for their offspring.
- 1.69** What you can see here, if you look at this first graph, this is looking at the percentage of offspring surviving until age x based on whether the parents are together, which is the top line, or whether the parents are divorced, which is the bottom line. What you can see is that the probability of still being alive at age 15, if you have a dad around, is about 75%, and if you have a dad who's not around, it's less than 50%. This is something that provides a huge cost to a female because, again, she's the one who has this nine-month investment plus time spent lactating, and then to have such a low probability that that child even survives into adulthood—ouch. That's very evolutionarily costly.

- 1.70** On the right-hand side, this is another figure looking at the importance of having an investing father around in terms of the survivorship of offspring. You can see that the darker-colored bars are the families where the dad is not around because the couple is divorced. The lighter-colored bars is where the family is intact. This is the percent of death from each cause by age 15.
- 1.71** What you can see is that for every cause that they recorded, except for birth, because of course, dads can do a lot, but they can't do everything; they can't necessarily save you from death during childbirth—but for every cause outside of that, whether it's sickness, accidents, homicide, or being captured, which thankfully is much less of a problem for people in our environment than it was in these types of environments, death from all these causes is almost double in all cases if you don't have a dad around.
- 1.72** Again, the benefits to a female from short-term opportunistic sexual behavior is lower, the costs are higher. But if you're a male, the costs are lower, and the benefits are higher, and this creates these asymmetries in the benefits related to casual sexual behavior. So we should expect to see that over the long course of evolutionary time, ancestral females, females who are our direct ancestors, would have benefited from being more sexually coy and cautious. Those females who had a tendency to be coy and cautious and delayed sex until they knew that they would be able to get investment, those females would have been able to do better by having better performing children than females who were more opportunistic and were willing to consent to sex despite not knowing whether somebody was going to stick around.
- 1.73** But on the other hand, for men, men may have benefited from, in addition to being willing to engage in long-term relationships, which we do know is true; humans are a long-term pair-bonding species. But nonetheless, males would have benefited from being more open to short-term sexual opportunities than females because males who had that tendency to be willing to capitalize on a sexual opportunity when it arrives, they would have probably performed better over evolutionary time, passing that tendency onto a greater number of descendants, given that that tendency would be related to a greater number of reproductive opportunities, which means a greater number of opportunities for passing that tendency onto offspring.
- 1.74** Do men's and women's sexual psychologies differ in ways that we would predict based on Trivers' theory? Are men more sexually opportunistic than women? One set of studies that was done to look at this question was a study that was done at the University of Florida, more than 20 years ago now, where the researchers hired an attractive male and female actor and actress to go out on campus and proposition people of the opposite sex and say to them, "I've been noticing you around campus and I find you very attractive. Would you be willing to—" and then they asked them one of three things.
- 1.75** Would you go on a date with me? Would you come back to my apartment with me? Or would you go and have sex with me? Then they simply recorded the number of people

who said yes or no to their propositions. Here's what they found. When it came to asking on a date, what they found was that about 50% of the men and women propositioned agreed to go on a date with the person when they were asked. There was really no sex difference here.

- 1.76 When it came to "come back to my apartment," we're already getting some sex differentiation in a pretty serious way, with men actually being more likely to consent to going back to the apartment than they were to going on a date. Again, this idea that men are open to opportunistic sexual activity when such an activity presents itself. For women, less than 10%—it was about 3% to 5% of women—when they've done replications of this, say yes to this arrangement. Very low.
- 1.77 When it comes to saying yes to sex, again, men remain very agreeable to this request, with about 80% of men consenting to this and 0% of women consenting to sex in this context. Again, this just goes to show sex differentiation of our opportunistic sexual psychology, with men being favored to be more opportunistic than women, women being favored to be more sexually cautious than men, and this creating some sex differentiation in our sexual psychology.
- 1.78 One of the things that I've noted is the fact that humans are really cool and interesting to study simply because males typically do invest way beyond the minimum levels of investment necessary in order to reproduce. Human males are characterized by a lot of parental caregiving. Because of this, and because Trivers' theory makes this really nice prediction about choosiness acting as a function of needed investment in offspring, what we tend to see is that men's mating psychology is very different depending on whether they're thinking about somebody as a long-term mate, somebody with whom they might have children, or when they're comparing that to somebody that's just a short-term sexual encounter.
- 1.79 We get a strong difference in how choosy men are. This is because, as Trivers' theory tells us, choosiness acts as a function of investment. The more you have to invest, the more choosy you're going to be because of the investment. You don't want to be wasting that investment. It's like that hand of poker. If you put in 100 bucks, you want to get that back. So you're going to be pickier about the decisions that you make when you're in it for the long haul than you are in the short haul. This is particularly true if you have a male body.
- 1.80 What we tend to see is that when you ask men about what their minimum standards are when it comes to all sorts of different qualities that you might ask somebody about, whether it's physical attractiveness or how kind and understanding they are or how interesting of a personality they have, what you tend to see is that if you average their preferences across all these different traits, what you see is that men tend to be just as picky as women.

- 1.81** If we look at this, this is looking at the minimum acceptability threshold for the presence of traits that people tend to value in partners—again, things like interesting personality, physical attractiveness, kindness and understanding. What you can see is that for women who are the pink bars, women become increasingly choosy with increasing commitment across the board. Men also become increasingly choosy with increasing commitment. They're the blue bars.
- 1.82** But where men are not terribly picky is when it comes to casual sex and one-night stands, because that is a low investment opportunity for men. It's never a low-cost opportunity for women because of the possibility of pregnancy. So you get this sharp sex differentiation, specifically in the context of low-commitment sexual opportunities, with both men and women being equally choosy or nearly equally choosy when it comes to long-term mate preferences.
- 1.83** Again, investment begets choosiness. The more you invest, the more choosy you are, and then vice versa. Although it is generally true that females are the ones who have to invest more obligatorily in the production of offspring, this isn't always true. It's true for something like 98% or it's probably 99.5% of species that are out there, sexually reproducing species, where females are the ones who also, in addition to having the larger initial investment, continue to have the larger minimum investment. It's not always true. There are a couple of species where the males actually invest more obligatorily.
- 1.84** This over here is the African jacana that you can see over here on the left, and on the right is a pipefish seahorse. In these two species, the males actually invest more obligatorily. What happens in the case of the jacanas is that the females will lay eggs really rapidly after having sex with the males, and then the males have to sit on the eggs. The males are the ones who sit on the eggs. This actually frees up the female to go get access to a lot of food, so she's able to turn around and be able to produce more eggs more rapidly. Then she's able to solicit males, have sex, have another brood of eggs that she has that male sit on, and it leads to this gender role reversal.
- 1.85** When you have males as the greater investing sex, they're the ones who are pickier, and the females have to compete for access to them. Even though it's generally true that females have the larger minimum investment, in cases where they don't, we get these really remarkable gender role reversals. The same is true for the pipefish seahorse. The females lay the eggs, the male carries the eggs around in a pouch, and then when the little seahorses come out of the eggs, they swim away. It's the male who has his reproductive resources tied up with reproduction instead of the females. In these sexes, we tend to see very choosy males, and we see very competitive females.
- 1.86** All of this is the exception that really proves the rule of Trivers' theory, and that is that choosiness tends to beget or tends to act as a function of investment. That also sets the stage for differences in the need for competition and the different benefits related to opportunistic sexual behavior. So we can understand differences between the sexes as

reflecting differences in adaptive problems that have confronted men and women, depending on the differences in their reproductive biology.

- 1.87** In areas where men and women have been confronted with similar adaptive challenges throughout evolutionary history, we should expect us to have no sex differences. For example, the preference for sweet and fatty foods that we talked about at the beginning of the lecture, that's something that both men and women throughout evolutionary time have had to worry about getting access to enough nutrients to survive, because most of our evolutionary history, there wasn't a Taco Bell on every corner.
- 1.88** We had to ensure that we were eating enough foods that would provide us with the energy that we need in case of energy shortfalls. Both men and women had to worry about getting poisoned from eating things that were bitter that they thought maybe were safe. So we should expect to see that there's no sex differentiation in terms of our flavor preferences, and there aren't. We tend to see that men and women are the same in these ways.
- 1.89** But when men and women have faced sex differentiated adaptive challenges stemming from their differences in their reproductive biology, then we should expect sex differentiation. We can use this as a framework for understanding men and women and the differences in our psychology. In particular, we should expect to see that women have inherited tendencies that help to promote survival and reproduction for people who have a female body, those who are the greater investing sex, and have to deal with the challenges not only of mating, but also of pregnancy and lactation.
- 1.90** We should expect to see that for males, that we should expect to see that they've inherited tendencies that have allowed them, those with male bodies, those with the lesser investing sex, and the cheap, metabolically inexpensive sex cells, to be able to survive and to reproduce. We can use this as a framework for understanding psychological differences that we see between men and women, as well as hormonal differences that we see between men and women.
- 1.91** I'm going to start by talking about some of the adaptive challenges that have confronted those who have female bodies. We'll use this, we'll use it as a talking point with this—we'll use mate preferences as an adaptive challenge that historically both men and women have needed to solve. Both men and women, over the course of evolutionary time, have been confronted with the problem of identifying a good mate. If you're going to have children and you're going to reproduce, that means that you need to find somebody that you're going to reproduce with.
- 1.92** It turns out that the different types of qualities that we most prefer in our partners should reflect each sex prioritizing those cues that historically would have helped to promote the survival of themselves and their offspring over evolutionary time. When we're thinking about this as an evolutionary psychologist or as an evolutionary

biologist, we ask ourselves: What are some of the unique challenges that human females historically would have been confronted with when choosing mates, specifically because they've had to deal with the adaptive challenges related to pregnancy and childbearing, which is something that has not confronted men.

- 1.93** We should expect to see, because women are the ones who have to bear the cost of pregnancy and have to bear the cost of lactation, we should expect to see that women, when they are choosing partners, should prioritize qualities that are related to provisioning and protection because pregnancy is a time of intense vulnerability for women. I don't know whether or not you have ever been pregnant or had children before, but it's not an easy time to get around. You imagine the types of environments where we spent most of our history—it would be very difficult to run away from attackers or run away from a sabre-tooth tiger when you're eight or nine months pregnant. It's just a very difficult thing to do.
- 1.94** Then, because women are also lactating, they're having to care for young offspring, that is also not optimal for hunting. If you can imagine a baby crying when you're trying to hunt, it's not going to be an optimal setup, or even worse, a two-year-old. Imagine trying to take a two-year-old hunting. It's probably not going to do you any favors in terms of being able to track down the wildebeest. The wildebeest is going to know that you're coming because the two-year-old is being silly.
- 1.95** So we should expect to see that when women are choosing long-term partners, again, people with whom they might be intermingling their genes into the future generations, that they should be interested in cues related to the ability to provide resources because, of course, because of pregnancy, women are going to go through periods of time when they're less able to get those resources on their own, somebody who's willing to invest and commit.
- 1.96** Of course, this is particularly important for females because they are the greater investing sex. The lesser investing sex doesn't need to invest. The male can just deposit sperm and leave, which we know that is something that, even in humans, sometimes happens. So females should also be really attuned to whether or not this is somebody who actually seems like they're going to stick around and continue to invest despite the fact that their minimum investment is so low.
- 1.97** We should expect to see that women also place a heightened preference on qualities related to investment and commitment, and also the ability to provide physical protection. That's something that women would have benefited more to a greater extent than men, again, because of the physical limitations that are created by pregnancy. It makes it a very difficult time. Women are in an intensely vulnerable physical state during that time. The expectation is that they would have, throughout most of our evolutionary history, relied very heavily on men for protection during this very vulnerable time.

- 1.98** Now we can just ask ourselves: Do we see that women, have they inherited this tendency to prioritize these kinds of qualities, would have benefited our ancestors in terms of their mate choice? We'll start by talking about whether or not we see differences in women's preference for access to resources.
- 1.99** Evolutionary psychologists—this was a study that was done back in 1989, so a very long time ago, but this was the very first study of cross-cultural differences and similarities in mate preferences that was conducted in countries around the world. This study was done in 1989 by David Buss, again, who was my mentor, and who's also one of the founding figures in the field of evolutionary psychology. He did a study looking at 37 cultures around the world—you can see them up there on the map with the little red dots of all the different cultures that he went and studied.
- 1.100** He asked men and women a series of questions about how important different qualities were in their choice of long-term romantic partners. When it came to what he called "good financial prospects—this is, "Does this person have the ability to get a good-paying job?" That's how we would ask that in our current environment. "Is this somebody who's able to provision resources?" is how we would ask that question in a more traditional hunter-gatherer type of society.
- 1.101** What he found was that of all the 37 cultures that he studied, there wasn't a single one studied where men valued this more than women. Women valued this to a much greater extent than men in every single culture that was studied. This is something that has since been replicated. Every decade, they go back and collect data, and it continues to replicate now, with women placing a much greater emphasis on these qualities than do men in partner choice.
- 1.102** This was a more recent study where researchers were looking at women's preference for resource access in their partners by studying dating profiles and looking at the extent to which men and women were receiving responses to their dating profiles. What they found is that women's responses to men's dating profiles increased as a function of the man's educational level. Those dark-colored bars are looking at the frequency of responses that men got based on their dating profile in the education level that they stated.
- 1.103** What you can see is that men with a high school education got a greater number of responses than men who had only an elementary school and GED education, and men with a university education had significantly more responses than men with a high school education, and so forth. When it came to men's responses to women's profiles, no difference is based on their educational level. If anything, it was trending such that women with greater levels of education were getting less responses, although that difference isn't statistically significant. It looks like it's heading that way, but it's really not significantly different.

- 1.104** Again, just going to show that women are really interested in cues related to access to resources. This is cross-culturally ubiquitous. It continues to play out today. Here is—this is looking at age preferences. In addition to differences in education level with women having a heightened preference for men with greater levels of education, we also tend to see that women have a greater preference for men who are older than themselves. Age is one of those qualities that's related to resource access, and this is certainly true now.
- 1.105** As you'll learn, you will earn more in your 40s than you do in your 30s, and you earn more in your 50s than you do in your 40s. This is also true in a historical context because, with people, with age also comes wisdom when it comes to hunting and when it comes to surviving and being able to create the types of social networks that help to promote survival of your kin group.
- 1.106** What we tend to see is that men tend to prefer partners who are younger than themselves, and we'll talk about that and the reasons behind that in a couple of moments. But with women, they generally want somebody who's a little bit older than themselves. It shows this, that there is—and this is another one of these preferences that tends to replicate cross-culturally. It's replicated over time, and it tends to be very constant with women desiring somebody who's older than themselves. Again, because this is a cue related to resource access.
- 1.107** Here's a recent replication. I told you that, every 10 years, they go and they test these sex differences and mate preferences cross-culturally, and now they're up to 45 different countries. It started off at 37, now they've got 45. What this is just showing you is that women, to a greater extent than men, value good financial prospects—that's the second quality from the bottom, showing that women have a heightened preference for that relative to men, and that women have a heightened preference for somebody who's older, while men have a heightened preference for somebody who is younger. Again, going to show that many of the qualities that women desire, at least some of the qualities that women desire in their partners, tend to converge around cues related to resource access.
- 1.108** When we pick up in our next lecture, we're going to talk about what the origins of these preferences are because even though it was proposed and it was hypothesized that we would see sex differentiation in the desire for access to resources and that this is something that has evolutionary roots, it has been argued that this may not be evolutionary at all, and that this might just be a byproduct of sexism, and that structural inequalities that allow women to have less access to resources may be actually what's responsible for sex differentiation in the desire for resources. We're going to talk about some of the data that have been collected and what they tell us about the degree to which these tendencies are evolutionarily inherited or reflect cultural biases in resource access due to sexism.

Lecture 2: Hormones and Mating

At a glance — Hormones and mating — how hormonal state shapes attraction and behavior.

- 2.1 When we left off, we were talking about the fact that researchers find that, no matter when you study it and no matter where you study it, that women tend to express a greater preference for partners with access to resources relative to men. And evolutionary psychologists propose that women have inherited this predisposition toward resource preference in their mates because, historically, women who had this preference would have been having more successful offspring than women who didn't have this preference.
- 2.2 But others have proposed that maybe this doesn't have evolutionary roots after all, and maybe it's just the result of what they call structural powerlessness. This is where women, because they have fewer resource access gaining opportunities, are forced to be reliant on men for access to resources. And if we take that situation away, and when we look at women in a context where resource access is more egalitarian, then we might expect to see that the preference for resources also vanishes. So, does it?
- 2.3 The first test of this is what's known as the cross-cultural test. This is just looking at whether or not women's preference for good financial prospects—that's what GFPs is—whether women's preference for good financial prospects varies as a function of female resource access across cultures. And to test this, researchers went back to the original 37 culture study.
- 2.4 This is just a snapshot of some of those data, but they looked at all 37 cultures, and then later on, the additional cultures that they added and looked: Is there a relationship between women's access to resources and their preference for resource access in men? The answer ends up being no. The answer is no. Women's preference for resources does not vary based on their own limited access to resources on their own.
- 2.5 This suggests that women's preference for resource access is not simply a byproduct of the fact that many societies have a sexist structure where women are less able to gain them themselves. What the researchers found is, there was no relationship between the amount of egalitarianism within a society and women's preference for resources, suggesting that this preference may operate independently of their own resource access.
- 2.6 The second test that they wanted to do, to test whether women's preference for resource access in mates is just a byproduct of the fact that they are not able to get resources of their own, is what's known as the within cultures test. This is where researchers wanted to look at the mate preferences of women who have resources.

- 2.7 Because if it's true that the only reason that women desire resource access in their partners is because they're not able to get resources on their own, we should expect to see that women who do have them, who manage to get a job that pays well enough that they don't have to rely on a man for resources, that these women will no longer have the preference for resource access in partners. This also was not supported.
- 2.8 In fact, it turns out that women who have high incomes actually value good financial prospects on the part of their partners to an even greater extent than women without high-paying jobs. Women are what we call hypergamous. Hypergamous means that you like to mate up. Women generally have a preference for mating with somebody who has at least as much education and resource access as they have themselves.
- 2.9 Both of these pieces of evidence are consistent with the idea that we may have inherited this tendency to have a heightened preference for partners with resources because, by so doing, this has provided our female ancestors an advantage in being able to provide for themselves and for their offspring, and that we are the result of this evolutionary legacy of inheriting this tendency, that we've inherited from our female ancestors, who were able to be successful because of the mates that they chose.
- 2.10 We, again, can use this framework, this meta-theory of sex differences in mate preferences, as a way to understand some of the differences that we see between men and between women. We spent some time focusing on women's mate preferences, and we talked about one type of preference in particular: the preference for partners with resources. But now I want to turn the tables a little bit and talk a little bit about men's preferences.
- 2.11 Because this meta-theory on sex differences in mate preferences predicts that men and women should have similar preferences when the adaptive challenges that they've been confronted with are the same, and they should be different in areas in which the adaptive problems associated with mate choice have differed. Now we're going to be talking a little bit about some of the male-specific adaptive problems inherent in partner choice.
- 2.12 For men, one of the things that they've had to worry about when choosing mates that women haven't had to worry about is identifying mates that are fertile and have what we call high reproductive value. Fertility refers to the momentary probability of being able to conceive at any given moment in time. For women, fertility peaks around age 25 and then begins to fall. You can see, on this figure on the left, looking at female fertility over years in life, and you can see that it's at its highest right around age 24-25, and then it begins to decline relatively slowly until age 34 and 35, which is when things really start to decrease pretty dramatically for women.
- 2.13 And so, for men, being able to identify a mate who's still fertile is something that men have had to worry about, and women have not. Because, of course, men's fertility goes on almost into perpetuity. Men's fertility generally doesn't start to decline until they're

in their late 60s or 70s. So the male reproductive lifespan is significantly longer than the female reproductive lifespan. Reproductive value is a very similar construct to fertility, except that it refers to—it's a theoretical construct that refers to the number of offspring that an individual can potentially have into the future.

- 2.14 For women, this is also something that is tied very tightly to their age. For a woman, reproductive value peaks at the time that she gets her first period when she becomes reproductive. At that point, she's at the point in her life where she's survived long enough to be able to have children. And this is the point in her life where the number of potential children that she can have in the future is at its absolute highest.
- 2.15 So let's say that the greatest number of children that a woman can have in her lifetime is 20—I'm just making up a number. So let's say that it's 20. With each year of life, that goes down by almost 1, because it takes women so long to reproduce, and then you hit menopause. So women's fertility—as we know from that graph figure—women's fertility begins to decline in their mid-30s, late-30s, in pretty dramatic ways. And so, here you have menopause constricting female reproductive value.
- 2.16 Further constricting female reproductive value is the fact that each pregnancy takes nine months. So women have very limited reproductive value, meaning that they only have so many offspring that they can potentially have in the future, just because of the limitations on their own bodies. This isn't true for men. Men's reproductive value is almost infinite. Because if a man, with our thought experiment that we did in our first class, where we talked about a man having sex with 500 different women, that could be 500 potential reproductive opportunities.
- 2.17 Thankfully, most men don't take this opportunity, and most men wouldn't be able to get away with it anyway, but this is something that makes men and women different from one another, is that they have these differences in fertility and reproductive value, with men's reproductive value and fertility being long-lasting and abundant, and women's fertility and reproductive value being constricted and relatively limited.
- 2.18 What this means is that, for men choosing a mate, they've had to worry more about a potential partner's fertility and reproductive value and have to pay attention to cues related to these qualities to a much greater extent than do women. Women didn't have to worry about whether they chose a partner who is older than themselves, for example, because if they choose a partner who's older than themselves, not only will they probably have more status and resources, they're probably more mature and able to settle down.
- 2.19 If a man chose a partner who's older than themselves, that could mean that she's not able to reproduce anymore. That could mean further limitation of her already-smaller reproductive value. We should expect to see that, when men are choosing long-term mates, they should be really focused, to a much greater extent than women, on qualities that are related to fertility and to reproductive value.

- 2.20** Now, obviously, a person's fertility and reproductive value isn't something that is tattooed on their forehead. We don't know what somebody's fertility and reproductive value is just by looking at them. But in some ways, we kind of do, because fertility and reproductive value are both related to youth and many of the other standards of attractiveness that we tend to have for women.
- 2.21** One of the things that we talked about in the last lecture is this idea that there's nothing inherently delicious about a hot fudge sundae. It's that our brains find those molecules pleasurable. Our brain rewards us when we have sugar molecules on our tongue. The same is true for a beautiful woman. There's nothing inherently beautiful about a beautiful woman. It's that our brain rewards us from looking at certain types of qualities and finds those qualities desirable.
- 2.22** What are those qualities? Well, some of the standards of beauty that we have for women are things like having full lips, clear skin, clear eyes, lustrous hair, having good muscle tone, health, youthfulness. All of these qualities that we tend to find beautiful in women are also related to fitness. They're also related to their fertility and their reproductive value. We know that, as women age and their fertility declines, we know that their hair becomes less lustrous, their muscle tone begins to decrease, the fullness of their lips begins to decrease.
- 2.23** Of course, we have all these procedures now that can allow women to try to postpone these things into perpetuity as well. But essentially, it's because we're trying to converge on these standards of attractiveness that we have for women, and these standards have evolutionary roots, because these qualities for most of our evolutionary history were related to fertility and to reproductive value.
- 2.24** Currently, they're not—not as much as they used to, because we do have all these potions and different types of cosmetic procedures that can allow women to show these qualities, demonstrating the cue, even though the cue is no longer necessarily tied to fertility and reproductive value. But for most of our history, it was. Because of this, we should expect to see that men, to a greater extent than women, should value cues related to physical attractiveness and beauty than do women when they're choosing their long-term mates.
- 2.25** Now I'm going to turn back to that 37 culture study, that very first cross-cultural investigation of men's and women's mate preferences, and talk about some of the results there. And again, here I'm providing you only a snapshot of the different cultures that were studied. But what the researchers found in this particular study and in subsequent replications of this study was, this is another one of these qualities where there wasn't a single culture studied where women valued the physical attractiveness of their partners to a greater extent than did men.
- 2.26** Men reliably preferred physical attractiveness and said that physical attractiveness is more indispensable in their choice of long-term partners to a much greater extent than

women do. So men value this more than women. And again, this is because, historically, those men who had the preference for qualities related to what we consider beautiful in women, those men would have passed those preferences down to a greater number of offspring, because those preferences are related to fertility and reproductive value. It's a very inheritable set of mate preferences to have.

- 2.27** Men who had a heightened preference for cues related to being postmenopausal wouldn't have passed down very many genes. They wouldn't have passed that preference down to many offspring because their partners would be non-reproductive. We also—again, this is that replication that was recently done, where they've now expanded their data collection pool to 45 different countries. Again, here I want you to be focusing on the green dots.
- 2.28** What you can see here is that the green dots, when you see something clustered on the right-hand side of the bar, that means that it's something that's preferred by males. When it's clustered on the left-hand side of the bar, it means that it's preferred more by females. What you can see here is that men, to a much greater extent than women, tend to value physical attractiveness. They also tend to value youthfulness.
- 2.29** That bottom set of dots, that red cluster, is looking at male minus female preferences for age choice. And men have a heightened preference for younger mates, whereas women tend to have that heightened preference for older mates. Again, this is because, even though, to a man, choosing a mate, a long-term mate who's older, would also come with some advantages. Because as you get older, you do have more resource access, you have more wisdom, you generally have a larger social circle, you have more social status, and that would have provided benefits to females.
- 2.30** But for men, even though they would get those benefits, too, it would be grossly outweighed by the cost of choosing a mate who is infertile. So men tend to have a heightened preference for younger mates, because this would increase the probability that they're choosing somebody as a partner who has high fertility and has a long reproductive value or high reproductive value.
- 2.31** We can understand the differences between men's and women's psychology. This is true for mating and extends beyond mating. It extends into the psychology of competition, and it extends to the psychology of friendship, and it extends to the psychology of coalition-building, by understanding the sex differences and the adaptive challenges that men and women have been confronted with over evolutionary time, stemming from the differences in their reproductive biology—those really small differences in the size of their sex cells, stemming out to create these two different evolutionary paths that lead to sex differentiation, not only in physical traits, but also psychological traits.
- 2.32** And our sex-differentiated reproductive biology is also at the heart of hormonal differences that we see between men and between women. So men and women differ

in their sex hormones. The reason for this is that men have one job to do in reproduction, and women have two. What this means is that men only need to have one primary sex hormone. The job that men's bodies need to do in order to successfully reproduce is to be able to attract a mate and then have sex with them. This is one set of jobs. Because of that, men have one primary sex hormone, testosterone.

- 2.33** For women, the female body has to do two different jobs to reproduce. The female body has to attract a mate and have sex, but then they also have to allow for the implantation of an embryo and allow for pregnancy. These are two qualitatively different things that the body needs to do. Because of this, women have two primary sex hormones instead of one.
- 2.34** Rather than having one hormone that's directing the jobs related to sex, only the female body has to cycle between being in a state that's optimized for attraction and for sex, and then shifting into a state that's optimized for implantation and for pregnancy. Before we start talking about some of these differences and the differences between the male primary sex hormone, testosterone, and the female primary sex hormones, estrogen and progesterone, we're first going to talk about what a hormone is.
- 2.35** Because hormones are oftentimes misunderstood and mischaracterized, blamed for things like having an ill-timed pimple and all of the awkwardness of puberty. But a lot of people don't actually really know what they are and what they do in the body. So I just want to take a couple of minutes to give you a crash course on hormones. A hormone is just a signaling molecule that communicates messages throughout the body.
- 2.36** A hormone, you can almost think of them as working in the body like a loud speaker. It's sending an announcement throughout the entire body from head to toe, including your brain. Our brain has receptors for hormones all over the place. But when hormones get released, they're sending a message throughout the entire body to get all of the parts of the body working together to do whatever it is that the brain wants the body to do.
- 2.37** For example, we have hormones that regulate sexual behavior. When our brain says, "We're going to have sex," hormones get released and it sends messages throughout the body, preparing every cell in the body for the possibility of sexual activity. Hunger—hunger hormones are things that get released, and it gets all of our body, all the cells in our body that are involved in feeding behavior, ready for the possibility of feeding.
- 2.38** Growth hormone—that gets released when our body wants to grow. It sends messages throughout the body from head to toe, telling cells, "It's time to divide. We're growing." It's like a loudspeaker making announcements throughout the body, so that way all the cells in the body that are involved in whatever it is that's being announced are able to do their job and chip in and help to get whatever that job is done.

- 2.39** Hormone release usually starts in the brain—our brain is the ultimate arbiter of what goes on in the body. Most hormone—not all hormone release, but most hormone release—starts in the brain. The brain tells the pituitary gland, which is a little gland that lives right below the brain, it's about the size of a pea, the brain tells the pituitary gland when it would like some hormones to be released. And then the pituitary gland, its job is to communicate with endocrine organs in the body—an endocrine organ is just any organ in the body that produces hormone.
- 2.40** So the brain tells the pituitary to tell whatever organ it is that's producing the hormone that the brain wants released, that it's time to release some hormones. Let's say, for example, that the brain says, "Oh, my goodness, I'm so stressed out. We need to release some stress hormone." It will tell the pituitary that. The pituitary will tell the adrenal glands, which are responsible for releasing the stress hormone cortisol. It will tell them to start releasing cortisol.
- 2.41** Let's say that cortisol starts being released. The hormones get dumped into the bloodstream. What makes hormones such an effective communication strategy in the body, and the reason I liken them to a loudspeaker, is, they do get into the bloodstream, and then they travel everywhere that blood goes, which is everywhere. Blood goes everywhere, which means hormones go everywhere, and they get picked up by every cell in the body that has a receptor for the hormone.
- 2.42** Cells in the body will have receptors for a specific hormone if the cell has something special that it's supposed to do when the hormone is being released. For example, in the context of stress hormones, we know that one of the things that has to happen is that fat and sugar have to get dumped into our bloodstream. Stress hormone, that's one of the things that it does, and it's doing that to prepare our body for the possibility of needing to make a fast getaway.
- 2.43** If you're feeling anxious about something, stress hormone, one of the things that it will do is, it will communicate with the cells that are responsible for dumping glucose into the bloodstream. Then those cells will go and do that thing because that's their job when the body is stressed. That same stress hormone will get picked up by other cells in the body and do other things. Any cell that has a job to do in response to stress is going to have receptors for the stress hormone.
- 2.44** That's going to change what the cells do. They're going to run the stress program. Or if growth hormone is released in the body, the cells that are responsible for growing, they're going to have receptors for growth hormone, and when growth hormone is released, they're going to run the growth program. Hunger, they run the hunger program. Bonding, they run the bonding program. It's almost like software. You can think about hormones as being a lot like software, like a software program that you run, and it's running different objectives that your brain would like your body to follow.

- 2.45** Hormones get picked up by all cells that have receptors for the hormone, and this causes them to change what they're doing. When we're talking about sex hormones, which we also call reproductive hormones, they are in charge of reproduction. When our body is releasing sex hormones, they lead our body, and all the cells in our body that play a role in either sexual development or sexual behavior. It causes them to change what they're doing and to do whatever is necessary for the body to be able to have sex.
- 2.46** These have two different types of effects in the body. When we're talking about sex hormones, they can have both what we call organizational and also activational effects in the body. The organizational effects of sex hormones are the changes in our body that are permanent as a result of the release of sex hormone. These are the things that happen when sex hormone is released during development.
- 2.47** The first time that sex hormones play an organizational role in the body is in the uterine environment. This is actually responsible for sex differences that occur between little boys and little girls when they first come out of the womb. When little boys and little girls first come out of the womb, there are already brain differences between the two. One of those differences that we know is that the male brain is more lateralized than the female brain. This just means that the female brain is more interconnected between the two hemispheres than the male brain.
- 2.48** One of the reasons for these differences in lateralization is because of the release of the sex hormone testosterone during development in male brains. This leads them down a male brain trajectory, which leads to developmental changes that are permanent. You get that exposure to prenatal androgens or prenatal testosterone, and that leads to greater lateralization of your brain, and then you've got that for the rest of your life. It's a developmental change, and it's not going away.
- 2.49** We also get another shot of sex hormones that cause organizational effects in our body that happen during the pubertal transition. As you're going through puberty and your sex hormones are going crazy, as we all remember probably, during that time our brain is totally reorganizing itself, from a childlike phenotype into its grown-up phenotype. You're no longer a child, you're turning into a grown-up, and your brain has all kinds of changes that it undergoes during this period of transition.
- 2.50** This is also something that is coordinated by our sex hormones. We can have these organizational effects—organizational effects are called that simply because they are more or less permanent. There's always a little bit of plasticity in the brain; the brain remains a very plastic organ throughout our lives, but some of that plasticity has limitations around it, depending on these organizational effects. We know that sex hormone has these organizational effects—again, with the two big periods of time that are really critical in terms of their organizational effects being the uterine environment and then during puberty.

- 2.51** Sex hormones also have what are known as activational effects. Activational effects are the "now you see it, now you don't" actions of hormones. For example, right now, if I were to take a bunch of testosterone, I would probably be really energetic and be more aggressive and more quick to fly off the handle. I'd be more sexually aroused. There's all kinds of things that would happen in response to that. And as soon as that testosterone was out of my body, the effect would be gone. I'd be back to myself.
- 2.52** These are the activational effects of hormones. They're transitory. This is where you're coordinating sex-differentiated—when we're talking about sex hormones, this is where you're coordinating sex-differentiated patterns of behavior, many of which are related to reproduction. Again, when we think about hormones and the effects of hormones, most of the effects that we're thinking about are these activational effects. These are the effects that you're experiencing as your body is releasing sex hormones right now.
- 2.53** When we're talking about hormones and talking about men's primary hormone testosterone and women's two primary hormones, we're going to be talking about them primarily in terms of these activational effects. And when we're talking about activational effects, these are also the same types of effects that you get in the context of the birth control pill, and even in the case of things like steroid use in men, and in menopausal hormone therapy in women. These are activational effects, where it's happening now, and then when the hormone leaves your body, the effects go away.
- 2.54** So now we're going to take a little minute, and we're going to get to meet our hormones. We're first going to talk about men's primary sex hormone, which is testosterone. As we noted, men only need to have one primary sex hormone because men's bodies only have one job that they need to do, in order to successfully reproduce, and that is mating. Mating is a big—that's a big job. It's not just having sex. Mating is also about attraction and about attracting mates. But we would consider that that falls in the single bucket of mating.
- 2.55** We can think about testosterone as being the hormone of mating effort, meaning that it is related to the effort that people expend, and especially men, expend in the service of sex. This can be sex itself. Testosterone motivates actual sexual behavior, but it also motivates the competitive spirit that's required to gain access to the types of resources that women desire, that will attract them as mates. Testosterone, we consider this the hormone of mating effort, and we tend to find that it's related to mating effort in men.
- 2.56** Just to give you some of the outcomes that we know from research that higher levels of testosterone are related to: higher levels of testosterone predict higher levels of sexual desire. This is actually, interestingly enough, true in both women and men. It's also related to a greater interest in short-term mating. We talked about the fact that men tend to have a mating psychology that is more sexually opportunistic than the female mating psychology.

- 2.57** The tendency to think in those terms of opportunistic sex and be interested in opportunistic sex is something that is positively related to testosterone levels in men. It's also related to flirtatiousness. It's also related to the number of sex partners that a person has in their lives. So men with higher levels of testosterone, on average, have a greater number of lifetime sexual partners relative to men with lower levels of testosterone.
- 2.58** It's also related to a greater desire for sexual novelty, that is, new partners. It's also related to greater competitiveness. So men who have higher levels of testosterone also tend to be more competitive, and this can either go all right or all wrong. For example, we tend to see that men who have higher levels of testosterone tend to be overrepresented in areas like being top executives within businesses, where it goes all right, that competitiveness leading you to do these things that are going to give you resources and access to women in socially sanctioned ways.
- 2.59** But in other groups, where people don't have access to traditional pathways for being able to compete, you also see it related to delinquent behaviors, with higher levels of testosterone being related, for example, to a greater probability of landing in prison, if you're of low social status, than not. It tends to be related to more competitiveness, which can either, as I said, go all right, or sometimes it can go all wrong.
- 2.60** Testosterone is something that varies across men, across the day, and across context. First, let's just talk about across men. If I had a marker right now, I would draw for you a nice curve, a nice normal distribution, nice bell curve, showing you that there's variability in men's levels of testosterone. You'll have some men who have super high levels of testosterone. You have some men who have super low levels of testosterone, but most men live in the middle. Most men have average levels of testosterone, by definition. By definition, most men are in the middle.
- 2.61** And so, there's differences between men. You see that some men's is higher, some is lower, and these differences predict some of those outcomes that we just talked about. But we also get differences within each individual man. Within each man, there also lives a bell curve, where each man will have some times during different contexts where testosterone levels are super high, he'll have some times when testosterone levels are super low, and most of the time he lives in the middle.
- 2.62** What are some of the things that influence men's levels of testosterone on a day-to-day type of a basis? Well, one of them is the time of day. Testosterone has a circadian rhythm, which means that it changes on a 24-hour clock, and men have the highest levels of testosterone in the morning when they first wake up, and then it declines over the day.
- 2.63** We also find that testosterone changes—in addition to having that nice circadian rhythm—it also changes based on the costs and benefits of exerting mating effort. Testosterone is the hormone of mating effort. We should expect to see that, when men

are in contexts where the benefits of mating effort are particularly high, they should get a little shoot-up of testosterone. Their body should release more testosterone.

2.64 If they're in a context in which that mating effort is unlikely to be successful, we should expect to see that testosterone levels decrease, to keep them from trying to compete when they have a really low probability of winning whatever it is, whether it's the girl or the competition. Indeed, that's exactly what we tend to find. Men's levels of testosterone change in response to environmental factors that impact the costs or potential benefits related to mating effort.

2.65 For example, we know that one of the contexts that increases men's levels of testosterone is seeing a beautiful woman; that will increase men's levels of testosterone. On the flip side, men's levels of testosterone will decrease when they lose a competition. If a man is involved in some sort of sporting game, and even when they're playing video games, when men lose, their testosterone levels decrease. Again, this is because this provides a signal that this is not a good time to be further competing and exerting mating effort.

2.66 Another powerful predictor of changes in men's levels of testosterone are changes in their relationship status and parenthood status. This is really fascinating research, particularly in our current cultural context, where it seems like everybody's obsessed with having enough testosterone. There's testosterone clinics everywhere. You see them next to every Starbucks. There's a Starbucks and a testosterone clinic. Testosterone's everywhere.

2.67 But it doesn't always make sense for men's bodies to have their foot on the gas of testosterone. The reason for this is, testosterone is the mating effort hormone. Any time you're exerting effort toward one thing, it means you have less effort to be able to invest in other things. Not only then, it's not always a good idea to be exerting mating effort, particularly if you have a long-term relationship that you're trying to maintain.

2.68 And so, because of this, researchers have investigated the relationship between levels of men's testosterone and changes in relationship status, and then also changes in fatherhood. What they find is that men's testosterone tends to take a couple of hits, and these things happen on purpose. Research finds that, when you look at differences between men's testosterone levels between men, unpaired men, so men who are not in relationships, have averagely higher levels of testosterone than do men who are in relationships.

2.69 When men are in relationships, testosterone levels go down. Then, when men are in committed relationships, it goes down even further. This first graph that you can see here on shows, you can see that the men who are unpaired in that sample have the highest levels of testosterone; the men who are married with children have the lowest levels of testosterone; and the men who are in serious relationships are somewhere in the middle.

- 2.70** In the graph on the other side of it, you can see that men who are single and men who are dating in a new relationship, where you still get the benefit of having to exert greater amounts of mating effort, have the highest levels of testosterone. But then, when men's relationships become serious and they're in a committed relationship, men's testosterone levels decrease.
- 2.71** This happens on purpose, and there's nothing wrong with it. This isn't something that's happening where suddenly a man is going to become a total pushover and not be able to protect his partner. Instead, it's just taking a little bit of a back seat to allow for men to be expending effort in other things, like maintaining their pair bond or helping to care for children.
- 2.72** This was some original research that was done examining this issue. But since that time, a more comprehensive study that was done longitudinally looked at the impact of relationship status and fatherhood status on testosterone levels. What they found was really beautiful. This was a study that was done in Cebu, Philippines. The researchers tested, and they followed more than 600 men over the course of four years.
- 2.73** They measured their testosterone levels at time one, when most of the men were unmarried, and none of them had children. Then they measured them four years later when the majority of them were married, and some of them had children. They wanted to look at how changes in relationship status and fatherhood status influence men's testosterone levels over time.
- 2.74** The first thing that they found was that the men who had the highest levels of testosterone at time one were the ones who were most likely to be in relationships at time two, which again goes to show that, when testosterone, this mating effort hormone, is high, these men are exerting mating effort and voilà, they're also successful. The men who had the highest levels of testosterone were most likely to be in relationships at time two.
- 2.75** The other thing that they found was that, for all men, testosterone levels were lower at time two than they were at time one. This is also something that's to be expected. Men's levels of testosterone decline with age. This decline with age happens pretty early. It starts to happen in the late 20s with testosterone levels beginning to decline over time. They also found that.
- 2.76** Then they also found that there were really distinct differences between fathers and non-fathers in their sample. What you can see here on the figure, the red line shows zero change in testosterone. That would be, if you measured a man at time one, and then four years later when you measure him, he has zero change in testosterone, so there's no change at all, he would be right at that red line.
- 2.77** But what you can see is that all men in the sample had a decrease in testosterone over time because that's what testosterone does. It declines over time. But now, the real interesting question is looking at the magnitude of the decline, based on whether or

not somebody has children that they're caring for or not. Because again, this is the type of a context, if you have children, where it doesn't necessarily make sense to have your foot on the gas of mating effort.

- 2.78** You need to be taking the foot off the gas a little bit, so that way you can invest effort away from looking at the sexy neighbor and direct it toward provisioning and being able to care for your offspring and take care of your partner. What they found is that there were major differences between those men who became fathers and those who didn't. On the far left, you can see the bars of the testosterone levels and the change in testosterone levels between men who do not have children at time two.
- 2.79** The differences between the light-colored bars and the dark-colored bars are the differences between morning and evening testosterone levels, because testosterone does have a circadian rhythmicity, and they were just interested in whether or not that testosterone levels change in response to—whether the morning testosterone or evening testosterone was affected in more immediate ways based on fatherhood status.
- 2.80** When you look at the left-hand side, you can see the never-fathers. You can see that their testosterone levels decreased, but not that much, especially when you compare it to the other categories of men, each of which are fathers. The first one that you see next to the never-fathers are fathers who have a newborn baby. This is a baby that's less than a month old. You can see that these men really have their foot on the brake in terms of testosterone levels.
- 2.81** This is a time when children are at their most helpless. It doesn't make sense to have a man producing high levels of testosterone that are going to be distracting him away from provisioning and caregiving, and also more likely to respond aggressively, because that's another thing that testosterone does. As a hormone that's related to mating effort and competition, it also increases aggressiveness. And when you have a newborn baby at home, not necessarily a great time to be flying off the handle in response to being annoyed. So, we see this really big decrease in men who have newborns at home.
- 2.82** But then, even when you look at the fathers that have older children, so the second category of fathers there is fathers who have an infant of one month to one year old, and the last one is fathers who have a child who's older than a year old. But you can see that all of them, even those who have a child that's older than one year old, their levels of testosterone are significantly lower than those who are not fathers.
- 2.83** This, again, is one of these really beautiful shifts that shows the function of our sex hormones and how sensitive they are to the environment and how the changes that we experience in our hormones are oftentimes really functional. We tend to think about, "Oh, low testosterone, that's bad." Not so fast. It might not be bad. In some contexts, it's actually really good. Having hormones that adjust to things that are going on in our environment is incredibly adaptive.

- 2.84** We have to remember the fact that the job of hormones is to coordinate the activities of all the cells in our body and getting us to do whatever the thing is that we're supposed to be doing. In the case of fatherhood, that's not aggressively competing and having sex with as many people as you can get access to. It's a different time. There's a different set of adaptive challenges that are being encountered. That's one of the ways that our body adjusts and helps us to behave in a way that's maximally adaptive, given our context.
- 2.85** This is just one last slide from that study looking at the differences in testosterone levels with fathers. This was one where they were interested in—again, the light-colored bars are the morning testosterone, and then the darker-colored bars are evening testosterone, because they broke down their results that way, although they didn't end up making much of a difference regardless of morning or night.
- 2.86** But here they were looking at whether the amount of childcare that a man did or performed was related to the decrease in testosterone levels. What you can also see here is that the men whose testosterone levels decreased the most were the ones who were having to invest the most in childcare. Again, going to show this really nice relationship between what the body is trying to do and changes in the adaptive challenges that are being confronted.
- 2.87** When there's a need for increased childcare, men's bodies will further tap the brakes on testosterone levels to be able to direct their energy toward childcare, instead of being directed toward things like competition and mating. So men have one primary sex hormone, because men's bodies have one primary job to do in order to reproduce. But the same is not true for women. Women have two primary sex hormones because our bodies have two sets of jobs that we have to do to reproduce.
- 2.88** On the one hand, just like men, we also have to have sex, and we have to do all of the things that are required in order to have sex, meaning attracting and being able to choose partners—that's one set of jobs. The other set of jobs that women's bodies have to do in order to reproduce is they have to allow an embryo to implant and allow for the possibility of pregnancy. These two jobs, because they're qualitatively different and require different things of the body, this is why women need to have two primary sex hormones that cycle, instead of one primary sex hormone that does not.
- 2.89** Women's two primary sex hormones are estrogen—estrogen is the female sex hormone whose job it is to coordinate all of the activities related to attraction and to sex. Then our other primary sex hormone is progesterone, and this is the sex hormone whose job it is to coordinate all of the physiological and psychological changes related to implantation and pregnancy.
- 2.90** To introduce you to these hormones, we're first going to have a quick refresher on the menstrual cycle. It's probably been a minute since you had to hear about the menstrual cycle. It was probably an awkward class. For us, it was a gym class. The girls went into

one room, and the boys went into the other. This here is a map of the female menstrual cycle and the two primary sex hormones that women have and how they cycle.

- 2.91** To orient ourselves to the figure, I want you to look at day 1 of the cycle, which is on the far left-hand side. Day 1 of the cycle, women's hormone levels are really low. This is actually the day that women get their period. The first day of your menstrual cycle is the day that you start bleeding on your period. At this time, hormone levels are at their lowest across the cycle.
- 2.92** Now, when hormone levels are low in the body, this actually gets picked up on by the brain, specifically the hypothalamus, which is always monitoring hormone levels to figure out what needs to happen next. When hormone levels are low, this tells the brain that it needs to tell the pituitary that it needs to tell the ovaries that it needs to start working on egg development because they're not pregnant—if hormone levels are low, it means your body's not pregnant, and it also means that your body's not currently trying to produce any egg follicles.
- 2.93** So low hormone levels tell the brain the ovaries need to get to work and start producing egg follicles. So that's exactly what happens. So in the early days of the cycle, the brain starts telling the ovaries through the pituitary gland, "We need to start working on some maturing egg follicles, because we should go ahead and try to ovulate, to see whether or not we can't get pregnant," because ultimately, that's what our bodies are always trying to do.
- 2.94** And so our egg follicles will start maturing, and as they start to mature, they cause the release of the sex hormone estrogen. When estrogen levels start to rise, that helps shut off menstruation. So generally, women's periods will last 3-5 days, and that ceases when estrogen levels begin to increase. As the egg follicles continue to mature, and eventually, one will become the dominant follicle. Whichever one is the healthiest and it has the strongest development, it will allow to continue to develop, and the other ones actually get reabsorbed into the body. It's almost like its own little evolutionary process within each cycle.
- 2.95** So, a dominant egg follicle will take over. It will begin really maturing and maturing rapidly, and at this point, estrogen levels really start to rise. So the dominant follicle takes over. Estrogen levels start to really climb around days 6, 7, 8, 9 of the cycle. This is when the egg follicle is getting really close to being ready to have this egg being released. Ovulation generally happens, if we're talking about a 28-day cycle, which I have pictured up here, ovulation generally happens on day 14.
- 2.96** Ovulation is when the egg gets released from the egg follicle. It makes its magical journey down the fallopian tube, where it can meet up with sperm and potentially get inseminated, and lead to a pregnancy. But that event, ovulation, generally happens around day 14. And right prior to that period, estrogen levels are peaking. During this

period of peak estrogen release, this is also when the probability of conception from sex is at its highest.

- 2.97** We actually call this period of time the fertile window. Usually, days 9 to 14, we consider this the fertile window, about five days prior to ovulation and then the day of ovulation itself, because this is a period in time, in the cycle, when the probability of sex is at its highest. And this is when estrogen is at its highest in the cycle.
- 2.98** After ovulation occurs, which, again, we're just going to say is on day 14, because that's on average when it happens, although everybody's cycle is a little different, after the egg is released, that empty egg follicle actually becomes a temporary endocrine structure—again, an endocrine organ or an endocrine structure is just anything that releases hormone. It becomes a temporary endocrine organ. It's called the corpus luteum, and its job is to release progesterone, and to release progesterone in steady levels for the next 12 to 14 days.
- 2.99** If a pregnancy takes hold, then the embryo will begin producing, or lead to the production of progesterone. If not, progesterone levels fall, and then you start your period. But progesterone gets released from this temporary endocrine structure, and it is its job, rather than preparing the body for the possibility of sex leading to conception, this hormone's job is to coordinate the activities related to the possibility that an embryo might implant and that pregnancy might occur.
- 2.100** We can think about our cycle as being made up of these two distinct halves. The first half, the follicular phase—that's the phase of the cycle when estrogen is being released; this is where the body is rolling out the red carpet on everything related to attraction in sex. Then the second half of the cycle is called the luteal phase—this is characterized by the release of progesterone. Progesterone's job is to get everything working together for the possibility of implantation and pregnancy.
- 2.101** Each cycle represents a shift between these two biologically distinct states. These hormonal changes, as women are experiencing them across the cycle, change the way that women think, feel, and experience the world around them. Because as we noted when we were talking about testosterone levels in men, that's exactly what they're supposed to do. The job of hormones is to get everything in your body, all the cells in your body, working together towards some goal.
- 2.102** What this means is that, as we experience these hormonal changes, it's going to change the way that we experience the world. We'll start by talking about the things that we know about estrogen and how estrogen changes the way that we experience the world. We're going to spotlight the follicular phase, or these first 14 days of the menstrual cycle. In particular, we're going to be talking about this fertile window that I talked about, because this is the period in the cycle when estrogen levels peak.
- 2.103** Just to look at this figure, just to show you why we call this the fertile window, this is a figure showing the probability of conception on each day of your cycle relative to

ovulation. And so, if ovulation is occurring on day 14, which you can see up there, that's related to a 20% probability of conceiving if you have sex on that day. The day before it, it's 42%, the day before that, 33%, then 27%, 18%, and so on. We call the fertile window generally either days 10 to 14, or 9 to 14, depending on who's slicing and dicing it. Everybody's got their own ways of doing it.

- 2.104 But the takeaway message is that estrogen levels peak when the probability of conception are at their highest. This generally occurs in the couple of days surrounding ovulation and then the day of ovulation itself. Estrogen cues in the body the probability of conception from sex. And so, it's absolutely no surprise to learn, when we consider the fact that we've inherited this brain from our ancestors that was able to be able to successfully reproduce in a continued line for generations, that our brain becomes primed for sex when estrogen is high across the cycle.
- 2.105 What research finds is that, when estrogen is high across the cycle, and as women move into the fertile window, this is related to an increase in sexual desire. Women regularly report feeling more sexually desirous during the fertile window of the cycle. They also tend to have more sex.
- 2.106 This here is a drawing of data that was found where researchers had women keeping diaries, and they were keeping daily diaries of whether or not they had sex that day, and also where they were in their cycle, by just simply asking, "Did you get your period today?" Then the researchers themselves were able to go back and track and figure out where women were in the cycle. They mapped all the women's data on this little map that we see up here.
- 2.107 What you can see, this is looking at the proportion of women who had sex on any given day of their cycle. That's what you see on the Y-axis. What you can see is that there's a sharp increase in the proportion of women who are having sex in the fertile window, which is something that's illustrated there by the dark bar, and that it decreases precipitously in the second half of the menstrual cycle, during the luteal phase, when sex is no longer conceptive, and the body is shifting gears into a state that's optimized for implantation and pregnancy.
- 2.108 It's going to show that women's sexual desire and sexual behavior has a cyclicity to it, peaking when pregnancy is possible and being lowest in the cycle at times when sex will not lead to pregnancy. We also see that women experience increased sexual pleasure during this time. So women also report that sex is more pleasurable, and they find sexual climax more easy to achieve in the fertile points in the cycle compared to outside of it as well.
- 2.109 We also find that estrogen impacts the types of qualities that women find most attractive in their partners. There's been a lot of research now looking at the role of women's estrogen levels and their preference for cues related to high genetic quality in men. One of those cues is levels of testosterone. The logic goes a little like this.

- 2.110** Testosterone is immunosuppressive. We know that testosterone, when levels of it are up, suppresses the functioning of your immune system because they share the same resource bank. Just like you can't spend a dollar twice, you can't spend the same dollar twice, you can't spend the same energy on immune function and testosterone production and all of the things that it entails. We know that testosterone is immunosuppressive.
- 2.111** A man who has a lot of masculinized male features, who has the higher levels of testosterone and has some of these things like the chiseled jaw and the deep-set eyes, this tells the viewer that this man's body is in such good shape that, despite the fact that testosterone is immunosuppressive, his body was able to release testosterone reliably throughout development to allow him to grow these masculinized male features, and therefore, he must have good genes.
- 2.112** He must have good immune genes to be able to have developed that chiseled jaw and those deep-set eyes, despite the fact that his immune system was being suppressed throughout childhood, because our bodies will not allow our immune system to be suppressed and release testosterone if we can't afford it. It's an honest indicator of genetic quality. What this is telling us here, what we're looking at, is women's preference for testosterone markers in male faces across the cycle as a function of fluctuating estrogen levels.
- 2.113** What research finds is that when women are entering into the fertile window—again, this period of time when estrogen is peaking and when conception from sex is possible—that women have a heightened preference for cues related to facial, vocal, and behavioral masculinity to a greater extent than what is preferred outside of this window. Essentially, it just makes women more attuned to these cues.
- 2.114** Even subtle qualities related to differences in testosterone levels—so subtle differences in facial masculinity, for example—women are better able to detect those differences at high fertility in the cycle, when estrogen is high, than they are outside of it. We know that women's preferences also seem to shift across the cycle, prioritizing some of these qualities that are hypothesized to be related to good genes at times when conception is possible from sex.
- 2.115** And then with women orienting themselves toward other qualities that women also desire in their long-term partners outside of the fertile window, for example, the desire for partners' provisioning ability and their fatherhood potential. It's just that women tend to be particularly interested and attracted to these testosterone markers in the fertile window in a way that they don't seem to have as much of a preference for outside of it.
- 2.116** Interestingly, in addition to the fact that estrogen levels increase women's attunement to cues of genetic quality in men and increases their interest in sexy types of qualities, the types of qualities that the female brain finds sexy, it also impacts how attractive

women feel and how attractive they are to men. To illustrate this point, I'm going to show you these two photographs of the same woman. These two pictures were taken of the same woman, once at low fertility and once at high fertility across the cycle.

- 2.117** Here's the same woman. She took her makeup off. In one of these pictures, she's right in that fertile window when estrogen is high. Then in the other picture, it was taken when she's in the luteal phase, at a time in the cycle when sex cannot lead to conception. The researchers show these photographs to people who see one, the other, or they see both, and they have to choose which one she looks more attractive.
- 2.118** What the researchers found is that people reliably converged on finding the woman most attractive when she's at high fertility across the cycle, compared to low. In case you're wondering which one is which, the red shirt is the high fertility photograph, and the black shirt is when she's at low fertility across the cycle. On average, most people tend to converge on rating her more attractive when she's in the red than in the picture when she's in the black.
- 2.119** They did this with a variety of different stimuli, the range of different attractiveness levels. What they tend to find is that, on average, we tend to converge on preferring the look of women when they're at high fertility. There's also been some interesting studies that were done using a naturalistic observational study that was conducted at a gentlemen's club. The researchers did this because this is a really good non-reactive way to gauge people's interest in women based on their fertility status across the cycle.
- 2.120** The researchers went in, and they formed a relationship with a group of dancers who worked at this club. What they noted was that about half of them were on hormonal birth control, which prevents ovulation, so you never get that nice big bump in estrogen levels. The other group were naturally cycling; they were cycling through women's two primary sex hormones the way that naturally cycling women do, with estrogen being high and rising during the first half of the cycle, and then being low, with progesterone being high and rising in the second half of the cycle.
- 2.121** They simply had the women record whether they got their period that day and how much they earned in tip money. They also were able to statistically control for time of day and the number of hours that the women worked. After they controlled for all of those different variables, they plotted the women's tip earnings across the days in the cycle. What they found was that, for naturally cycling women, which is shown with the solid line, the women peaked in the tips that they earned at the gentlemen's club right near high fertility.
- 2.122** Right during that fertile window, you can see that women are earning more tip money than they are in other points in the cycle. These women are earning more than the pill-taking women across the board, because women who are on hormonal birth control have their estrogen levels kept really low. Again, just showing evidence that there's this relationship between estrogen and these cues that—it's like, most women don't even

know when they're ovulating. Yet, nonetheless, they are exhibiting these cues that tend to make them appear more attractive to potential partners.

2.123 We're going to be picking up, talking about this and some of the other things that we know that estrogen does in our next lecture. Then we're also going to be learning about women's other primary sex hormone, progesterone, and what it does in terms of changing women's psychology in the way that they think, feel, and experience the world.

2.124 So, you've talked a lot about the evolutionary benefit to everything that we've developed as humans. Has it been explored what the evolutionary benefit is for menopause? I know age is, we used to die a lot sooner, but wouldn't it be advantageous to have a larger fertility window for both genders?

2.125 Right. Really great question. For a really long time, people thought that menopause was just a byproduct of the fact that we lived longer. But it turns out that it's actually being actively selected for. You can actually do these equations. It's called the Hardy-Weinberg equilibrium equation. It shows you when a trait is actively being selected for. It seems like menopause happens intentionally.

2.126 The leading hypothesis for the evolution of menopause is what's known as the grandmother hypothesis. It's this idea that, as you're transitioning out of fertility, it's essentially—kind of like we talked about how men's bodies tap the brakes on testosterone production when it's not advantageous to be investing in direct reproduction, but instead, they should be orienting their resources toward their partner and their kids.

2.127 The same is true for women with menopause, where it's like, OK, now is the time to start sunsetting your reproductive career and instead take the effort that you would be investing in direct reproduction and instead invest it in your existing offspring and then ultimately your grand-offspring and so forth. It's essentially your body's way of getting you to shift gears away from investing effort in direct reproduction toward augmenting the success of your existing children. Yeah, really interesting question.

2.128 It really will be interesting to see, because everybody's getting such a later start in life, we would expect—I don't know, even 1,000, 2,000 years from now, I'm guessing that we're going to see the age of menopause to keep getting delayed because people are reproducing later. That means that's going to be selecting for genes that are related to the ability to reproduce later, because people who aren't able to reproduce later won't be passing those genes on. I think that we're going to start to see evolutionary change even in the next, however many generations, like six, seven generations, I think, that we'll expect to see the average age of menopause begin to tick up. It's a really interesting question.

2.129 When you were talking about men's testosterone being lower at certain times and stages in their life, well, if a man goes in to get bloodwork and they're only measuring

those labs by their age, do you think there should ever be a qualifier like, "Hey, are you married? Do you have kids? You're 10% lower, so you're normal now," because we're really not factoring that in. That driver might be increasing this idea that men need to go to get testosterone or the injections.

- 2.130** Yeah, no. I think that's such an interesting question, and it's something that nobody's asking because everybody just assumes more is better. There's some things about men's low testosterone levels in our current environment that aren't good, because we know that there's all the xenoestrogens and all these other things that are making testosterone levels lower in a way that's not functional. But at the same time, yeah, these are important caveats, I think. I think that nobody in that clinical world is asking those questions, and I think that they need to, because we just have these really simple ideas about, well, more is better, and maybe it's not always better.
- 2.131** In general, when we use the word hormone, maybe in pop psych or just comment on the street, I would say that it is maybe more towards a negative connotation, not even positive or neutral, for instance. Like, "Oh, they're hormonal," or "She's so hormonal." Why do you think that we use a negative connotation in general when, literally, it might be the single most important factor or driver to continue humanity?
- 2.132** It is so interesting because not only is it that they're talked about in a negative way, but they're externalized. You know what I mean? Like, "Oh, it's her hormones." But you are your hormones. If it's with a man, it's his hormones. Honestly, I think that one of the reasons that they tend to be talked about negatively is because for—and this isn't true, but for a long time, people assumed that it was a female thing. It was just another way, I mean, essentially, of just saying that women are less than.
- 2.133** It's like, "Oh, she's hormonal. She can't think rationally. She's hormonal," without understanding that men are just as hormonal as we are, and their hormones change more frequently. I mean, ours are cyclical but predictable. Men's hormones, it's like, a beautiful woman walks by, and their hormone levels change. That doesn't happen to us. Our hormones, our primary hormones, are pretty non-reactive.
- 2.134** But I do think that there's this tendency to talk negatively about them, because I think, for a long time, it was assumed that it was female. Then it was a way of creating this lens that the female way of being is less rational, and all these other things. Yeah, really interesting. It is so interesting, though. I've always found that interesting, and also just this idea that they're so externalized. It's like, "Oh, it's my hormones." I mean, nobody would say, "Oh, it was my neurotransmitters. It's my synaptic patterns. I'm sorry." Nobody would say that.

Lecture 3: The Menstrual Cycle

At a glance — The menstrual cycle and estrogen's role in orienting the body.

- 3.1** When we left off, we were talking about all of the things that estrogen does to help orient the body toward attraction and sex. Just as a reminder, women have two primary sex hormones, estrogen and progesterone, because our bodies have two jobs that are required for reproduction. That, of course, is attraction and sex, and then pregnancy and implantation.
- 3.2** When we left off, we were talking about the fact that, in addition to the fact that women tend to have a heightened preference for all things sex and sexual activity when estrogen is high and rising across the cycle, that it also seems to make women more attractive and desirable to mates. We talked about the results of this study, which found that women are found more attractive by both women and men at points in the cycle when conception is possible, so during the fertile window of the cycle.
- 3.3** It's also a time when women who are dancers earn more money. Other studies have also found that women smell better, sound better, and move in ways that men find sexier at high fertility in the cycle, compared to low fertility in the cycle. This suggests that, among all the other things that estrogen does, is that it may make women more attractive to men, particularly at times in the cycle when sex can lead to conception.
- 3.4** It also seems to play a role in impacting women's motivation to attract mates. Again, when we're talking about estrogen rising in the cycle during the fertile window, this is a point in the cycle when sex can lead to conception. And so, in addition to the fact that women's bodies are rolling out all the stops in terms of looking maximally attractive, it also shifts women's motivational states in ways that are consistent with the desire to attract mates.
- 3.5** Research has found, for example, that women tend to put more effort into their appearance and dress sexier at high fertility in the cycle compared to at low fertility in the cycle. Just to tell you about one study that was done, this was a study that was done at the University of Texas, and it was done when I was a graduate student there. It was done by one of my contemporaries, Dr. Kristina Durante. She had women come into the research lab at two different points in the cycle.
- 3.6** She had them come into the lab once during the cycle when fertility was high. This was something that was confirmed using a urinalysis test, which tests for the luteinizing hormone, which is something that gets released right prior to ovulation in very high levels. And so they had women coming into the lab when luteinizing hormone levels were high, which means that they're at a fertile point in their cycle and their body is preparing to ovulate.

- 3.7** Then they had them come in a week later, after ovulation had occurred, and women's bodies had shifted gears into the progestogenic phase of the cycle, which is a time when conception is not possible. She had them come into the lab to do a variety of tasks. But one of the things that she also did is, she just took a picture of what the women were wearing. It wasn't a special day. This is just what women were wearing to the research lab to show up to do their experiment. What they found was that, reliably, women were wearing sexier outfits and revealing more skin at their high-fertility session compared to the low-fertility session.
- 3.8** They tested this over nearly 150 women, and they tested this within subjects, meaning that they looked at women twice, once at low fertility, once at high fertility, and made these comparisons. What they found was that women did show significantly more skin at high fertility compared to low fertility, and that people also evaluated their outfits as being more subjectively sexy at high fertility compared to low fertility, suggesting that women may feel sexier and be more motivated to dress in sexy ways at times in the cycle when estrogen is high and when conception is possible from sex.
- 3.9** In this same study, she also gave women—and again, these women came into the lab twice, once when estrogen is high, and fertility is high, once when progesterone is high, and fertility is low— and she told them to imagine that they were going to a party that night and that it was going to be over at a friend's house, and there was going to be a lot of attractive members of the opposite sex there, as well as a lot of attractive members of their same sex there. Then she told them to draw what they would like to wear to the party that night.
- 3.10** She, again, had them do this twice. She had them do this once at high fertility and once at low fertility across the cycle. Then they scored the outfits according to subjective ratings of sexiness, so having a group of independent researchers, who were blind to the condition that the women were in, evaluate how sexy the outfits were. They also measured the square millimeters of flesh that were revealed in the outfit choice that was drawn by the women.
- 3.11** What they found was that, at the high-fertility session, women drew outfits that were significantly sexier than the outfits that they drew at the low-fertility sessions, and they exposed more skin in the photographs. So here you can see some of the results on this little figure. You can see that the women revealed significantly greater number of square millimeters of flesh in the high-fertility drawings, compared to the low-fertility drawings.
- 3.12** Here we have these two outfits. They're both pretty cute, but the one is obviously significantly sexier and shows more flesh—this was her high-fertility drawing, and the one on the left was the low-fertility drawing. This is something that's also been replicated using a shopping paradigm, where they had women come into the lab and do some online shopping with a little storefront that they had set up in the research lab.

- 3.13** They had the women buying things and selecting the items that they'd most like to buy, if they were shopping in that moment. They had them do this at both low and high fertility. And lo and behold, yet again, women were choosing significantly sexier clothes at high fertility compared to the low-fertility session. If you've ever had that outfit in the back of your closet, that sexy outfit that you bought, that you're like, "Why did I buy this?", this might be why. It could be that was your high-fertility outfit, and you didn't even know that it was going on.
- 3.14** The first half of the cycle, the first 14 days of the cycle, what we call the follicular phase of the cycle, when estrogen is the primary sex hormone, this is when our body and our brain are oriented toward attraction. Again, we find these increases in women's preferences for cues related to testosterone levels. We find that women are more interested in men in general. Women are more interested in sex. They're more, they enjoy sex more, they have more sex, they dress sexier, they smell sexier, they sound sexier.
- 3.15** Everything, all the red carpet is being rolled out on everything related to attraction and sex during that time. Then after ovulation occurs and women shift into the luteal phase of the cycle, this is when sex can no longer lead to conception. The job of women's second primary sex hormone, progesterone, is to redistribute all of that effort that was being geared toward attraction and toward sex, and to shift that effort toward preparing the body for the possibility of implantation and pregnancy.
- 3.16** This is what happens during the last two weeks of the cycle, which is known as the luteal phase. This is something that's characterized by the release of the sex hormone, progesterone. Progesterone essentially is orienting all of the cells in the body to prepare for the possibility of an embryo implanting and for the possibility of pregnancy. This means that, as women are shifting into the luteal phase, they're going to experience both physical and psychological changes that help to promote these types of outcomes.
- 3.17** Essentially, if we think about this in terms of the different types of adaptive problems that women have to solve in order to reproduce, what you see is, when women are shifting between these two hormonal states, is that they're shifting toward a body that's solving one set of adaptive challenges and then into another. What are some of the differences that we tend to see when women shift into the luteal phase relative to the follicular phase?
- 3.18** Well, one of the changes that women experience—and this is by far the one that women notice the most— is that women become more sensitive to signs of threat. This isn't just physical threat, it's also social threat. Women's brains become more primed to consider things as threatening during this time compared to other times in the cycle because our bodies are in a state of heightened vulnerability.

- 3.19** As our body is preparing for the possibility of pregnancy, we're preparing for being in a state in which we're very reliant on our partners for things like protection and provisioning, as they're preparing for the possibility of pregnancy and childcare. Because our bodies are more vulnerable during this time, it actually changes the threshold that our brain uses before considering something a threat. This is because of something that we call the smoke detector principle.
- 3.20** To introduce you to the idea of the smoke detector principle, I want us to think about our smoke detectors for a minute because we all have at least one in our house somewhere. We've all probably had the experience of burning some food or you're making microwave popcorn, and then the smoke detector goes off. The reason that our smoke detectors tend to go off, even with little slights like making popcorn or burning a pizza, is because smoke detectors are wired to be intentionally oversensitive to signs of a fire.
- 3.21** The reason for this is that it's better to be safe than sorry. You'd much rather have a smoke detector that goes off when you're making popcorn than you would one that doesn't go off when there's actually a fire. And so, when smoke detectors are designed, they're actually designed to be oversensitive, to be super-sensitive, so that way they don't miss an actual threat. Our brain is actually wired the same way when it comes to threatening stimuli.
- 3.22** When it comes to things that can kill us, either from the outside or the inside, so things like pathogens, those things kill us from the inside, or things like predators, which are things that kill us from the outside, we tend to overgeneralize fear and disgust. For example, if it's a dark night and you hear scratching from outside, you tend to err on the side of assuming there's a real threat there, because our brain is wired to err on the side of caution in the way that would have historically led to a greater probability of being able to pass down our genes.
- 3.23** It's better to be safe than sorry, and it's better to assume that something is threatening and actually have it not be threatening than to make the reverse mistake—assume that something is not threatening and it actually is. Our brain is wired to err on the side of caution when it comes to things that can kill us. This is why fear tends to overgeneralize. We don't like all snakes, and we don't like all spiders. We don't discriminate. We don't say, "Oh, no, that's a good one, and no, no, he's a nice one."
- 3.24** We're also that way with disgust. We don't care that you sneezed because of allergies. I don't care. I don't want that on my arm. We overgeneralize with things that are potentially threatening to us. The degree to which our body overgeneralizes with things like fear and disgust changes depending on the vulnerability of our body. The more vulnerable we are to a threat, the more quick we are to overassume that even minor slights are the sign of danger.

- 3.25** Enter the luteal phase brain, and you have a brain that is primed for fear and anxiety. A lot of women report having heightened anxiety, especially relationship anxiety, in the second half of the cycle. It's because our brain, using its brilliant smoke detector design, knows that here we are in this really vulnerable state. Our body is preparing for this really vulnerable state of pregnancy, in which women have historically been very dependent on their partners, and any sign that there might be some sort of trouble in paradise with their relationship seems like a sign of alarm.
- 3.26** This is the reason that a lot of women can feel totally comfortable with the idea that their partner has to stay late at work one night. But then two weeks later, when she's in the luteal phase, all of a sudden, the alarm bells are going off. "Wait a minute. What's going on? Why are you staying late? Does he not care enough about me, or is there something worse going on?" Women's brains are more quickly to trigger the alarm bells during the luteal phase than they are in the follicular phase because we're in a heightened, vulnerable state.
- 3.27** The more vulnerable we are, the more quickly our alarm bells sound; the less vulnerable we are, the less easily our alarm bells sound. This is also why chihuahuas bark more than big dogs. The little dogs bark a lot because they're vulnerable, they're small. The big dogs don't need to bark as much because they can protect themselves. And so is true with our brain.
- 3.28** And what we find from imaging studies is that, during the luteal phase of the cycle, the amygdala in the brain, which is the fear center in the brain, actually becomes more interconnected, functionally connected, to other areas of our brain than what it is in the follicular phase. So our brain actually becomes more wired to detect threat coming from any different area of the brain, and it's also more quickly to sound the alarm bells on threat to all of the different areas of the brain. So we become more sensitive to signs of potential threat, which, of course, can lead to heightened levels of anxiety and also can lead to distress, especially within close relationships.
- 3.29** We also tend to find that, as women are transitioning into the luteal phase, that this is a time when women experience worsening mood. It's usually a depressed mood. When I say depressed, I'm not talking about clinical depression, not the clinical depressed mood, but rather that their mood, rather than being outward-facing and energy-exerting, tends to be more inward-facing and energy-conserving.
- 3.30** Women are generally less motivated to be going out and seeking pleasure during the luteal phase of the cycle compared to the follicular phase of the cycle. This shift in our motivational states, where we tend to be less optimistic and out to go and conquer the world, but instead tend to be a little bit more subdued in focusing inward, is part of the metabolic shifts that we experience as we transition between the follicular phase of the cycle and the luteal phase of the cycle.

- 3.31** What research finds is, women's motivational states and their moods tend to be less sunny and optimistic and outward-facing, and tend to be more inward-facing and almost low-key, subdued, because their body is shifting their motivational states in ways that are going to conserve energy, instead of having them go out into the world and doing things that could potentially burn calories or be dangerous.
- 3.32** Indeed, one of the changes that we tend to see as women are transitioning into the luteal phase of the cycle is that women do tend to have lower mood. This is the result of a study that was done cross-culturally. You can see all of the countries that they collected data on when you look at this map. But this is a premenstrual mood map, and this is mapping women's emotional states as they are in the last two weeks of the cycle compared to the first.
- 3.33** All of the regions that are colored are the areas that they were actually able to collect data from. You can see that they collected this from places around the world. What you can see is that there wasn't a single country that they studied in which women reported feeling happier in the second two weeks of the cycle than they did in the first two weeks of the cycle. The first two weeks of the cycle, when estrogen is the primary hormone, we tend to be motivated to go out and socialize and do things that are going to put us into proximity with potential mates.
- 3.34** We also are looking really attractive, and it's a good time to be showing that off in terms of mate attraction. We have less of a metabolic load because our body isn't preparing for the possibility of pregnancy. And so women tend to feel lighter, happier, and more energetic in the first two weeks of the cycle than they do in the second two weeks of the cycle, with many women reporting feeling more subdued and almost sad during the second half of the cycle compared to the first.
- 3.35** Another change that happens in the second half of the menstrual cycle, and this is one that women aren't really told about, is that women burn more calories during the second two weeks of the cycle than they do during the first. When the body is preparing for pregnancy, which is what it's doing in the luteal phase of the cycle, it is requiring a lot of energy to create and lead to cell differentiation in the endometrial layer, to prepare for the possibility of implantation and pregnancy.
- 3.36** This raises women's metabolic rate by between 7 and 11% every cycle, in the second two weeks of the cycle. If you're a woman who eats about 2,000 calories a day, this is an additional 140 to 200 calories a day that your body burns just by virtue of the fact that you're releasing progesterone. This is the result. What you're looking at here is a figure. This was a study that was done on women, where they measured their basal metabolic rate as they slept, and they did this at different points in their cycle.
- 3.37** What they found is this very shift that I told you about, where women on average, and it ranged between about 7 to 11%, but when progesterone is being released, our body, it just revs up the metabolism. It's because creating that endometrial layer and

differentiating those cells requires a lot of tissue-building, and this requires a lot of energy and a lot of amino acids. It's a big, expensive job for the body.

- 3.38** It's also a time when our basal body temperature increases. One of the ways that women are actually able to track their cycles, if they're trying to figure out which hormone and what cycle phase they're in, is by tracking their basal body temperature across the cycle. Because after progesterone starts being released, so after you ovulate and you're in the luteal phase, your body temperature, your core body temperature, increases by almost a full degree. So you become warmer.
- 3.39** And as your body becomes hotter, this is part of your metabolism revving up. This requires more energy to create that thermic event that's allowing you to be hotter. What we find is that, even when women are sleeping, you get greater energy expenditure in the second half of the cycle than you do in the first half of the cycle. This might not surprise you, but given that metabolism is increasing, women's hunger also increases during this time.
- 3.40** Here, this is a figure looking at changes in women's sexual desire and food intake over the course of the cycle. This was data that was collected on a large sample of women, and they followed them across two cycles, and they used hormone tests to confirm their hormone levels. What they found was that, when you look at sexual desire, the pattern follows what we talked about when we were talking about the follicular phase and how estrogen rises at times in the cycle when conception is possible.
- 3.41** What you can see is that sexual desire is highest right near ovulation. In those middle bars, you can see that sexual desire is very high. Then when you look at that light-colored bar next to it, you can see that food intake is actually really low. Interestingly enough, you do get this sort of trade-off between sexual desire and physical hunger, where the more sexual desire we feel, that that oftentimes comes at the expense of having lower hunger, and then vice versa. When we're more hungry and need to eat more, our sexual desire decreases.
- 3.42** Part of this refers to this idea that we talked about a little bit earlier, which is just that you can't spend the same dollar twice. Your body has a limited energy budget, and you can only do so many things with that energy budget. If you're directing energy toward mating, that's not going to give you as much energy to be able to find food and digest food and do all the things that your body needs to do to get food in your system.
- 3.43** We make these different types of trade-offs across the cycle. One of those trade-offs is that, when we're close to ovulation, we get an increase in sexual desire, and we tend to get a decrease in food intake. Then, when we get into the luteal phase of the cycle, when progesterone is being released, and sex can no longer lead to conception, during this period, we see that food intake increases, but that sexual desire decreases.
- 3.44** There's also been a number of studies showing that, in addition to hunger increasing and women's basal metabolic rate increasing, that women also tend to report feeling

more subjectively hungry during this time. And part of the things that we tend to think about as problematic, many women don't love the way that they feel during the last two weeks of their menstrual cycle. And one of the reasons that women don't like the way that they feel is because they feel like they're hungry and they're having food cravings all the time.

- 3.45 And the reason that women are hungry and having food cravings all the time is because nobody tells them about that. They don't tell them that their body is increasing the number of calories that they're burning in the second two weeks of the cycle compared to the first. Instead, women are given a one-size-fits-all set of nutritional guidelines where they're told, "You need to eat X number of calories a day. If you're eating more than that, well, then that's just too much."
- 3.46 So then, here's a woman whose body is increasing its energy expenditure by 7 to 11% and thinking, "I'm starving. I'm so hungry." They're feeling tired and they're feeling grumpy, and then they're wondering why their son's Halloween candy looks so good. It's because they haven't been eating enough. This is just another one of these changes that we experience as our body is preparing for pregnancy.
- 3.47 Pregnancy is an incredibly physiologically costly event, and our body has to rev up our metabolism to get our body prepared for pregnancy, while also trying not to lose some of the stored body fat that we have on ourselves that's also going to be supportive of a pregnancy. During the luteal phase of the cycle, we tend to see this increase in hunger and the decrease in sexual desire.
- 3.48 Here's just some more research demonstrating the decrease in sexual desire. This is something that, again, is very common, and it's something that many women are not told about—and that is the fact that, yes, sexual desire does increase in the fertile window of the cycle. When estrogen levels are high and rising, this corresponds to an increase in sexual desire. But as women are transitioning into the luteal phase, this is also associated with a decrease in sexual desire.
- 3.49 Progesterone is actually antagonistic to sexual desire. Many women have had the experience of—and I've been asked this a number of times by women, too— where they'll get an increase in sexual desire that happens as soon as they get their period, and they're wondering what that's all about, because that's usually not a time in the cycle when fertility is high, but women notice that their libido increases when they get their period, and it's because, when they get their period, progesterone falls, and progesterone is actually sexually inhibitory.
- 3.50 The reason that progesterone is sexually inhibitory is that sex is physically and physiologically costly. Sex is physically costly, just because it requires effort to do, and your body is in a state when it's conserving resources because its resting metabolic rate is higher and it needs to conserve resources for the possibility of pregnancy. But it's

also a time when exposure to some of the germs that you get in your body when you're having sex is a little bit less of a good idea.

- 3.51** The reason for this is twofold. One, the maternal immune system gets functionally dampened in response to progesterone release. One of the changes that happens when we transition into the second half of the menstrual cycle is that we shift from a pro-inflammatory type of an immune response into an anti-inflammatory type of an immune response. We call this shifting from a Th1 response to a Th2 type response.
- 3.52** The reason that our bodies shift our immune responses across the cycle is because our body needs to be able to allow for an embryo to implant without attacking it. The fact is, an embryo has an awful lot in common with things like cancer cells and things like parasites because they have rapidly dividing cells that are not totally genetically related to the self, and it's trying to take a bunch of resources from the body. That's something that the immune system would normally flag as being either a parasite or a cancerous cell, and its job would be to search and destroy.
- 3.53** But of course, we don't want to do that if we have an embryo implanting. So a workaround that the body has made is that it functionally turns down the volume on the immune system during the luteal phase of the cycle. This allows for the possibility of implantation, but it does open up the possibility of opportunistic infections. This is a time in the cycle when women need to be a little bit more cautious when it comes to exposure to germs.
- 3.54** Research finds that, when progesterone levels are high, that women's disgust sensitivity increases, so women become more easily disgusted. This is a form of our behavioral immune system being activated. Our behavioral immune system is psychological changes and behavioral changes that keep us away from things that can make us sick.
- 3.55** Our behavioral immune system tends to be most active during times when our physical immune system is suppressed, which is the case when we're in the luteal phase of the cycle and during pregnancy. What you tend to find is that, when progesterone levels are high and the immune system is suppressed, that this is related to increases in disgust sensitivity. We see this both in pregnant women and we see this in women during the luteal phase of the cycle.
- 3.56** The second reason that women tend to experience a decrease in sexual desire in the second half of the cycle is because, whenever you are having sex, the material that's entering your body is genetically non-related to you. That's something that the immune system is designed to go after and attack. Any time that the immune system is attacking something, it creates inflammatory activity.
- 3.57** And so, if you are having sex in the luteal phase of the cycle, and particularly if you're having sex near what's known as the implantation window, and this is that window in time in the luteal phase when the endometrial layer is receptive to an implanting

embryo, during this time, having inflammatory activity near the cervix and the vaginal canal is suboptimal for implantation. And so, another reason that women tend to get a downregulation in their sexual desire in the second half of the cycle is because they are essentially minimizing the probability of having sexual activity at times when sex comes with a cost.

- 3.58** Now, this doesn't mean that women shouldn't be having sex in the second half of the cycle, because we are actually really an interesting species in that we are one of the only species that exists where females have sex outside of the fertile window. This serves a really important function in terms of promoting pair bonds.
- 3.59** But instead, it just turns the volume down, because the cost of sex is greater. The cost of sex is greater in the luteal phase, and the benefits are lower in the luteal phase, than they are in the follicular phase. Because of this, we tend to see that female sexuality is cyclical, with it being highest near the fertile window in the cycle and being lowest in the luteal phase.
- 3.60** These hormonal changes that women experience across the cycle, they're pretty pervasive and they have an impact on everything. We've talked about some of them. They impact everything ranging from the way that we smell to the quality of our sleep. Our energy levels change across the cycle, attraction to partners. We talked about the fact that, when women are at high fertility, that they have this heightened preference for testosterone markers, and that kind of goes away when they're in the second half of the cycle.
- 3.61** It can affect things like the types of symptoms that they have and the severity of the symptoms that they have from chronic conditions. For example, research finds that women who have asthma oftentimes have worsening of symptoms in the second half of the cycle compared to the first. The reason for this is, among the whole suite of changes that our body experiences as they're preparing for pregnancy, our respiratory drive increases, meaning that our bodies need to breathe; that feeling that we have where we need to breathe, increases.
- 3.62** It does because our respiration rate increases in the luteal phase, because our body is having to work harder to prepare for the possibility of pregnancy. This leads to worsening of asthma symptoms, for example. You get differences in mood regulation, you get differences in responses to therapy, and the list goes on and on and on. There's a whole range of changes, as our body is shifting between the state that's optimized for attraction and sex, to the state that's optimized for pregnancy and implantation that our bodies experience, and we experience these on a monthly basis.
- 3.63** This has led some people to really bristle at the idea of speaking too loudly about the fact that hormones matter. People have suggested that talking about hormones and the way that hormones impact women is something that can potentially be dangerous, because women have a history of being treated as overly emotional, less-rational

versions of men. Sometimes it can feel like acknowledging that sex hormones have this huge range of effects that they actually have can feel really risky.

- 3.64** It can feel like something that can be bad for women because it can be used to argue that we shouldn't be able to vote or we shouldn't be able to own land or we shouldn't be in positions of power because we're just too hormonal. Is this idea dangerous? I'm going to start this—I'm going to answer this question by saying that, while it is true that women's hormones vary cyclically, they do not vary unpredictably.
- 3.65** One of the arguments that gets put out there about why it's problematic that women have these cycling hormones is that, "Oh, women are fickle and they're overly emotional and unpredictable." The fact is that our hormones do cycle, but they cycle in a way that is incredibly predictable. I always tell people, if you tell me a woman's age and you tell me the first day of her last menstrual cycle, I will be able to tell you with a pretty good degree of accuracy what her primary sex hormone are doing.
- 3.66** If she tells me that she's this age, and the first day of her last menstrual period was 12 days ago, I know that she is in the estrogenic phase of her cycle. I can pretty much make predictions about what she's going to be feeling. If she tells me that she's on day 24, I can make a different set of predictions. It's very predictable. So cyclical, not unpredictable.
- 3.67** Men's primary sex hormone, testosterone, on the other hand, is a little bit unpredictable. We know that men's primary sex hormone changes in response to things in the environment that suggest either it's a good idea or a bad idea to be competing or exerting mating effort. We know from research that men's testosterone levels change in response to age. They change in response to the time of day. We know that they're higher in the morning than they are in the evening.
- 3.68** They change in response to relationship status. So when men get married, their testosterone levels decrease. They decrease further when they have children. We know that if their favorite sports team wins, their testosterone increases. If their favorite sports team loses, their testosterone levels decrease. If their favorite political candidate wins or loses, their testosterone levels change.
- 3.69** Their testosterone levels even change if they see weapons. There's been studies where they put men into a room, and sitting next to them at a table, there's either guns or there's tennis rackets. When men are in the room with the guns, their testosterone levels increase. If I were to make a prediction about what a given man's levels of testosterone are, I would need to know a lot more information than just how old he is and when is the first day of his last menstrual cycle.
- 3.70** I'd need to know what time of the day it was. Is he married? Is he considering not being married? Has his favorite political figure just won or lost? Has he seen a weapon? Is there a beautiful woman in the room? Men's primary sex hormone actually is a little bit

unpredictable, but not unpredictable. It just changes more frequently, and it changes in response to environmental factors. These things are good, and they make us better.

- 3.71 The idea that men's hormones change and that that changes how men think, feel, and experience the world, it's supposed to. We talked about what happens when men become fathers, and how testosterone decreases and how that's functional, and it helps to redirect men's effort toward doing the things that are going to help promote their relationship and their reproductive success.
- 3.72 The same is true for women. As women's hormones change across the cycle, they're orienting women's brains and women's bodies toward doing those things that are going to help promote their reproductive outcomes, orienting them toward attraction and sex during the first half of the cycle, and then orienting themselves toward pregnancy and implantation during the second half of the cycle.
- 3.73 This is something that doesn't make us worse, it just makes us different. When we start talking about some of the biological differences between men and women, including the biological differences in our hormones, we need to not be afraid of having these conversations, because we can have these different paths, and it doesn't mean that one is better or worse than the other. The fact is that men and women differ in ways that extend far beyond the shape and nature of their reproductive organs.
- 3.74 We can't really understand the experience of being female without understanding our cycling sex hormones. The fact of the matter is, ignoring just how important our hormonal changes are across the cycle, ignoring that has set the stage for the birth control pill. The birth control pill, while it's not bad, it does make women different. When we change women's hormones, we change women.
- 3.75 Without having these open discussions about the ways that hormones shape the way that women's brains work, and the way that they think and feel and experience the world, we can be really cavalier about things like changing a woman's sex hormones. We can be recommending things like the birth control pill for things like acne and cycle irregularities. When we're doing that, we're not only changing a woman's proclivity to acne or their tendency to have irregular cycles, but we're changing the women themselves.
- 3.76 One of the things that we're going to be talking about as we start to move forward is, we're going to be talking about how some of the promises that we were made about denying the importance of sex and denying the importance of hormones has really been dangerous for women. Because throughout most of history, and particularly in not-so-recent history, but in the time of my mom's generation, there was very much this idea out in the world that, in order for women to be treated equally to men, that the thing that we needed to do was completely deny our biology.
- 3.77 There was this idea that the only way that we can prove that we are equal to men and that we're deserving of the same rights as men and that we should be able to vote and

own land and have jobs is by saying, "Our hormones don't matter, and we're just like you, except for the differences in our reproductive organs." But even though we've all been promised that denying our biological sex would be this great deliverer for women and would allow us to transcend sexism, it's actually had the opposite effect.

- 3.78** And by ignoring biological sex, and trying to ignore the fact that our bodies and our brains differ from those of men in ways that have absolutely nothing to do with our reproductive organs, we've actually created a lot more harm than good for women's mental and physical health. And so next time, when we pick up, we're going to be talking about some of the ways that biological sex denial has ultimately created and laid the groundwork for bad health care for women, but then also laying the groundwork for dismissing the importance of women's hormones and being really cavalier about the birth control pill.
- 3.79** My question is, if someone is taking a hormone replacement like progesterone, so now it's a consistent level all the time of what they're getting in their body, essentially, do they get stuck in the luteal phase then, along with the things that come with being in the luteal phase?
- 3.80** Yeah, that's a really interesting question. If you're not taking estrogen and you're just taking progesterone, your body is going to be running—like, we talked about hormones are sort of like software—it will be running the progesterone program. But depending on what your levels of estrogen are like, because if your levels of estrogen are really high and the progesterone levels are lower, you'll also be running the estrogen program.
- 3.81** And so, what you feel like and the physiological impact that it has on your body is going to depend, really, on the ratio of those two hormones. But if you are in a state where it's always progesterone is the dominant hormone, then you will feel some of those differences. You'll feel your body running the progesterone program.
- 3.82** What steps should fathers take for their daughters who are coming around that age, 13, 14 years old, past knowing this information to help guide their daughters through this? Because I feel like it's such a thing of, they stay away. But obviously, that's a really important time for a daughter to have her father in life. Can you speak on that?
- 3.83** Yeah. No, that's a really interesting question. I think that all parents can really be supportive by having open conversations about hormonal changes and some of the psychological and behavioral and even physical changes that happen in response to that, and not having it be something that is seen as a liability. I know in my household with my daughter, that when she first started cycling, she was having really crazy mood changes, and they were really bad.
- 3.84** We just talked about what her body is doing and talked about the fact that especially teenagers, their cycles also tend to be unpredictable, and sometimes they're really heavy. It's because their brains and ovaries are learning to communicate with each

other, and so they don't always get it right. Just helping them understand and have a language to understand what's happening, I think is so helpful.

- 3.85** I think, I look at my situation with my daughter because she went through years where her cycle was, like, she couldn't predict it, and it was heavy, and just even understanding that her brain and her ovaries are learning to talk to each other, and that it's going to get better. Then seeing that it was getting better, and over time things did get better. And with her mood, just being able to support those changes and say, "This is really common to be feeling this way right now, and it feels like the bottom is dropping off."
- 3.86** But your brain is also adjusting to that, because as you go into the luteal phase in particular, and your amygdala is becoming more interconnected with these other areas of your brain, and you're not used to that, it feels like, "Why does everybody hate me?" Because you don't realize that you're going to ebb and flow between how sensitive you are to signs of threat. Just even giving her a language to be able to understand what's happening, I think, can really do a lot and really help daughters as they learn to understand their bodies.
- 3.87** Past knowing the information, having the conversations, what proactive steps should husbands take with their wives in knowing this information as it relates to their sex life and expectations?
- 3.88** I think there's been such a disservice to women by not being taught about this. Again, we talked about this with calories, but the same is true as sexual desire. It's like, we've all been told it's, like, one size fits all, and you should have this amount of sexual desire, and you should have this amount to eat. Meanwhile, women are like, "This doesn't fit me. Is there something wrong with me? Is there something wrong with my relationship?"
- 3.89** I think that having a partner and teaching your partner about this, and then knowing that yourself and having those open conversations about this, I think is a total game-changer. I think that it's really the key to true intimacy, is being able to discuss these things and having these be issues that are shared within the relationship. I know that, within my own relationship, my husband is very tuned in to—he knows my cycle better than I do. He's like, "You're day 19." I'm like, "I am not." Then I look at my, and it's, like, day 19. I'm like, "All right, fine."
- 3.90** But having those conversations and understanding it, it's being able to—he knows not to take things too personally. If I react differently in the luteal phase, for example, if something seems threatening to me, he understands now why the fact that he loaded the dishwasher wrong can seem like a threat, almost like, "If you loved me more, you would do it the right—" which is silly when you say it out loud, but your fear-primed brain, that's ultimately where you're at. Being able to have open communication about that stuff, I think, is hugely important.

- 3.91** Then with the sexual desire, also same thing, because I think that very many men, because men don't cycle like women do, if their partner has low sexual desire, they think that there's something wrong with them or there's something wrong with their relationship. Because for men, oftentimes, that's what causes low sexual desire, not the fact that you have this other hormone that's turning the volume down. Having that conversation also, I think, changes men's expectations about what's normal. I think it's a huge relief.
- 3.92** I know that I've gotten so much feedback from men who read my first book, where I talked a little bit about cyclicity and desire, and they were like, "I have been feeling stressed out about this my entire relationship, because I was feeling like there was something wrong with me or something wrong with my relationship. I didn't realize, and I was putting all this pressure on my partner, and I didn't realize that—and the reason I was putting pressure on my partner was that I wanted validation that there's not something wrong with me and not something wrong with the relationship."
- 3.93** Having that open conversation and realizing that, it takes the pressure off men, and then they don't put as much pressure on women because a lot of the pressure, I think, that comes from men, is men feeling like they need to validate their partner's attraction to them because they don't understand what's wrong. I really think that having open conversations about these changes is, like I said, I think it's a real door-opener with true intimacy.
- 3.94** When a woman is testing her hormones when she's getting older, and the doctor only wants her to be tested at day 17 or day 18, how does that work for the rest of the cycle in knowing what's happening?
- 3.95** I could write a dissertation on this topic. So one-shot hormone measures, where it's like, and it's usually day three, because that's when they want to look at FSH, which is follicle-stimulating hormone, and then something like day 17 or 22, because they're trying to look at progesterone release, when they take these one-shot hormone measures, they can be kind of useful.
- 3.96** So if I measure you at day, let's say, 17 or 22, and there's no progesterone, I'm just going to say, "OK, those are really, really low levels. You probably didn't ovulate this cycle." But even with that, you might have ovulated the cycle before, and it just might be this one cycle that you're not. And so these one-shot hormone measures are oftentimes not very informative, and they're also not very informative because the range of what normal is, and I'm sure any of the women in this room who have had their hormones measured, the range of what normal is, it's huge.
- 3.97** It's like saying, "A normal income is a difference between \$30,000 a year and \$2 million a year. If you're somewhere in here, you're normal," and that's just not helpful at all. And so there's so much range, because there's so many individual differences. And so, obviously, the best solution to this is that you test regularly across the lifespan. But if

you can't do that now, what I would recommend is, they do have these at-home tests now where you can measure your hormones every day across the cycle.

- 3.98** Like, Mira Fertility has one that's really nice. You do it at home, which is nice. You do these little strips and you stick them in a device, and it gives you your hormone levels. Do that for one or so cycles and you don't need to bankrupt yourself buying hormone strips. But then learn the physiological indicators, too, because when you start to notice, like, if you measure your body temperature and there's differences in cervical mucus that you can pick up on, and you'll also notice, there's oftentimes just some other physical changes that you can monitor that will help to—you can give those a label.
- 3.99** If you do it one or two cycles, where you're measuring your hormones and you're feeling around for cervical mucus and you're measuring your temperature, you'll start to learn what your body, like, "Oh, OK, this is what it feels like when I start to release progesterone. OK, got it." And then, "Oh, OK, this is what it feels like when I'm normally ovulating." Then after you do that, a couple of cycles, you don't need to buy the hormone test anymore. You'll just be able to tell. That, I think, is really useful, and especially for women as you get older.
- 3.100** I mean, I'm in my late 40s now, and I've been measuring hormones more or less, and just tracking my cycle and everything for a long time. It's really useful, but of course, they don't— because now I'm like, OK, I can see what's happening here. But a lot of women don't know to start, and so then they're in this situation where they feel like they're in a hormonal no man's land. I really do like those daily hormone testers, and then learning to identify within your own body when you're ovulating and when you're not.
- 3.101** So one last thing I want to say about that is, if your hormone levels are normal and your doctor is telling you that there's not anything wrong and you're not in perimenopause or you're not whatever because your hormone levels are normal, that's not appropriate, because a normal, like I said, \$30,000 and \$2 million. If you're somebody who owns—let's say that you have a \$2-million-a-year salary, and then they measure your salary and it comes back at \$30,000 and they say, "You didn't lose any money," no, that's not true.
- 3.102** You did. You just lost over a million bucks, but it's just because you started at, your normal was high and then you're low, you're still within the range of normal. If they don't have a baseline and they tell you, "Well, your hormones are normal," but you don't feel like things are right, then I would find a different doctor who's willing to listen to your symptoms, instead of just relying on a hormone test.
- 3.103** I have a question about progesterone, as it relates to lowering or dampening the immune system, you had mentioned—how people with asthma might get worse during the luteal phase. It can increase chronic illness symptoms. But I'm having trouble

understanding where they also will put people on immune suppressants to manage those types of symptoms as well. So you'd think progesterone would actually make those symptoms better.

- 3.104** Yeah, it depends on what they are. For example, because even some types of autoimmune disease are the result of an overacting of the Th1 immune response, and some are the result of an overacting of the Th2 response. We find that certain autoimmune conditions get worse when progesterone is high and some get better. So some of it just depends on what the condition is, because there are conditions that get worse—like I said, asthma, because your respiratory drive increases, and that means that your feelings of breathlessness are oftentimes going to get worse.
- 3.105** But there are other conditions where women feel better. And so, migraine headaches, for example, is one where, many times, although some women get worsening migraines in the luteal phase, it's like, the way that all of this works is not something that's totally well understood, just because women have been understudied for so long that we know very little about these things. But many women experience relief of migraine headaches during the luteal phase of the cycle, just because of some of the neuro-modulatory effects that progesterone has on the brain.
- 3.106** It actually increases what's called a GABAergic activity in the brain, so it slows the brain down. And one of the things that tends to cause migraines is, it's like overexcitement in the brain. It's too much excitement. And same with epilepsy. So many people who have epilepsy get relief also in the luteal phase of the cycle. So you get some things that can get worse, and some things that can get better.
- 3.107** How you said one size doesn't fit all in this phase for nutrition, do you think we could correlate, as well, one size does not fit all for mental health guidelines?
- 3.108** Fascinating. No, absolutely. Absolutely. One would expect, and we do tend to see premenstrual worsening, or it's called PME's, or premenstrual exacerbations, of different types of mental health conditions. But you can also get it where you get premenstrual exacerbations of just everyday bad mood that can push itself into a diagnostic level, if it's measured in the luteal phase, which I hadn't even thought of, but is really fascinating.
- 3.109** And so I would think that it might be helpful to do, to administer the DSM criteria at different cycle phases and see whether you got significant differences, and then where you land with your diagnosis. That's fascinating. Another just really quick thing, because this is where I thought you were going, was, there is some research that finds that extinction training that you do for people who have, or extinction therapy, for people who have PTSD—this is where you try to, it's like exposure therapy, where you try to get somebody to extinguish their fear of some sort of an event or a stimulus.
- 3.110** It works less well in the luteal phase than it does in the follicular phase. There's been several studies now that have found that to be true, and they find this to be true both

in rats and in humans. There's also probably a really important role for cycle phase, in terms of helping different types of therapeutic interventions as well.

- 3.111** Like, for instance, OCD and exposure therapy and things like that.
- 3.112** I would bet that would be exactly the same thing. I'm thinking that would be exactly the same thing.
- 3.113** Last question would be understanding that, during this time or this phase, the luteal phase, the immune system is more dampened or suppressed. At a time right now where there's so many different health things out there as far as detox, parasite cleanse, intense exercise, and all that, would you actually recommend not doing those things during that phase?
- 3.114** Yeah. I mean, the body is working harder during that time. If I was doing this across the cycle, I would put anything that's going to stress your body out, do it in the follicular phase. I find that there's also been some research, and I didn't talk about this, because I thought I wasn't going to have enough time. I'm like, "I should have added that in there." But there is also research showing that we don't put on muscle mass as well during the second half of the cycle as we do during the first half of the cycle, because estrogen actually primes a lot of pathways that help to build muscle tissue.
- 3.115** It's anabolic, meaning that it likes to build tissue, and progesterone is catabolic, and it likes to break tissue down and differentiate cells. They've done research where they have women doing the same amount of exercise at just, heavy loading during different points in their cycle. What they find is that, when you have women exercising or doing more of their resistance training in the follicular phase of the cycle, that they build more muscle mass than they do when they do the exact same number of workouts, but are doing most of their heavy lifting in the luteal phase.
- 3.116** Anything that's going to stress the body out and make it work harder than it already is, I tend to heavy-load those things in the first two weeks of the cycle and try to be more gentle with myself in the second two weeks of the cycle. It really makes a difference because—and again, if you wear an Oura ring, it's really easy to see, your recovery scores are always worse in the last two weeks of the cycle. It's just that your body needs more time.
- 3.117** It's working so hard. Your basal metabolic rate is up. You're differentiating a bunch of tissue, getting ready for the possibility of pregnancy, and you're just you're strapped. Just making a point of understanding this and going to bed earlier and trying to manage your stress and not treating your body like an amusement park. I can go out with my friends, we can drink a bottle of wine and whatever, and I can go to bed late in the first two weeks of my cycle, I'm fine. I wake up and I'm right as rain; I'm ready to go for a run.

3.118 I do that in the second half of the cycle. It's terrible. It's like I drove off a cliff. My recovery is terrible, my heart rate never dropped. This is very common for women in the second half of the cycle, because the luteal phase is a time of this incredibly metabolically stressful period in the cycle. Anything that puts stress in the body, first two weeks. You're welcome.

Lecture 4: Sex Differences

At a glance — Sex differences in psychology and physiology.

- 4.1 Hi. We're today going to be picking up where we left off talking about women, hormones, and the birth control pill. We spent the first couple of lectures talking about the ways that men and women are different from one another. We're different from one another because we've each been confronted with unique adaptive challenges that are specific to our own biology. That makes us psychologically different, it makes us hormonally different, and it makes us physiologically different in a lot of ways from head to toe.
- 4.2 Now, today, I'm going to be talking about the way that science has chosen to ignore biological sex differences, and the way that this has created problems for both women and for men. Despite the fact that men and women have each been confronted with unique evolutionary paths, stemming from the fact that we have sex differences in our minimum levels of obligatory investment in reproduction—we have different roles in reproduction, we have different-sized sex cells—science and medicine has been very slow to acknowledge this.
- 4.3 The fact of the matter is, the way that women have been treated in science and medicine is—they've been treated as if they're just small, less-hairy versions of men. It has been long believed that the only things that really made women different from men when it came to medicine and biology were the parts of the body that get covered by a bikini. Hence, the term "bikini medicine", which is this idea that if you give a drug to a woman that was tested in a man, the woman's body should respond exactly the same way as a man's does, unless it has something to do with her breasts or her ovaries.
- 4.4 But this just isn't true. We've been talking about all of the different ways that men and women differ from one another. But this, for a long time, wasn't really appreciated within medicine and within science. Up until 1993, which wasn't that long ago, most biomedical research was conducted almost exclusively on men and male research subjects. When I talk about male research subjects, I'm usually talking about nonhuman animals.
- 4.5 Up until 1993, there were still papers being published on things like the normal course of human aging, where it was a study done completely on men. So we're trying to make assumptions about what women's bodies are going to do, based on science conducted on men. Now, in 1993, scientists started to call for the fact that biological sex does matter and that it does influence outcomes.
- 4.6 And so the NIH made it necessary for clinical trials to include women in research, if they're going to get federal funds. And so science has gotten a little bit better, in terms of including women in research, but they're not doing it in a way that actually allows for

the test of sex differentiation. And this is a problem. I'm going to show you now the results of some research that's been done to look at the proportion of research studies that are and are not differentiating between participants' sex.

- 4.7 So this first study I'm going to be talking to you about is a research study that was done analyzing research publications in the field of neuroscience and psychiatry. We know, especially in psychiatry, that a number of different outcomes that are studied are sex-differentiated. Women, for example, are two times as likely as men to be diagnosed with depression and anxiety. Men are about three times as likely as women to be diagnosed with autism spectrum disorder.
- 4.8 We know that there's a lot of sex differentiation in these outcomes. Neuroscience serves as the foundation for understanding these differences. Neuroscience is the study of the brain and the study of the processes that create the experiences that the brain creates. The research that's done in the field of neuroscience informs what we learn in psychiatry.
- 4.9 A study recently that was done in 2020, so not that long ago, they looked at changes in the reporting and inclusion of the different sexes in research studies in these two fields by surveying 10 years of papers in six of their most famous research journals. What they found was that things haven't gotten that much better since the time that the NIH started requiring women to be included in research. Even when making comparisons between the years 2009 and 2019, over that 10-year span, things didn't improve much.
- 4.10 If you look at the figures that I have posted above, the first two are just showing whether or not both men and women are being included in their research studies. What you can see is that, of all of the different studies that they analyzed, the majority of studies, particularly in neuroscience, didn't include both males and females. The graph that you can see on the left-hand side shows that, in the field of neuroscience and psychiatry, when you have studies that only include one sex, that sex is usually male.
- 4.11 So those top bars are showing you the percentage of studies that included male-only participants. Those bottom bars are showing you the percentage of studies that included female-only participants. You can see that, if only one sex was studied, the sex that was typically studied is usually male. If you look down at those bottom bars, this is looking at the proportion of studies that actually include both sexes, so both male and female research participants are being included.
- 4.12 And what you can see is, in particular in the field of neuroscience, the majority of studies are not conducting any analyses based on sex. Even though they're studying both males and females in the case of nonhuman animals, or both men and women in the case of humans, they're being lumped together, and they're never being looked at separately to see whether there's sex differentiation in the outcomes.

- 4.13** If you look at the bottom left figure, it shows that 60% of studies in psychiatry do analyses based on sex, which means 40% of them do not. When we look at neuroscience, only 20% of studies are testing for differences between the sexes. This is insane when we consider the fact that we know that males and females are different from one another. We've been confronted with two different sets of adaptive challenges. The process of evolution by selection has shaped different sets of adaptive solutions to those challenges, and it means that our bodies and our brains are sex-differentiated.
- 4.14** This, of course, is going to create problems. Before we get to those problems, I'm just going to show you the result of one more set of studies that were looking at, the degree to which sex differentiation is tested for by scientists across different fields of science. This was another study that was done in 2021. Here, the researchers looked at all of the papers that were published in top journals within the fields of behavioral ecology, physiology and neuroendocrinology, endocrinology, general biology, immunology, neuroscience, physiology, and reproduction.
- 4.15** They looked at all of the papers that were published in the top journals in all of these fields in the year 2019. They pulled out a total of 720 articles that they looked at, and they wanted to see which of these papers, of all these papers that they pulled, how many of them included analyses looking for sex differentiation, and how many of them found sex differentiation when they were looking for it.
- 4.16** What they did is they pulled these articles, and there were 720 articles altogether. What they found was, of these 720 articles, only 352 of them included both sexes. This is less than half. Less than half of the research studies that they pulled included both sexes. Less than half of that half actually analyzed the data by sex. In these other studies, the 352 that included both males and females, or both men and women, they had both men and women in the study, but they never tested for sex differentiation.
- 4.17** In less than half of them was sex even analyzed and included in the research analysis, to see whether sex differentiation exists. What the researchers then found was that, when sex was compared, in that minority of studies where researchers actually test whether men and women or male and female subjects differ from one another in the outcomes, more than 70% of those found a sex difference. When sex is included and sex is tested, oftentimes sex differences exist.
- 4.18** Yet, sex is typically ignored. This is a huge problem because so many conditions are sex-differentiated. As noted, when you look at things like the rates of autism, it's diagnosed three times more frequently in men than in women. ADHD has a male-to-female ratio of about two and a half males for every female. Dyslexia is about two to one, men to women. When we look at other conditions like dementia, anxiety disorders, depression, and eating disorders, these are conditions that are asymmetrically present in women.

- 4.19** This tells us that men and women are not the same. Our brains are vulnerable to different types of pathology. The development of that pathology is oftentimes different, and the treatment for that pathology is oftentimes different. And yet, in most of the studies that are being done, looking at the basic science of the mechanisms that contribute to both health and disease, males and females are lumped together, and nobody's even bothering to ask the question of whether or not there are differences.
- 4.20** And this is a problem even for conditions where there's not sex differentiation in the rates of diagnosis. When we look at something like heart disease, for example, we know that this is the number-one killer in the West of both men and women. Both men and women are equally as impacted by heart disease. But the mechanisms are different. Even though we have the same disease, the path to get to that disease is oftentimes sex-differentiated.
- 4.21** For men, for example, two of the biggest risk factors that can be used to predict the probability of developing heart disease are hypertension, which is high blood pressure, and obesity. These things are not as diagnostic in women. For women, the biggest predictors of developing heart disease are having diabetes, being a smoker, and being exposed to chronic stress.
- 4.22** When we lump men and women together and don't bother to look for sex differences, we wouldn't know these things. We would only be making assumptions that are based on either mixed data, where both men and women are combined together and nobody's bothering to test for differences, or research conducted exclusively on men, which is the way that research was done for a very long time, just simply because men are easier to study than women, because you don't have to worry about: Is she pregnant or is she not? Is she menopausal or is she premenopausal? What phase of the menstrual cycle is she in?
- 4.23** Researchers avoided asking these questions and having to deal with the difficulties of studying women, because you have to keep track of these differences, simply by studying men and then applying the insights that they learn from men to make predictions about what's going to happen in women. But when you do this, it's not always going to work. You would be looking, in the case of heart disease, for whether a woman has hypertension and obesity. But that's not going to predict the probability that she gets heart disease. That's not going to work for her.
- 4.24** In fact, this problem is so ubiquitous in slowing the progress of what we know about health and disease to such an extent that researchers have argued that the huge amount of progress that we've seen in the development of effective treatments for breast cancer and prostate cancer, the reason that those two are doing so well in terms of research advances, compared to some of the other types of cancers like lung cancer and pancreatic cancer, is because the sexes are necessarily studied separately.

- 4.25 When you study breast cancer, you're studying women. When you're studying prostate cancer, you're studying men. And researchers believe that the reason that they've been able to make so many great strides in research in these areas is because they don't have all the murkiness that happens when you lump males and females together, assuming that their bodies are going to be responding the same way to a disease, assuming that their bodies are going to be responding the same way to the same treatment, and assuming that the mechanisms that create the disease are going to be the same.
- 4.26 We just can't make that assumption. It's not reasonable, once we actually take the time to understand the fact that men and women have confronted very unique adaptive challenges, and this has led us down different evolutionary paths. This is also a problem because, when we lump together data that's collected from both males and from females, it oftentimes can mask treatment effects. This is particularly true when we see that baseline levels of a disease are sex-differentiated, as many diseases are.
- 4.27 We talked about the fact that, for example, autism spectrum disorder is diagnosed significantly more frequently in males than it is in females. Now, I want you to imagine that we're conducting a study where we have a treatment that's supposed to ease some of the symptoms of autism spectrum disorder. We're going to be testing this in a group of both men and women, not sex-differentiated.
- 4.28 We bring our group of men and women into the lab. We randomly assign half of them to take a placebo treatment. We give half of them an actual treatment, and then we measure their outcomes. What we see looks like this set of figures or this set of bars that you see on the left-hand side. It looks like there's no treatment effect. The participants who took the placebo or the control treatment have the same amount of disease impact as those who took the actual treatment.
- 4.29 So we might scrap the treatment and say, "This treatment doesn't work. We shouldn't use it." But if you actually took the time to divide the sexes and look at each sex separately, what you might learn is that in men, the base rates of autism spectrum disorder are significantly greater than what they are for women. You can see that by looking at the differences in the black bars between males and females in the control condition, because these are participants who took a placebo treatment, which isn't having any impact on the disease. What you can see is that, in the absence of any treatment, men have significantly higher disease impact than women.
- 4.30 But then, when you give the men the treatment, they actually have improvement in symptoms, while women are made worse. There's so many cases where this has been what has happened, when medical research is being conducted; is when we lump men and women together and assume that the processes that lead to the disease and that the processes that can be used effectively to treat the disease are the same, it oftentimes creates murkiness, where it looks like there are treatments that might be effective in one sex but not the other.

- 4.31 It makes it seem like those treatments don't work. They're abandoned, despite the fact that they may be beneficial to one sex but not the other. This, of course, creates problems. One of the most well-publicized examples of these problems is what happened in 1992 when the drug Zolpidem was cleared by the FDA for commercial use. Zolpidem is more commonly known by its brand name Ambien.
- 4.32 When Ambien first hit the store shelves in 1992, after being tested in clinical trials, which were populated primarily by male participants, just like most clinical trials—because even though the NIH said that women need to now be included in research, that just usually means "some women, wink, wink." It doesn't mean that you need to be testing women in equal numbers as men. It just means that there needs to be some women.
- 4.33 In these clinical trials, testing Ambien's effectiveness, soon after it hit the store shelves, doctors' offices were being flooded by phone calls from women who were having negative side effects from the drugs. These negative side effects were ranging from anything from drowsiness to having pretty severe cognitive impairment. Further research, as the researchers began to dig into their data, they learned that women actually metabolize this drug differently than men do.
- 4.34 For a woman who takes a 10-milligram dose, which is a standard dose and one that's recommended for both women and men, after taking the drug, the levels of the drug in their system were twice that of men's because they metabolized it more slowly. And so, ultimately, this led to a revision in the way that Ambien is being prescribed to men and women, with dosage differences now being sex-differentiated. But this sort of thing goes on all the time. This isn't unique to Ambien.
- 4.35 Women are two times more likely to have adverse effects from prescription medication. And a recent study revealed that 8 out of 10 new medications are regularly pulled from the shelf during their first year of use because of unanticipated negative side effects in women. This is 80% of drugs being pulled from shelves during their first year of use because of unanticipated side effects in women. We can't assume that women's bodies are going to respond to things the same way that men's bodies do.
- 4.36 Pretending that they do isn't good for women. It's not good for anybody. It contributes to these sorts of issues. It also leads to routine diagnostic delays that we see between men and women. And because men, historically, were the ones who were studied, again, because they're simpler to study, because they don't have as many hormonal changes due to things like pregnancy status, menopausal status, and cycle status, most of the recommendations and most of our understanding of what the symptoms of disease are have been grounded in research conducted on men.
- 4.37 And not surprisingly, a recent study where researchers looked at 1,000 different health conditions that are suffered by both men and women found that women routinely experience diagnostic delays in more than 700 of these different health conditions. And

these are health conditions that are ranging from anything from a bladder infection all the way to things like cancer. This is just a snapshot of some of the data that they showed. If you look at this figure, the purple dots represent women, and the green dots represent men.

- 4.38 On the Y-axis, you can see a list of some of the conditions that they studied, so looking at things like sepsis, which is a blood infection, looking at heart failure, pneumonia, tuberculosis, and so on and so forth. And what you can see is that almost every single disease studied, and everyone that was studied that's shown on this particular graph, you can see that it takes women significantly longer to receive an accurate diagnosis than what it takes for something to be diagnosed in men.
- 4.39 And this is a relic and a legacy of the fact that men were studied first. Most of our understandings about the mechanisms that contribute to disease, the symptoms of a disease, and the treatment of disease were historically tested only on men. Applying these things to women creates a situation like this, where women end up being subject to not only adverse reactions from treatments, but also diagnostic delays.
- 4.40 This also harms women in other ways. Research has now found that everything ranging from seat belt design to what's considered the optimal room temperature in an office were something that were created by men. If anybody has ever, if you're a woman and you've worked in an office, you know that you're always cold. If you're a man and work in an office, you know that your female colleagues are buried under lots of blankets or wearing a winter hat with winter gloves on.
- 4.41 The reason for this is that the ideal office temperature was based on the metabolism of a 40-year-old man in a business suit. Men's metabolism runs hotter than female metabolism because they have a greater amount of muscle mass, and muscle burns energy, and that creates heat. What is considered even a reasonable office temperature is something that's very much based on a male ideal of what's normal. We can't assume that what works for men is also going to work for women. This is true for room temperature.
- 4.42 This has also recently been found to be true for seat belts. Seat belt design was originally created with male bodies in mind, and they've since had to revise the way that seat belts are designed, so that way they can accurately and appropriately protect both male and female bodies. We also find that even legal standards that exist, and these are legal standards that are currently in place, have been designed with the idea of men in mind, in ways that can be really harmful to women.
- 4.43 I want you to consider for a moment what's known as the reasonable person standard that exists in the law. There are some laws that exist for things like stalking and sexual harassment that invoke what's known as the reasonable person standard. The idea is this, is that the jury and judge are asked to consider, "Would a reasonable person consider this behavior threatening?" So in the case of stalking, for example, it's like, if

you say that somebody is stalking you because you're receiving unwanted phone calls, somebody is sitting outside your apartment at night, would a reasonable person feel threatened in response to these behaviors?

- 4.44 This is how it's determined whether or not the behavior is actually stalking or not stalking. With sexual harassment, the reasonable person standard is often also applied. Would a reasonable person find this behavior offensive, hostile, or abusive? The problem is that what a reasonable person is differs depending on whether you're a reasonable man or a reasonable woman. We know from the lectures that we had previously that one of the things that our brain does is that we have a very sensitive threat detector.
- 4.45 We know that our threat detector tends to be especially heightened in cases of physical vulnerability. Women are more physically vulnerable than men because we're not as physically formidable, and we're also more likely to be the victims of injury, particularly from a current or former romantic partner. This means that women's smoke detectors, when it comes to behaviors that could be seen as potentially threatening, are very sensitive to the possibility that something like getting a bunch of unwanted phone calls from a former romantic partner, that this might lead to injury or harm.
- 4.46 Men don't have as sensitive of a smoke detector, because men aren't as vulnerable. When we're making a decision about whether a reasonable person would consider unwanted phone calls or somebody's parked outside your apartment in the late hours of the night, potentially threatening, the way that that's going to be perceived will depend on whether you have a male or a female brain.
- 4.47 And so, invoking the reasonable person standard and determining whether something is considered stalking, and whether the victim of stalking should get legal protections, will oftentimes depend on whether the jury is made up of reasonable men or reasonable women. This is something that is, of course, bad for women.
- 4.48 The same is true for sexual harassment cases. Men and women routinely differ in the degree to which they find different types of behaviors offensive, hostile, and abusive. This here is just an example of some different types of activities and what constitutes harassment. Women are much more sensitive to cues of sexual harassment than are men, which can create issues when we're trying to determine whether a reasonable person would consider a particular type of behavior harassing.
- 4.49 For example, we know from research, and here's just some data bearing on this issue, that getting unwanted sexual attention, women are more likely to see that as harassment than men, with almost 80% of women saying that that's always harassing when you're getting unwanted sexual attention in the workplace, and only 60% of men see that as potentially harassing. There's differences here.
- 4.50 What this means, if you're a woman who's feeling threatened or you're feeling uncomfortable in the workplace because of these types of activities, if you go to a legal

venue to try to be able to get protection for yourself, the fate of what happens next is going to depend on the sex of the jurors. Is this comprised mostly of reasonable men or reasonable women? This is something that we can address if people actually acknowledge that sex differences exist.

- 4.51 Because if we're able to tell people, "Just so you know, men and women see things differently," and allow for decision-making to take that into account, we can get very different types of outcomes than we can get when we pretend that there aren't sex differences in our perceptions of the world or, in the case of the medical issue, sex differences in our physiology. Also consider what happens with perceptions of sexual intent that oftentimes can muddy the waters in outcomes of sexual assault and rape cases.
- 4.52 So although juries are always instructed that, when they're trying to determine whether a sexual assault or a rape has occurred, that legal consent must be explicit—that is, it must be verbally explicit that sex was consented to and cannot be inferred—research nonetheless finds that perceptions of a victim's sexual interest oftentimes impact jury decisions, with people being less likely to say that something was sexual assault or rape, if they perceived that the woman was consenting to sex by way of her actions or behaviors.
- 4.53 This also can create issues, because men routinely overestimate sexual interest from a range of very subtle cues. The figure that you're looking at here is one of the original studies looking at this male sexual overperception bias. In this study, the researchers had each, men and women, evaluate the amount of sexual interest that a woman was communicating in a variety of different ambiguous scenarios.
- 4.54 For example, one of the scenarios is: imagine a man and a woman are at a party together, and the woman smiles at the man. Does this mean that he is sexually interesting? Does this mean that she wants to have sex with him? Or imagine a man and a woman are in the workplace, and they're walking down the hallway, and the woman brushes her arm up against the man. Is this communicating that she's sexually interested in the man or not? They read these different scenarios, and then each, men and women, evaluate the degree to which they believe the woman in the scenario is communicating sexual interest.
- 4.55 What they find is that, in every one of the scenarios asked, men over-infer sexual interest on the part of women, compared to what the women infer. This was later followed up on in a study where the researchers, rather than having people evaluate the actions of others by reading these scenarios that have a third-party female and a third-party male engaged in certain types of actions and having to make estimations on it, in this study, they wanted to study it in a more of a first-hand, first-person sort of way.

- 4.56 What they did was, they had men and women come into the research laboratory, and they had them imagine that they were on a date. Then they had them rate the degree of sexual interest communicated by the woman—so if you're the woman, it would be by you, and if you're the man, it would be by your date— of 15 common dating behaviors. Women were shown these different types of dating behaviors, so for example, smiling at your date. The woman was asked, "If I engaged in this action, it would mean I want to have sex with my date."
- 4.57 Then men read it as if they were the person on the date. So men read the question, so for example, "Your date smiles at you." Then men would see the question, "If my date engaged in this action, it would mean that she wants to have sex with me." Then men and women evaluated these things on a scale of negative 3, which means extremely unlikely, "There's no way she wants to have sex with me, or there's no way I want to have sex with him," to a positive 3, which means, "Yes, absolutely. That means that sex is forthcoming."
- 4.58 Here's what they found. Again, with every single behavior that was measured, they found that men over-inferred sexual interest to a greater extent than what women were actually communicating by a behavior. The women are the pink bars, the men are the dark-colored bars, and what you can see is that, when it comes to a kiss, for example, women say that that's communicating sexual interest at a level of about 0.05. Men are rating it at a 1.5. If your date tells you that she loves you, a woman is communicating that she'd like to have sex at about a 1, men estimate it at a 1.5.
- 4.59 Holding hands, both men and women say this is unlikely to be communicating sexual interest, but men are still more optimistic about what that might mean, that women are actually communicating. The most sex-differentiated one that they found was the invitation up to the apartment after the date for a drink, where men rated that as a 2 on sexual interest, on a scale of negative 3 to positive 3.
- 4.60 So men are believing that that is very much communicating sexual interest, with women's interest only being rated at a 0.5. So there's some mind-reading errors going on here, with men having, mistakenly, when they're making these guesses about what women are thinking and what women desire, that they're over-inferring sexual interest.
- 4.61 This has also been replicated in a speed dating paradigm. In one study, researchers had men and women come in. There was about 10 of each in each group. Each man would talk to each woman for a couple of minutes. It was three to five minutes, and then they would have them switch spots, and then they would talk to the next person, and then they would switch spots, and they would talk to the next person.
- 4.62 They would do that until each man and each woman had had a chance to interact. After the interaction, the men and the women were asked to report on the extent to which they were interested in each of the people that they talked to. Then they also had the

men and women report on the extent to which they believed the people they were interacting with were interested in them.

- 4.63 What they found was that, across all of these different sessions, men, again, over-perceived sexual interest on the part of the people that they were talking to, with women's level of actual sexual interest on a scale of negative 3 to positive 3, was about a negative 1, and men were perceiving it at about 0.75. Just showing that men were over-perceiving the interest of the people that they were talking to.
- 4.64 They also found in this study, which was really interesting, that the magnitude by which men over-perceive sexual interest was based on the attractiveness of the woman they were interacting with. The more attractive the woman is, the more the man believed that she was interested in him sexually. Why is it that men over-perceive sexual interest? Are they just creeps? No. This is the smoke detector principle, going at work again.
- 4.65 Whenever a person is making a judgment or making a guess about whether somebody is interested in them, it is always based on ambiguous cues. None of us is straightforward about what our interest is when it comes to our romantic pursuits. People don't just walk up to people that they're potentially interested in and say, "Hi, I'm sexually interested in you." Wouldn't the world be so much easier if that's actually what we did? But yeah, we don't. There's always these ambiguous cues. You smile, you look away, you text, but then you don't text back too soon.
- 4.66 There's all these ambiguous cues, and we're always having to make guesses about what somebody's intentions are. There's no perfect information. This is that good old smoke detector principle at play again. We have inherited a brain that tends to err when we're making decisions under uncertainty, when we don't know if something is threatening, or we don't know if something suggests sexual interest, our brain is wired to err on the side of making the mistake that has been least costly over the course of evolutionary time.
- 4.67 Because of the potential evolutionary benefits of a low-cost sexual opportunity in terms of gene transmission for men, men's brains are wired to err on the side of sexual interest, because the cost of missing a potential mating opportunity is far greater from an evolutionary perspective than the cost of being embarrassed because you assume that somebody's interested in you and you're actually not.
- 4.68 When we're making guesses about somebody's behavior, there's always a way that we can be wrong, and we're not always right. And so, our brains are wired in a way to make the least costly mistake. Given the gene transmission benefits of short-term mating for men, for men, this means erring on the side of sexual interest. This is something that reliably occurs.
- 4.69 Researchers have tested this in 800 different ways, and they find, time and time again, that men tend to over-perceive sexual interest. And yet, we assume that men and

women are the same, psychologically, when we're creating things like legal proceedings or even different workplace policies. There was a recent lawsuit from a popular fast food company, where the women were suing the company because the women were told that they needed to smile and engage in an overly friendly manner with all of their customers.

- 4.70 And what this was doing, ultimately, was making women subject to a lot of unwanted sexual attention from customers who were inferring sexual intent. When the women talked to management and said, "This is making us uncomfortable; when we do the things that we're supposed to be doing, it's leading to unwanted sexual attention," they said, "No, this is the policy." This has since been revised because of this research.
- 4.71 Once we understand the differences between the sexes, it creates a world where women are less likely to be subject to either unfair policies where they're unlikely to be able to get protection from things like stalking and things like sexual harassment. But it also protects men because, once they know that they do this, they also can check themselves, and then they're not engaging in these behaviors that they don't even know are making women uncomfortable.
- 4.72 It's bad for women, and it's bad for men to assume and try to pretend that sex differentiation doesn't exist. There are powerful differences that exist between men and women. This can, of course, create harmful misfires that can be harmful to men and can be harmful to women.
- 4.73 Another issue that we get into when we make the assumption that biological sex doesn't matter, and that men and women are essentially more or less interchangeable with one another, is that it takes anything about women that's female-specific. It creates this lens by which these things are perceived as problematic or pathological. The reason for this is, again, most of what we know about human behavior and about health and disease is based on research that was originally conducted exclusively on men.
- 4.74 This means that most of our views about what's normal are male. And this means that anything about you that, as a female, that's different, is seen as potentially problematic. And this includes our hormonal changes. I want you to consider for a minute this idea of PMS, this premenstrual syndrome. One of the things that we talked about in some of our earlier lectures is this idea that, when women's hormones change across the cycle, this changes women physically and it changes them emotionally.
- 4.75 These changes are an expected and normal part of being a human female whose body is shifting into a state that's optimized for implantation and pregnancy. But we talk about these things, and we experience these things, oftentimes as pathological. Some of the issues that we tend to see that get lumped under symptoms of PMS are things like increased sensitivity to threat. That's not a symptom; it's a sign of our inherited biological wisdom, is what that is.

- 4.76 We also have lowered energy levels and lower approach motivation. That leads to some of that depressed mood that we talked about in our other lectures. This also makes good adaptive sense when we consider the metabolic changes that are going on in the context of the luteal phase and the body preparing for pregnancy.
- 4.77 The physical changes that women experience in the luteal phase, like slower gut motility. So oftentimes, women who have conditions like irritable bowel syndrome will report that they have worsening symptoms in the luteal phase than they do in the first half of the cycle, because you do get slower gut motility during this time, because your body is grabbing onto nutrients at a greater rate.
- 4.78 It needs to absorb more nutrients because it's building a lot of new tissue and it needs those nutrients. It slows down gut motility on purpose. But again, this is experienced as a symptom, which is a sign of pathology, instead of this really nice piece of inherited wisdom we've gotten from our ancestors. Less sexual desire. We talked about that and the reasons that women experience lowered sexual desire. This isn't a symptom; this is something that's happening on purpose.
- 4.79 Our increased calorie needs. Many women report that they're hungry and that they're having food cravings during the second half of the cycle compared to the first. There's nothing wrong with this. This is actually functional because our bodies are burning more calories. We need to eat more. We have this lens where, any time that we experience something that hasn't been described and explained to us, based on a male model of what's normal, is that it seems problematic instead of being part of our design.
- 4.80 This is something that creates, by ignoring it, in addition to the fact that we tend to see some of these functional changes as pathological, it can also create pathology when we ignore them. One of the things that we know that women oftentimes tend to experience in the luteal phase is that they experience food cravings, and they're at a greater risk of binge-eating. And the binge-eating is oftentimes triggered because women aren't eating enough, and their body is responding poorly to that.
- 4.81 They end up not eating enough, and then they have a binge episode, and then they create this narrative about themselves where their body is the enemy. So many women have this narrative about themselves, where they feel like their body is the enemy and is standing in the way between them doing all of the things that they want to do. And a lot of this is based on the idea of what they're told their bodies are supposed to do has been created based on research done on men. It was never designed with a female body in mind. This is problematic.
- 4.82 Ignoring female differences from men and trying to minimize them, it does create this lens that pathologizes the things about us that make us unique from men. Even though we've been told that the thing that women need to do is try to ignore the fact that our hormone cycle, and try to ignore the fact that we're different from men, and that

minimizing these differences is the only way that we can ensure an egalitarian world where women are given the same opportunities as men, it does more harm than good. We need to move past this idea that different means unequal, and understand that different can be equal. It's just different. Different means different and nothing more.

- 4.83 It turns out that if we make the decision and believe that sex doesn't really matter, and the things about women that make them different from men are trivial and don't have much of an impact on the way that the world is experienced by each, this also sets the perfect stage for something like the hormonal birth control pill, which shuts off the production of female sex hormones, shuts off the thing about women that makes them specifically unique from men, and be able to use this as a treatment for things as minor as acne.
- 4.84 So even though when the birth control pill first came out on the scene, it was being touted as the key to women's freedom and the key to women's liberation, in ads for the birth control pill, which, of course, allows women to be able to have sex without the fear of pregnancy, this was seen as this godsend for women that would allow them to be able to go out and compete for the same resources for men, which it has allowed them to do, and so it's had a lot of really wonderful effects, but it's also really trivialized what it means to shut down a complex endocrine system.
- 4.85 When we pick up in the next lecture, we're going to be talking about the ways that the pill changes women, and the way that ignoring biological sex and ignoring the importance that our sex hormones have in creating the experience of who we are, has minimized the attention that's been paid to the range of effects that can happen when you take women's sex hormones, this key signaling property that helps create the experience of women being the people they are, what happens when we turn that off with the birth control pill.
- 4.86 What needs to be done at the clinical research levels to even the playing field for men and women to both be treated, because women have more—well, as you said, they put \$100 in, and a male puts \$5. So it feels like they are way more at risk to do a clinical trial, not knowing what it's going to do to potentially cost them reproducing later on. Does that make sense?
- 4.87 Yeah, yeah, it does. To start with the first point, I appreciate that, because women have been left out of clinical research. I mean, originally, one of the reasons that women were left out of clinical research was just good old-fashioned sexism. But then after that, it was because of legitimate concerns about what does this mean for the future of fertility. If you're testing a new drug, for example, and it has a lasting impact on something that can impact a woman's reproductive capacity, then obviously that's hugely problematic. And so, researchers are trying to protect women from these types of outcomes.

- 4.88 Really, I think that the solution to that is just, you need to do better preclinical work. Preclinical work is the work that you do to try to understand the mechanisms that contribute to a disease, and then also can contribute to a treatment. A lot of this preclinical work gets done in nonhuman animal models, but then it's also done in human models in a way where the stakes are lower, where you're just understanding how something works without actually going in and testing a new treatment, so that way you can know whether or not it's going to be safe for the female body.
- 4.89 I think that there's a way to do that, and I know that there's a way to do that, because really good preclinical work sets the stage for being able to go in and test things in humans in a clinical context and have it be relatively low-risk. But I think the big change that needs to happen is that it needs to be required that sex differences are assumed, and that if we're going to assume a lack of sex differences, that the onus is on the researcher to demonstrate that an outcome is not sex-differentiated.
- 4.90 Currently, the way that research is being done is that men and women are lumped together, and sometimes they'll test for sex differences, and sometimes they won't test for sex differences. The assumption is that the sexes do not differ, and they don't even have to prove that they don't differ in order to interpret the data with the sexes lumped all together. It's a mess. I think that what really needs to happen is that we need to start creating science and pushing for science that tests men and women in equal enough numbers, where we can test for sex differentiation.
- 4.91 Then the second thing that needs to happen is that we need to start studying women as women, because the other dirty little secret in science that I didn't—I was afraid I'd get too into the weeds, so I left it out of this, but is that, when researchers are studying women, especially in biomedical research, they'll only study them in the first couple of days of their menstrual cycle when hormone levels are really low. They do this to make women maximally similar to men, which allows the data to be easier to interpret.
- 4.92 Now, what this means, of course, is that we don't know what the treatment does to women's bodies when hormone levels are high, which is the state that women's bodies are in 85% of the time. In animal models, what they'll actually do when they're including females in research is, they'll ovariectomize the mice and the rats, and they're like, "Look, we're including women in the research. We're including females in the research," but they have no ovaries and no sex hormones.
- 4.93 Then they apply that to humans and say, "Oh, look, this is tested on females, so we know that it's A-OK." But that's not the actual state that a female is usually in. The other thing that I think really needs to happen is that women need to be studied as women and need to be studied under different hormonal contexts. We need to see, for example, if you have a new medical intervention, how does this work in the follicular phase when estrogen is the primary sex hormone? How does this work in the luteal phase when progesterone is the primary hormone?

- 4.94 Because we know that these hormones are immunomodulatory, and they modulate all of these different systems in the body. It would be very surprising if there weren't differences across the cycle, sort of like what you were talking about with the DSM-IV criteria. It's like, even measuring pathology may differ the outcomes that you get and the diagnosis that you get, depending on the cycle phase. To study women only in this narrow window of time when their hormone levels are low is also not doing a good service to women. There's a lot of changes that need to be made.
- 4.95 In order to study women at a more extensive level when it comes to treatments and also diagnosticians being able to diagnose, it's clearly evident that that process will take more time, more effort, and be more intensive and more costly to both the researchers and the diagnosticians. Would there be any sort of external environmental factor that you would use to persuade pharma and scientists that they should override profits in order to have this, let's say, medical research be in more alignment with the sexes?
- 4.96 I think that this is where policy comes in. I really do, because I think that allowing for free markets to do their things is really great, unless when everybody, by doing the thing that benefits them, if everybody loses, that's when you need policy. I really do think that it needs to be policy, because the fact is, the way that it's currently being done, it's left to the researcher's own goodwill to test men and women in equal numbers and account for cycle phase.
- 4.97 You can't do that because, for a scientist, if you're looking at the possibility of not getting tenure, you want to do research as quickly as possible, to get as many things published as you can, so that way you don't get fired. Then, for a diagnostician, the idea of having to spend twice as much time with your patients as somebody else, then you're losing money. It's really got to be a policy issue because, right now, when everybody is acting in a way that is rational and makes sense for themselves, we're creating a world that's bad for half the population.
- 4.98 And so, I really do think that it's a policy, because if it's not acceptable to publish papers that only include male subjects, or it's not acceptable for your paper to get published when it combines the data of the sexes and never looks at sex differentiation, people are going to continue to do that because it's easier and cheaper. When we make that unacceptable, that's when change is going to happen.
- 4.99 This has to happen at the level of the policy of research journals that are publishing these studies. It has to happen at the level of the funding agencies saying, "I'm not going to fund something unless we're actually testing men and women, or males and females, in equal numbers and testing for differences between them, so that way we actually understand how they work." Until those policies are made, it is unfortunate, but for many people, the rational decision to make is to do it the wrong way and to ignore women and ignore sex differentiation.

- 4.100** I talk about this in my book, in *The Period Brain*, and in particular with women in the luteal phase, because when you look at research and when you look at the way that women are often studied—I mentioned that they're generally studied in the first couple of days of their menstrual cycle when hormone levels are low, so that way, they're maximally similar to males. If I'm a woman—and I am— what I do is, I keep track of symptoms, and I keep track of responses to medications across the cycle, and especially in the luteal phase, because that's the time in the cycle when women are most ignored by science.
- 4.101** If you're going to have an unanticipated side effect from something, the probability is going to be greatest that that's when you're going to have it. It's this condition— progesterone creates in women a state that's most biologically unique from men. This is something where, if you're studying something in both men and women and not differentiating, or you're studying it in women, the dirty little secret, the way that we do it in science, where we only study them in the early days of the menstrual cycle, we're never even touching what this does in response to progesterone.
- 4.102** And so, really keeping an eye on things in the second half of the cycle, because that is— if the train is going to go off the tracks, that's when it's most likely to happen. And so, really keeping track of symptomology as they change across the cycle, keeping track of side effects across the cycle, and then communicating with your doctor, because there is some evidence that there are certain drugs, for example, that work better at some points in the cycle and not other points in the cycle. You might even be able to get your doctor to work with you to do things like cycle-specific dosing recommendations, which some doctors are doing with certain types of medications.
- 4.103** Along those same lines, you mentioned scientists have an incentive to track women only in a particular part of their cycle when including women in a study, and even then, it's at a limited capacity. Is there any evidence that they are also using people who only socially identify as women to weigh their studies? And as someone who would maybe be reading those studies, how can you understand? How can we infer transparency doing that, or interpreting those when they still count that as a woman within the—
- 4.104** Right. Gosh. People don't have time for this. The answer I'm going to give you is completely impractical and not something I would actually recommend to anybody in the world, but I mean, it's like, you've got to read the research. It's like, when you get into the research, you can see that, when they're including women. Actually, it's considered, like, the gold standard. If you publish a paper where you're only looking at women during the first couple of days of the cycle, you write that proudly on that paper, because you were really tightly controlling for the effect of their sex hormones on the outcome.
- 4.105** You can feel really good about cause and effect. You'll see it written in the paper. Then you can know, "All right, this might not necessarily apply to me every single day of the month." This is something that also, I think, is worth mentioning in conversations with

physicians, because I think that a lot of times doctors are under the impression that research is cleaner than it really is and that the results are more comprehensive, and the way that things are being studied is more comprehensive than what actually goes on in the science world.

4.106 I think that there's also—if you're experiencing, for example, let's say you're taking a medication and you find that you don't feel like the medication is managing your symptoms in the luteal phase of the cycle, and your doctor is like, "Well, I don't understand that," and then you can say, "Well, from what I understand, most research, especially biomedical research, that's conducted on women, is conducted in this really narrow window of the cycle, and I don't think I'm getting good symptom management now. Is there something else we can consider?"

4.107 Might there be something I can take in addition to this drug during this other time? Is there an alternative practice I can take or I can do during this time that's going to improve my ability to get treatment results? And having those kinds of conversations, but unfortunately, most people don't have time to get a PhD and whatever, medication, figuring out. And so trying to go and do that and read all these things firsthand is really challenging. But ultimately, I think it's the only answer at this moment, sadly.

Lecture 5: Birth Control Basics

At a glance — Birth control basics — how hormonal contraception works and its history.

- 5.1 Hormonal birth control entered the marketplace more than 50 years ago, and it entered as an opportunity for women to be able to have sex without having the fear of pregnancy. This has obviously been a huge advantage in women being able to do things like build careers, get educations, and be able to do these things without the fear of having to be unexpectedly benched because of a pregnancy that they weren't planning for.
- 5.2 However, the pill, even though it has the effect of blocking ovulation, which is good for preventing pregnancy, this also prevents women's bodies from producing their own sex hormones. So even though the pill has offered women a number of advantages and not having to have to fear the possibility of an unwanted pregnancy when trying to achieve life goals, it comes at a great cost.
- 5.3 And when it came to the pill, what we're going to be learning about as we start to move through the next sections of these lectures is that, when we change women's hormones, we change women in ways that are far more substantial than what first meets the eye. Just like the tip of the iceberg represents just a small snippet of a much larger berg of ice, so is true with hormonal birth control, by the changes that we make in women to protect from pregnancy, leading to larger psychological changes.
- 5.4 We're going to be spending the next sections of the lecture discussing the ways that hormonal birth control changes women. One of the things that we talked about in some of our earlier lectures is that our hormones, and our sex hormones in particular, are an important part and play an important part of creating the experience of being the people that we are. When we change those hormones, it means that we're going to change women.
- 5.5 To introduce us to these ideas, the first thing that we have to do is understand a little bit about how a natural menstrual cycle is regulated. One of the things that I mentioned when I first started talking about this is that ovulation is something that starts in the brain. It starts in the brain on the first day of the menstrual cycle. Here we are orienting ourselves, again, to a 28-day menstrual cycle.
- 5.6 Just as a reminder, during this 28-day menstrual cycle, you get estrogen rising in the first half of the cycle, and that's shown by the red line. And then you get progesterone rising and falling in the second half of the cycle, and that's the luteal phase, which is shown there with the blue line. The first day of the menstrual cycle begins because of a hormone crash that occurs at the end of a menstrual cycle, if a pregnancy doesn't take place.

- 5.7** And when hormone levels are low, as you can see them being illustrated in low, both estrogen and progesterone are low, this tells the brain that you are not ovulating and that you are not pregnant. The brain, being oriented toward reproduction, as a brain designed by the evolutionary process will be, thinks that this is a terrible idea and that it needs to start initiating the processes required for ovulation to occur, to see whether pregnancy might be possible on this next cycle.
- 5.8** So when hormone levels are low, the brain will start stimulating the pituitary gland using a hormone, a releasing hormone, called GnRH. It's gonadotropin-releasing hormone. The hypothalamus will release GnRH, which tells the pituitary gland—which again is this gland that lives just south of the brain, whose job it is to communicate with endocrine organs—the brain will tell the pituitary gland that the ovaries need to start stimulating egg follicles to prepare for the possibility of ovulation.
- 5.9** Then the pituitary gland stimulates the ovaries. The ovaries will start releasing estrogen, and that's what we see happening in the first half of the cycle. After an egg gets released, that empty egg follicle becomes that temporary endocrine structure, and it begins releasing progesterone. When progesterone starts being released by the corpus luteum, that signal, that progesterone signal, gets picked up in the brain. That is the brain's signal that ovulation has happened and that the body is waiting to see whether a pregnancy might occur.
- 5.10** Because of this, when the hypothalamus and the hormone receptors in the hypothalamus of the brain, when they start picking up on progesterone, they inhibit the release of GnRH, which means that the brain and the ovaries are no longer having that conversation, where the brain is telling the ovaries to do something. It quiets the communication between the brain, the pituitary, and the ovaries, allowing for nature to take its course, and allowing for this egg that's just been released to potentially be fertilized and potentially implant.
- 5.11** When the brain gets the signal of progesterone, that means that it does not stimulate the ovaries to produce egg follicles to ovulate. This is an important piece of information to know because the birth control mimics this action. The way that hormonal birth control works is that it provides women's bodies with a daily dose of a progestin. A progestin is a synthetic form of progesterone that gets picked up by progesterone receptors in the hypothalamus of the brain, and so the brain thinks that the body has just ovulated. It thinks that progesterone is being released.
- 5.12** This inhibits the release of gonadotropin-releasing hormone, which inhibits the release of the releasing hormones that lead the ovaries to start stimulating egg follicles. This is how pregnancy gets prevented. It prevents the brain from telling the ovaries to stimulate egg follicles, which means that eggs never develop, eggs never get released, ovulation never happens, and pregnancy never happens. So birth control works by inhibiting ovulation through the daily dose of these synthetic progesterone molecules

called progestins. So progestins mimic progesterone, telling the brain not to kickstart ovulation.

- 5.13** And this is the key ingredient in all forms of hormonal birth control, is the progestin. These synthetic progesterone compounds are what has the effect of inhibiting ovulation. This is why most birth control works. Now, this is important to note because there's lots of different types of hormonal birth control out there. Some of them have a progestin and then also have estrogen. Some of them have progestin-only, and then the different doses of progestin, and there's different types of progestin, and we're going to get into all of that.
- 5.14** But for now, the thing to note is that the thing that really makes hormonal birth control work, the money-making mechanism, is the progestin. This is because it does have this inhibitory effect on the hypothalamus, which prevents it from stimulating the ovaries, which means there's no ovulation.
- 5.15** Now, it's important to note that these synthetic progestins, these synthetic forms of progesterone that are manufactured by drug companies to create hormonal birth control, are not the same as progesterone. They're not the same in a number of different ways. These are really important and are going to end up informing our conversation about the impact of hormonal birth control on women's brains and bodies. We're going to be spending a little bit of time on this slide, talking about some of these differences.
- 5.16** Unlike synthetic forms of estrogen that are found in birth control pills, the synthetic progesterone or progestins that are in hormonal birth control are not synthesized from progesterone. They are not synthesized from the hormone that they're designed to mimic. Most forms of progestin are actually synthesized from testosterone, and chemists manipulate the molecules in ways that allow them to get picked up by progesterone receptors in the brain, which is why they inhibit ovulation. But they don't have perfect binding specificity, and they don't have perfect binding affinity.
- 5.17** And what this means is that they can also bind to things that aren't progesterone receptors, and sometimes they can bind to progesterone receptors and fall off. Another thing about them that makes them different is, because they're not structurally created from progesterone, they also get metabolized in the body very differently than progesterone. We're going to talk through some of these differences now because, like I said, this is something that becomes important.
- 5.18** Progesterone is in our first column here. When we look at the differences between progesterone and progestins on things like mental health, these are two totally different types of compounds. So progesterone, the real deal, the stuff that our body creates, and also the stuff that is in the products that contain micronized progesterone, which is sometimes used in menopausal hormone therapy, this is actually something

that's very positive for the brain. It promotes neuroplasticity, meaning that it promotes the cell's ability to quickly adapt to environmental changes.

- 5.19** It's something that is anti-excitotoxic, meaning that it calms the brain down, and it's actually even used in cases of traumatic brain injury. Because progesterone has this really nice calming effect on the brain, it is able to actually help people who have traumatic brain injury not experience excitotoxicity. The reason that progesterone has all these really lovely effects on the brain is because, when progesterone is broken down in the body, it converts to a compound known as allopregnanolone.
- 5.20** And allopregnanolone is a really potent neurosteroid that acts on GABA receptors in the brain, which are the calming centers of the brain. So GABAergic transmission in the brain is a form of neurotransmission that slows the brain down and calms you down. And so, just to give you some ideas of what this feels like, when your body is releasing GABA, and you're having this GABAergic activity going on in your brain, we get a lot of GABA activity when we meditate or when we do yoga.
- 5.21** Drinking alcohol stimulates GABA receptors, so that feeling that you get when you have a cocktail, that's your GABA receptors being released or that's your GABA receptors being stimulated, and your brain has a calming in response to that type of stimulation. And progesterone, when it's being broken down in the body, releases allopregnanolone, which stimulates these receptors. It has these really positive effects on the brain in that it is very calming. It calms neurotransmission, and it helps promote mood stabilization, and it helps the ability to be able to adapt to stress.
- 5.22** Progestins, which again, most of them are actually synthesized from testosterone, because they're not created from progesterone, when they get broken down in the body, there is no release of allopregnanolone. What research finds is that women who are using hormonal birth control, their levels of allopregnanolone are significantly lower than those of women who are naturally cycling, whose bodies are releasing this hormone that leads to this really potent neurosteroid release that has this very calming effect on the brain.
- 5.23** So, progestins do not convert to allopregnanolone. In fact, some of them have been found to potentially blunt mood and cause anxiety and irritability, instead of producing a calming state, which is what we see with actual progesterone. There's also differences between progesterone and progestin when it comes to neuroprotection. As noted, actual progesterone, the stuff that our body creates, supports neurogenesis, and this is the birth of new neurons. It also helps to support myelination and synaptic plasticity, allowing for greater flexibility and adaptability of brain cells.
- 5.24** And it shows neuroprotective effects. As noted in different models of both brain injury and also models of aging, progesterone has these really lovely effects in terms of preventing some of the damage that can occur both from age and from traumatic brain

injury. Now, progestins don't offer these same benefits. Many progestins not only do they lack neuroprotective actions, but they can also have glucocorticoid side effects.

- 5.25** What this means is that sometimes the progestins—because they've been monkeyed with, they're molecularly not the same as progesterone, and they're not the same as testosterone, which was what most of them are created from—they are a little bit of a Franken hormone that's cobbled together, and we know that it stimulates progesterone receptors, but it also will sometimes stimulate glucocorticoid receptors. That's a cortisol receptor.
- 5.26** This can actually create a state where women's bodies and brains believe that they're in a state of chronic stress, because their cortisol receptors are being stimulated by this imperfect binding capacity of these progestins. Progesterone is anti-inflammatory. We talked about that. It modulates the immune system, dampening the activities of inflammation to allow for the possibility of implantation, because otherwise, the body would attack an implanting embryo.
- 5.27** Progestins don't have those anti-inflammatory effects. In fact, there's been some research that finds that they're pro-inflammatory. Women who are on hormonal birth control that uses progestin oftentimes have higher levels of C-reactive protein, which is a marker of chronic inflammation. We also know that there's differences in their impacts on headaches. So women, naturally releasing progesterone, like real-deal progesterone, tend to calm the brain in ways that tend to be soothing for people who have headaches.
- 5.28** It can also prevent the occurrence of migraine headaches, whereas progestins tend to have the opposite effect. And although progestins are safe for women to take, even if they experience migraine headaches, which isn't always the case for estrogen, they nonetheless do not have these protective effects on headaches. We also see differences in their impacts on the HPA axis, which is our brain's stress axis.
- 5.29** We know that progesterone tends to calm down the stress response, and we tend to have a more adaptable stress response in response to the release of progesterone. This doesn't happen in response to progestin. Progestin, instead, because it has these glucocorticoid receptor-stimulating effects, meaning that it binds to cortisol receptors, making the body believe that there's something stressful going on, it actually can potentiate stress, increasing stress instead of calming stress.
- 5.30** Then lastly, there's differences in cancer risk. We're going to talk a little bit about this when we talk about menopausal hormone therapy, which is a topic that we're going to touch on briefly. But there's a lot of research that indicates that taking estrogen and micronized progesterone—this is the actual progesterone that your body creates, something that's biologically identical—is related to either a decreased risk, or at least no increased risk, of breast cancer, from taking these kinds of treatments.

- 5.31** But whenever they have hormone therapy that includes the use of progestin, which, again, is not biologically identical, this is associated with an increased risk of breast cancer. And this has created a lot of confusion in the world of menopausal hormone therapy, with many women who would otherwise benefit from taking it being afraid of taking it because of what we know about what can happen when you're using progestin instead of progesterone in the forms of treatment. We'll talk about that in just a little bit.
- 5.32** But the takeaway message here is that progestin in a birth control pill is very different than progesterone that a woman's body releases. When we take hormonal birth control and when we're preventing ovulation, we're not only preventing the possibility of pregnancy by making it so that an egg isn't released, we're also preventing women's bodies from producing their own hormones. This means that when you're taking hormonal birth control, which contains a progestin and estradiol, women aren't getting the benefits that we get from actual progesterone.
- 5.33** In addition to the changes that we get from the pills themselves, we also get the changes that are initiated by the fact that women's bodies aren't able to enjoy the benefits that come from the production of their own hormones. Essentially, what hormonal birth control does is it flattens women's hormone levels. They get the same daily hormonal message every day. It makes women hormonally different than what they would be if they were naturally cycling.
- 5.34** Just to do a little compare and contrast, for a cycling woman, what we tend to see is that estrogen is relatively high. It goes through that nice period of peaking in the first 14 days of the cycle. Whereas for a woman on hormonal birth control, estrogen levels are low and stable across the cycle. For a cycling woman, progesterone rises and falls on days 15 to 28 of the cycle. So the last two weeks are characterized by the rise and fall of this hormone that has all these really lovely neuroprotective effects.
- 5.35** And for women on hormonal birth control, there is no progesterone release. Instead, they get a high and stable level of progestin that they're consuming every day that's inhibiting the communication between their brain and their ovaries and inhibiting ovulation. In cycling women, in a woman who has a natural cycle, you get dynamic changes in testosterone that change along with estrogen across the cycle. We tend to see that when estrogen is rising across the cycle, so too does testosterone for women.
- 5.36** It also changes dynamically in a way that's very similar to the way that it changes in men, where women, when they're feeling sexually aroused, their testosterone levels will increase. When women experience a competitive victory, their testosterone levels will increase. Women who are naturally cycling have this nice dynamic testosterone response, the way that men do, and the way that helps to guide their behavior in ways that encourage mating effort at times when mating effort is likely to be beneficial.

- 5.37 Women who are on hormonal birth control, on the other hand, their testosterone levels are very low. Their testosterone levels are kept very low—one, because they're not getting the dynamic changes that occur across the cycle with the rise and fall of estrogen and progesterone. But they're also kept low because, when women are on hormonal birth control, their bodies start to release what's known as sex hormone-binding globulin, and it tends to grab onto and bind to women's testosterone, and it makes them biologically inactive.
- 5.38 And so what we see is that women who are on hormonal birth control have levels of free testosterone that are about 60% lower than that of a naturally cycling woman. And this, again, makes women who are on hormonal birth control hormonally very different than a naturally cycling woman.
- 5.39 And lastly, for women who are naturally cycling, their cortisol levels—and cortisol is a stress hormone. And even though we tend to think about cortisol as being a bogeyman, because we think about cortisol and stress, and stress is bad, cortisol, when it's being released in dynamic bursts in response to stress, it's actually a really good thing. Cortisol isn't what causes stress. It's what helps our bodies cope with and learn from stress. Cortisol, when it's being released in response to stressful situations, is actually a very positive thing, and it allows our bodies to adapt.
- 5.40 What we find for naturally cycling women is that they have relatively low levels of cortisol when they're resting. So their baseline levels of cortisol are relatively low, and then they experience nice big bursts and dynamic bursts of cortisol during times of stress. Women on hormonal birth control, on the other hand, tend to have elevated resting levels of cortisol, meaning they're getting chronic exposure to cortisol signaling. And then when they're experiencing stress, it doesn't rise very much.
- 5.41 It's because their bodies are adjusting or trying to accommodate the fact that they have these really high levels of cortisol already, their body starts releasing a bunch of cortisol-binding globulins, so these are called CBGs, and they make cortisol inactive, and this then prevents women from being able to experience a dynamic change in cortisol in response to stress. So hormonal birth control pills, on the surface, are making a small change, but these small changes cascade and lead to a number of different types of hormonal changes between naturally cycling women and women who are on the pill.
- 5.42 I'm going to be using "the pill" off and on as shorthand for hormonal birth control, but just know that I'm talking about all forms of hormonal birth control. One of the things that we'll be talking about in one of our last lectures is some of the differences in the different types of hormonal birth control in their mechanisms of action and their effects in the body. But for the most part, many of them are the same. If you hear me lapse into saying the pill, just understand that I mean all hormonal birth control.

- 5.43 What we tend to see is that women who are naturally cycling, they have these nice big rises and falls, these bursts of estrogen and then progesterone across the cycle, as their body is orienting themselves to be optimized for attraction and sex, and then optimizing themselves to be ready for implantation and pregnancy. Women on hormonal birth control have the same hormonal message being delivered every day, with the exception of the days that they take a placebo pill if they're on the birth control pill.
- 5.44 That message that they're receiving, that their body is receiving, is one where the synthetic progesterone or progestin is the primary hormone, and that's shown by the dark line on top, and that estrogen levels are relatively low. This, of course, means that women who are on hormonal birth control are going to be different than women who are naturally cycling. Just because women on hormonal birth control are going to be different doesn't necessarily mean that they're going to be worse. There are obviously some benefits to hormonal birth control, or people wouldn't be taking it.
- 5.45 The pill has a lot of benefits, including, most importantly, pregnancy prevention. One of the things that we're going to be talking about in our last lecture, when we talk about the ways that the pill has changed the world, is that it has really opened up doors for women to be able to dream bigger and achieve more than even the most optimistic feminists would have ever dreamed possible 50 years ago.
- 5.46 Women are able to go on and get advanced degrees. They're able to finish college degrees. They're building huge careers. And they're able to do these things because they know that if they put the effort in, they don't have to worry about being suddenly sidelined due to an unexpected pregnancy. That allows women to set bigger goals for themselves, achieve those goals, and I think that we can all agree that this has been a really wonderful invention for women in this way.
- 5.47 Another benefit that the birth control pill offers—and it's not as big of a benefit as the ability to regulate your fertility—it does do a really good job of controlling acne. It turns out that acne is one of the most common reasons that women are given the prescription for the hormonal birth control pill because it binds up that free testosterone, like we talked about with the release of those sex hormone-binding globulins. When testosterone is inactive, it's less likely to cause acne.
- 5.48 The reason that you get the clearing of the skin is that lowering women's testosterone levels is very skin-clearing. Androgens—testosterone is a form of an androgen— they're one of the number-one culprits in the, especially teenage acne, and so it clears up acne. It's also oftentimes used to help manage the symptoms of things like endometriosis. Endometriosis is the overgrowth of the uterine lining, where you can get growth that even goes outside of the uterus.
- 5.49 These women can oftentimes get painful adhesions throughout their abdomen of this endometrial tissue that's growing in places it's not supposed to be growing. When you

take hormonal birth control, because you're keeping estrogen levels really low, that prevents proliferation of uterine cells. This, of course, tends to be very palliative for women who have endometriosis because they no longer experience the pain from the overgrowth of these cells.

- 5.50** It's also used as a go-to treatment for PCOS, which is a hormonal imbalance that's created from an excess of androgens, so too much testosterone, that causes irregular menstrual cycles and can also cause androgenic-related effects, where women oftentimes will have excess body hair. They'll get changes in their body shape. They'll tend to put on weight around their middle. And taking hormonal birth control—because, again, it has these anti-androgenic effects in the body— it also is a treatment that's used for PCOS.
- 5.51** It's also used as a treatment for irregular cycles. So a lot of times, especially when women are first learning to cycle, their bodies are first figuring it out, their cycles are very irregular. Well, you'll have one cycle that's 35 days, and then one cycle that's 20 days, and then one cycle that's 42 days, and then one cycle that's 15 days. That can be really frustrating to women, especially young women, as they're starting to cycle and having to learn to manage all of the things that go on with having a period.
- 5.52** Oftentimes, these women are prescribed hormonal birth control because that, of course, is going to prevent that from happening. Because you're flat-lining a woman's own hormone production, you take a daily dose of a synthetic, your endometrial lining stays stable, so there's no breakthrough bleeding. Then, when you take the placebo pill for those few days that you take it, then there's a little bit of a breakthrough bleed, and then it all starts over again. It helps to mimic the experience of having a regular menstrual cycle.
- 5.53** It's also used to help soothe the symptoms related to premenstrual dysphoric disorder, or PMDD, and other types of issues that stem from hormonal sensitivities that women experience. Women who have PMDD, which, again, is a really intense form of what we tend to think of as PMS, this is extreme negative mood in the luteal phase of the menstrual cycle. It tends to be characterized by extreme depression, extreme anxiety, and many women experience suicidal ideation regularly during the luteal phase of the cycle when they have PMDD.
- 5.54** One of the leading causes, or one of the believed leading causes of PMDD symptomology, is extreme sensitivity to hormonal fluctuations, and the idea that some women are just really sensitive to the rise and fall, the rapid rise and fall, of progesterone in the second half of the cycle, and that that is what is responsible for creating their symptomology. And so the birth control pill, because it does level off and flat-line hormones, and there's not a lot of movement—every day is the same, hormonally—this actually prevents a lot of symptoms from PMDD.

- 5.55** So it can make things better, but it oftentimes comes at a cost. There's all kinds of things that are going to happen when you change a woman's hormonal profile. Just shutting off sex hormones and shutting off ovulation is going to have a whole range of effects because almost every cell in every system of the body is going to be sensitive to women's changing sex hormones.
- 5.56** The reason for this is that almost every system in the body needs to change what it does in response to pregnancy. Our body is wired to listen to our sex hormones. Then, when we shut that off, it means that all of these different systems in our body, even ones that seem to have nothing to do with sex, are going to be influenced by our use of hormonal birth control.
- 5.57** Just to give you an example of something that seems to be completely unrelated to sex, but we know is impacted by hormonal birth control, there's been some research that finds that using hormonal birth control can impair a woman's ability to put on lean muscle mass in response to weight training. They've done studies where they assign women to the exact same workout regimen, and half of the women are using hormonal birth control, and the other half of the women are naturally cycling, and they have them, over a period of three months, doing the exact same workouts.
- 5.58** And what they find is that the women who are using hormonal birth control are less able to build lean muscle mass in response to the training, relative to the natural cyclers, because the hormonal profile that's created by hormonal birth control isn't one that's conducive to muscle growth. Muscle growth is promoted by things like androgens, testosterone, which gets blocked in the case of hormonal birth control. We know that the sex hormone-binding globulins are eating up a lot of that testosterone, binding it up where it's not able to be active in the body.
- 5.59** There's differences in things like DHEA, which is something that's important in muscle synthesis, and that's also related to levels of androgen in the body. We know that that's lower in women, and that also inhibits their muscle growth. We also know that hormonal birth control pill use during the teen years can influence a woman's risk of developing osteoporosis. This is because the process of bone mineral acquisition is something that's—a really important player in that process is progesterone. This is another one of these things that progestin doesn't do.
- 5.60** Research has found that if women are using hormonal birth control, especially in the first couple of years after they start menstruating, their low estrogen levels and their low levels of progesterone are both responsible for not being able to lay down the bone minerals necessary in order to promote healthy bone density in adulthood. This just goes to show that when you change somebody's hormones, you can have this whole range of effects that go from head to toe in areas of the body that seem to have nothing to do with sex.

- 5.61** There's no such thing as making a small change in a biological organism. It's because everything is interconnected, because our bodies were not designed by an engineer, where this part does this and this part does that. Instead, with our bodies, everything does everything. It's a big, messy pot of spaghetti. This means that any time that we're going to be making a change with a medication, you will get the desired effects if you have a good medication, but it's also going to come with a number of hidden costs.
- 5.62** Now we're going to be talking and marching our way through some of the ways that we know that hormonal birth control use can impact the way that women think, feel, and experience the world. One of the things that we talked about in an earlier lecture is that estrogen, as it changes across the cycle, tends to correspond with changes in sexual desire.
- 5.63** We know that, at times in the cycle when estrogen is rising and is high, this is a time when women tend to have higher sexual desire. They tend to have more sex, and they tend to experience more sexual pleasure than what is experienced outside this period of time in our cycle. When estrogen levels are low, and especially when progesterone levels are high, sexual desire tends to decrease, as does sexual behavior.
- 5.64** We also know from research that we talked about that when estrogen is rising across the cycle, that this is related to an increased preference for cues to testosterone in men. We learned that women have a heightened preference for masculinized male faces when the estrogen levels are high, and then when estrogen levels are low, that this heightened preference is dampened. They no longer exhibit this heightened preference for masculinity cues. We also know that women are found more attractive by both men and women when they're at the point in the cycle when estrogen is rising and high.
- 5.65** So estrogen is related to a lot of physical and psychological changes that impact not only how desirable women are seen to potential partners, but how desirable women feel themselves toward other types of men, and then also how much sexual desire they experience. And so it's probably not surprising to you, given the role that estrogen plays in terms of promoting sexual desire and sexual function in women, that hormonal birth control use can decrease sexual desire and decrease sexual function.
- 5.66** One of the things that the pill does, and one of the most important things that the pill does, is that it inhibits ovulation. But ovulation is the primary way by which our body releases most of its estrogen. So estrogen levels are being really dampened in response to hormonal birth control use. And so what this is going to mean is that, of course, it's going to be leading to changes in women's sexual desire and their sexual function.
- 5.67** And not terribly surprisingly, there's been a number of research studies that have now found that hormonal birth control use does appear to lower sexual desire. It tends to lead to sexual arousal difficulties, and it can lead to changes in sexual pleasure that are not necessarily something that are enjoyed by the women who use them. In some

ways, it's the great irony of hormonal birth control use: most women go on it so that they can have a better sex life, and then it kills their sex life. This great facilitator of sexual behavior, not necessarily promoting sex.

- 5.68** What we're looking at here on the left-hand side is, we're looking at a study that was done looking at female sexual functioning based on whether or not they're taking hormonal birth control use or a placebo. This was a really nice experimental study where they actually had women blind to what they were taking. Everybody's having protected sex using a condom, but some of them are on hormonal birth control, and some of them are on a placebo.
- 5.69** What they found when comparing the sexual functioning and sexual desire of the women on the pill compared to the placebo is that the women who are taking hormonal birth control perform significantly worse in all of their different measures. As you can see here, women who are using hormonal birth control had lower levels of sexual desire. They had lower levels of sexual arousal. There were really no significant differences in orgasm, so at least that was left intact.
- 5.70** They reported less sexual pleasure. There was no differences in their concern about their sexual behavior. They had lower sexual responsiveness, although that wasn't a significant difference. There was trending but non-significant differences also in their self-image, with women who are using the hormonal birth control feeling a little bit worse about themselves than the women who are naturally cycling.
- 5.71** When we consider some of the things that we learned about in the lecture where we talked about the role that estrogen plays, in terms of how sexy women feel, it's not surprising that we might get that trending slope, even though, as noted, it was not statistically significant. But the takeaway from this figure that we see on the left-hand side is that women who are using hormonal birth control do seem to have less sexual desire, less sexual arousal, and experience less sexual pleasure than do women who are naturally cycling, taking a placebo.
- 5.72** They also looked at, when you average these things across different domains, you can see that there's differences here with women who are using hormonal birth control—this is the figure on the right—faring less well in terms of sexual function relative to women who are naturally cycling.
- 5.73** And so on the whole, what this research tends to suggest—and there've been several studies that have been done looking at this question. I liked this one in particular to highlight because it was experimental, where it was assigning people to receive either the hormonal birth control pill or a placebo, not knowing what they're taking, and then looking at its impact on sexual function. But there've been several studies since the time of this one that have found similar effects, testing it either cross-sectionally, so looking across women, and some other experimental work finding similar patterns as well.

- 5.74** This is a slightly different study. This one was looking at hormonal birth control use and physiological arousal to, in this case, pornography. In this study, the researchers were interested in trying to test whether there are differences in levels of vaginal lubrication, which is a really nice measure of physiological sexual arousal. They were interested in whether or not there are differences in physiological sexual arousal between women using hormonal birth control and then women who are naturally cycling.
- 5.75** They looked at hormonal birth control users in two different buckets. Some of the women who were hormonal birth control users were on what is known as an anti-androgenic form of hormonal birth control—this is a form of hormonal birth control where the progestins have anti-androgenic effects. What this means is that they actually block testosterone receptors in women's cells. Then there's the androgenic forms of hormonal birth control. In addition to stimulating the progesterone receptors, which is what they need to do to inhibit ovulation, they also stimulate androgen receptors, meaning that they can have testosterone-like effects in the body.
- 5.76** And what the researchers found when they had women—they measured their levels of vaginal lubrication prior to watching an erotic film, and then they measured it after watching an erotic film—and what they found was, the first thing that they found is that the differences in vaginal lubrication were significantly greatest among the naturally cycling women. You can see here, they are the ones who are marked by the circles. And the women, you can see that they had the highest vaginal response to the erotic video.
- 5.77** You can see that it's dampened in women, regardless of the type of hormonal birth control that they're on. Women who are on androgenic or anti-androgenic hormonal birth control, both types experience a dampened vaginal lubrication response, in response to the erotic video, showing lower levels of physiological arousal from using hormonal birth control, regardless of whether or not that form of hormonal birth control contains an androgen or blocks androgen receptors or not.
- 5.78** Why does this happen? Well, one of the reasons is simply, suppressing estrogen is suppressing one of the most potent forms of sexual desire that women's bodies create. Suppressing estrogen leads to lower sexual desire. It also leads to lower self-perceived sexiness. We know from research that women feel sexier when they're in the estrogenic phase of the menstrual cycle. They feel more attractive, they dress more attractive, they show more skin, they put more effort into their appearance.
- 5.79** Dampening this is something that's going to dampen the way that women feel about sex, but also the way they feel about themselves. This is something that's actually really important, because there's a lot of research on female sexuality that indicates that one of the strongest predictors of women's sexual desire is how they feel about themselves and how sexy they feel. It tends to be inwardly focused. When women don't feel sexy, oftentimes they don't feel like they want to have sex with their partner, regardless of how they feel about their partner.

- 5.80** We also know that women who are using hormonal birth control have higher levels of sex hormone-binding globulin. That's that SHBG. This is something that women's livers release in response to all these synthetic hormones. The byproduct of the release of sex hormone-binding globulin is that it binds up with women's testosterone, making a portion of it inactive.
- 5.81** And so what we tend to find is, when we look at levels of what's known as free testosterone— and it's called free because it's unbound and it can be biologically active and stimulate cells, so that way they run the sex program—women who are using hormonal birth control, their levels of free testosterone are about 60 to sometimes as high as 80% lower than that of a naturally cycling woman. And testosterone, in addition to promoting mating effort and sexual desire in men, also contributes to mating effort and sexual desire in women.
- 5.82** Lastly, progestins, even though they're not biologically identical to progesterone, they do still stimulate progesterone receptors. One thing that we know about progesterone from our lecture on the luteal phase of the cycle is that it tends to dampen libido. It tends to turn down the volume on sexual desire because having a lot of sex during a time when pregnancy is possible isn't necessarily the best use of physiological resources and potentially can lead to an inflammatory event near the site of implantation.
- 5.83** The hormonal profile of a woman who's using hormonal birth control isn't one that is very facultative of sexual desire. Instead, it's one that's pretty inhibitory to sexual desire, which is why we tend to see these types of patterns. Now, it's important to note that when we're talking about sexual desire and we're talking about hormonal birth control use, the effects are relatively idiosyncratic, meaning that there's a lot of individual differences in these types of patterns.
- 5.84** This is important to note because this is something that a person should be knowing about. They should know to be looking out for the possibility of sexual dysfunction in the context of hormonal birth control, but it's not something that's always going to happen. And one recent review of all of the published literature on the impact of hormonal birth control use on sexual function and libido found that there is a lot of heterogeneity.
- 5.85** Some women have problems with libido from using hormonal birth control; other women have no changes from using hormonal birth control; and some women actually report improved libido. Those improvements are oftentimes the result of the fact that women feel more able to relax and enjoy sex because they're not worried about getting pregnant.
- 5.86** We can also learn a lot about the impact of the pill and the idiosyncratic nature of the impact of the pill on libido by looking at differences in women's return, like changes in women's sexual desire once they discontinue the birth control pill. What this research

finds is that, for many women, after they discontinue using hormonal birth control, they experience an increase in libido that corresponds to the return of their natural menstrual cycle. This makes sense.

- 5.87** As soon as you start ovulating again, and your body starts giving you that nice, beautiful boost in estrogen and your testosterone is returning, this is going to correspond to an increase in sexual desire. Indeed, this is what we find in a large group of women. About half of women, in this particular study, it was 44.8% of women, after they discontinue hormonal birth control, experience an increase in their sexual desire and their libido and also sexual behavior that corresponds to the return of their cycle.
- 5.88** But then there's also another half of women who don't experience those changes, or experience a decrease in sexual desire after discontinuing hormonal birth control. This, again, just goes to show that there's a really idiosyncratic and complex relationship that's going to be created in the body when you change hormones. Because for some women, when you go on hormonal birth control, it's going to cause huge changes in what your pre-existing hormone levels are.
- 5.89** But for another woman, those changes might not be as severe. They might not be as pronounced because their starting or their beginning hormonal profiles were really different from one another. The way that we metabolize these things is different. There's many different formulations of hormonal birth control. Anytime we're talking about any of these effects, there is always going to be a great deal of heterogeneity. When we're talking about changes in sexual desire, this is no exception to this rule.
- 5.90** When we pick up, we're going to be talking about the impact of the pill on sexual attraction. Because we do know that using hormonal birth control dampens our release of estrogen. We know from several decades of research now that, when estrogen is rising in the cycle, that it tends to be leading us to have a heightened preference for cues related to masculinity and testosterone levels. So what happens to women's partner preferences when the volume is turned down on estrogen, and that's no longer being released in high volumes across the cycle? We'll be talking about that next time.
- 5.91** This is in relation to one of your last lectures, where you said that progesterone lowers the immune response in the body. It dampens it.
- 5.92** Yeah, it dampens it.
- 5.93** So does progestin do the same thing, or how does it impact the immune response of a woman's body?
- 5.94** Yeah, so that is not very well known, but we do know that, and I can tell you about some research that we had going on in my lab, and then I can also tell you about the few things that we do know. One thing that we know is that women who are on hormonal birth control do tend to have elevated levels of, or these elevated markers of chronic inflammatory activity in the body. And so they tend to have higher levels of C-

reactive protein, relative to women who are naturally cycling, and that's a marker of chronic inflammatory activity in the body.

- 5.95** The idea is that because they're not experiencing these immunomodulatory effects of progesterone, they're not having that brake tapped monthly, that this may lead them to just be in a more exaggerated inflammatory state on average across the cycle, because they're never going through that phase in the cycle where they're going into an anti-inflammatory state. That's one thing that we know.
- 5.96** Another thing that we know—and this is from some research that came out of my research lab— is that, when you stress women out—and stress is a context that elicits an inflammatory response in the body, because the body is preparing for the possibility of injury, and so it'll start to release inflammation just in case something happens and they get hurt—what we find when we look at the release of inflammatory cytokines—cytokines are the signaling proteins that are released by the immune system that cause inflammation—when we look at the different types of cytokines that get released in response to stress, and we're looking at women who are using hormonal birth control and then women who are naturally cycling, it's actually different. They have different inflammatory profiles in response to stress.
- 5.97** And the type of inflammatory profile that we tend to see in women who are naturally cycling is more of an anti-inflammatory profile. There's some of cytokines that have both pro- and anti-inflammatory effects. We tend to see those in greater volumes in women who are naturally cycling. Then, in women who are using hormonal birth control, we tend to see a more unilaterally pro-inflammatory type of a response. This also suggests that there might be some differences there.
- 5.98** There is some evidence that women—certain types of autoimmune disease that are related to chronic inflammation, you do tend to see a greater risk in women who are hormonal birth control users compared to women who naturally cycle.
- 5.99** So, for a woman with endometriosis that finds herself years into hormone replacement therapy, like progesterone, to stop the period, or on birth control to try to manage those symptoms, but then is learning all this, and is like, "Wait, I want my natural cycle back," how does a woman do that in a healthy way so that it doesn't freak out the system, so that she can still work her way back to a natural cycle in a safe way?
- 5.100** Right. So, I just want to make sure I understand your question. Do you want to know about how to safely transition off of hormonal birth control, or what you need to do specifically in the context of endometriosis or both?
- 5.101** OK, both. More of like, if somebody has endometriosis and the result from the doctor is, or the solution is, "We're going to put you on progesterone, no more period, or let's try to maintain it with hormone birth control," but they want to stop that and work their way back, how do you even begin that process?

- 5.102** The good news is that you can discontinue hormone use, including hormonal birth control and hormone therapy use, safely without having to worry that your body is going to absolutely lose its mind, which is really good. I think that there's some fearmongering that goes on in the internet, in particular, where they make you believe that you need to buy this pile of supplements to ease, and it really doesn't need to be like that. If you want to discontinue, you can safely discontinue and not really worry that anything terrible is going to happen.
- 5.103** That being said, it is a little bit of a transition. When you saw the little picture of what a natural cycle looks like, and you have these rise and fall of these two hormones, and then compare that to what happens when you have this hormonal flat-lining, the cells in your body are going to have to learn how to adjust to rising and falling levels of hormones. That can lead to a couple of rocky months. A lot of women will experience one or two, sometimes three months, where they don't quite feel great. But it's not so terrible that you need to call your therapist to be on speed dial.
- 5.104** Everything is generally OK, but it can be a period where you'll have a few more breakouts. I've talked to some women who had their sexual desire go so crazy that they're just totally obsessed by sex. They're like, "This is all I can think about. All I can think about is sex. I'm like, what am I going to do?" You can get these kinds of things. But I would try to work with a doctor and especially somebody in restorative reproductive medicine. There's these groups of physicians, and they work essentially to try to fix women's health issues, instead of masking women's health issues.
- 5.105** Things like PCOS and endometriosis, oftentimes doctors will give you hormonal birth control or these hormones to try to prevent the symptoms. But then, if you go off of them because you want to have a baby, then all of a sudden you can't have a baby, because you have this issue you never dealt with, because you were given the pill. If you go see one of these doctors who do restorative reproductive medicine, they try to actually get and solve the problem.
- 5.106** And so, they can work with you to try to manage, if you have PCOS, manage your PCOS. In the case of endometriosis, manage your endometriosis. Sometimes that can mean surgical excision of the endometrioma. But oftentimes, these can end up being really effective when you either have a surgery or you put in place some of their different protocols for trying to restore your fertility and restore your health that way, where you no longer need to rely on medication.
- 5.107** A quick question around ovulation. If you're on hormonal birth control, are you saying that you're actually not ovulating?
- 5.108** Mm-hmm.
- 5.109** OK. Then does that mean you're actually not getting a period?

- 5.110** Yeah. It's a really interesting question, because an actual period is the shedding of your endometrial layer that got built up during this process of ovulation. When you're ovulating, your body is stimulating egg follicles, and then as that happens, estrogen is being released. One of the things that estrogen does is it causes proliferation of the endometrial lining, which is why they try to keep estrogen low in the case of endometriosis, is they don't want that over-proliferation.
- 5.111** You get that proliferation, and then when your body is releasing progesterone, that leads to further proliferation of cells and differentiation of those cells to prepare the endometrial lining for the possibility of pregnancy. Then, when you have a period, all of that sheds and leaves the body, and that's a true period. When you're on the pill, and your hormone levels are really low, it's like, there's going to be a little bit of cell proliferation, but not much, because the primary driver of cell proliferation in the endometrial area is estrogen, and it keeps estrogen levels really low, so you don't get that proliferation of the lining.
- 5.112** Then you don't get the release of progesterone, and so you don't get the differentiation and maturation of those cells creating an actual endometrial layer. When you get a bleed on the period, it's not actually a period because you're not really shedding an endometrial lining. Instead, what you're getting is bleeding that's created by a hormone crash that's very similar to the one that happens at the end of an actual cycle.
- 5.113** Because your period starts—you know, when we look at those graphs and you see in the second half of the cycle and you see the rise and fall of progesterone, when sex hormone levels start to get really low, that causes a destabilization of the endometrial layer and causes you to bleed it out. In the case of hormonal birth control, you're getting that same daily hormone message, and then you take the placebo pill for whatever it is, 5-7 days.
- 5.114** When you take that, all of a sudden your hormone levels crash, and that causes, almost as a reflex, the shedding of any material that's there. But there's really not an endometrial lining to shed, which is why, when women are on the pill, they have a much lighter period, because there's a little bit of bleeding, because when hormone levels crash, that causes those cells to shed, and there's going to be some blood, but there's not really a true endometrium to shed. It's not a true period, but it's a hormone withdrawal bleed that gets initiated by the hormone crash that you get when you're on the placebo.
- 5.115** I appreciate the clarification because I think a lot of times we feel misled as women, saying, "Oh, let's put you on the pill or put your daughter on the pill because it'll regulate the period."
- 5.116** Regulate the period.
- 5.117** It's not regulating a period. Just kind of like in science, where it's like, the brain has a biochemical imbalance. Some of those things have been debunked. The same with this

idea of, it's not a period. I think women might think differently if you're telling them, "You're completely blocking your ovulation, which stops the period."

- 5.118** Yeah, you're stopping ovulation. You're stopping your own hormone production. There's so much more to it than the idea of, "Oh, you're just not going to get pregnant and don't worry about it." I think that that is exactly what we're told, and, "Oh, it's going to regulate your period," or, "Oh, it's going to fix your endometriosis." It doesn't fix your endometriosis. It doesn't regulate your period. There's still whatever underlying issue existed that was leading you to have irregular periods or leading you to have endometriosis or leading you to have PCOS are still going to be there.
- 5.119** Unfortunately, what happens is, a lot of women don't learn this until they want to get pregnant. Then they go off of it, and then they realize that, "Oh, I have this reproductive health issue that I need to take care of, and it hasn't been solved." Then they have to go—there's a lot of wasted time that oftentimes can lead to women having to look for alternative fertility treatments. It's sad.

Lecture 6: Birth Control Effects

At a glance — Effects of hormonal birth control on sexuality and desire.

- 6.1 When we left off last time, we were talking about the ways that hormonal birth control use can impact sexual desire and sexual function. Of course, we know that, as women's hormones change across the cycle and estrogen levels rise and fall, that this, for a naturally cycling woman, also corresponds to a rise and fall in sexual desire, which corresponds to the rise and fall of conception risk, with women experiencing highest sexual desire at times in the cycle when sex can lead to pregnancy.
- 6.2 Now, these same changes can also potentially lead to changes in partner attraction because we know from decades of research now that, not only does the rise and fall of estrogen correspond with an increase in sexual desire, but it's also associated with an increased desire and heightened attention to cues related to genetic compatibility and high genetic quality in partners. This has led researchers to propose the idea that hormonal birth control use may impact partner attraction and even mate choice.
- 6.3 In one set of studies, researchers had women come into the research lab, and they had them do a digital face-making task. And what they did is, they showed women photographs of a face, and then they were able to use a computerized slider by sliding a mouse left and right that would make the face more masculinized or more feminized, depending on which way they moved the mouse. Move the mouse to the left, and the face becomes more masculine. Move the mouse to the right, and the face becomes more feminine.
- 6.4 They showed the women how the software worked, and they first showed them a photograph of a male face, and they said, "Create, using the mouse slider, the most attractive and desirable male face." The women did that, and they chose a face. Then they had them do the same thing with the female face. They said, "Now we would like you to use this slider to create the ideal and most attractive and desirable female face." And so the women did this. Then they had the women come back three months later.
- 6.5 But the trick of the study was that half of the women, after study one, started using hormonal birth control, and the other half of the women stayed naturally cycling. At the time of the first study, they were all naturally cycling, so no women were using hormonal birth control. They participated in the first study session, and then afterward, half of the women went on the pill. The other half stayed off of it. Three months later, they came back to the lab, and they repeated the same activity.
- 6.6 They told the women, "Create the most attractive and desirable male face; create the most attractive and desirable female face." You can see the results over my shoulder. We'll start by talking about the women in the control group. Those are the blue bars. Here, our control group is the women who did not have a change in hormonal status

between time one and time two. They were naturally cycling at time one; they were naturally cycling at time two.

- 6.7** When we look at the change in masculinity preference in the faces that they created, you can see that, when they were creating an opposite-sex face, there was no difference in how masculine they made the ideal face that they found most attractive. There also wasn't a difference in the masculinity of the female face that they created. Even though it looks like it became a little bit more masculine, it wasn't statistically significant.
- 6.8** Now, let's look at the experimental group. This is the group of women who, at time one, were naturally cycling, and then, at time two, came into the lab after having been on hormonal birth control for three months. What you can see is that, when the women were creating the most attractive and desirable male face, the male face they created was significantly less masculine than was the face that they created at time one.
- 6.9** Just like it was with the naturally cycling women, there were no differences when they were creating a same-sex face. This suggests that what women find most attractive in male faces may become less masculinized when women are using hormonal birth control. This is actually very consistent with what we know about the impact of estrogen on women's preferences for facial masculinity.
- 6.10** There's been a number of studies that find that, when estrogen is rising across the cycle, women's attunement to cues related to testosterone levels—so things like masculinized faces, masculinized voices, like deep male voices, and masculinized behaviors, so socially dominant behaviors—tend to be particularly attractive to women at high fertility in the cycle, when estrogen is high and rising, and that women tend to be less interested in these things when estrogen levels are low.
- 6.11** This research was one of the first studies to find that women may exhibit a dampening of their preference for masculinized male faces in response to using hormonal birth control. In a follow-up study, researchers were interested in whether using hormonal birth control, in addition to changing the preference for facial masculinity, might it actually lead women to choose partners, men, with less masculinized faces than what is chosen by women who are naturally cycling?
- 6.12** To test this, the researchers had a large sample of men come into the research laboratory. They took their pictures, and they asked the men to report about whether, in their current romantic relationship, was it begun at a time when the woman was naturally cycling, or was she using hormonal birth control? Then they divided the men's pictures into two piles: a pile of faces of men who were chosen by women who were using hormonal birth control. They created a pile of pictures that was pictures of men who were chosen by women when they were naturally cycling.

- 6.13** They had the pictures evaluated for subjective masculinity, so having people evaluate, looking at the photographs and saying, "How masculine is this face?" But they also used computer software that's able to measure facial dimensions that are related to levels of testosterone. Things like the ratio of facial width to height is one of the indicators of testosterone levels. They also are able to look at things like how deep-set the eyes are, using the computer software. So they were able to get objective computerized ratings of facial masculinity, as well as subjective facial masculinity scores.
- 6.14** What they found was that the faces of the men who were chosen by women who were using hormonal birth control were significantly less masculine than were the faces of those men who were chosen by women who were naturally cycling, which, again, is very consistent with this idea that hormonal birth control use may dampen women's interests in attunement to testosterone-related cues to men, potentially leading them to favor a less masculinized male face.
- 6.15** Now, it's important to note, when we're looking up here at this picture—this is showing facial composites of faces, average male faces, chosen by women who are using hormonal birth control and women who are naturally cycling— that these aren't huge differences. This isn't a thing where you would look at a person's partner that they chose when they're on hormonal birth control compared to off of it and say, "Oh, my gosh, how in the world did you land here? This is so totally different than what we would expect if you were naturally cycling."
- 6.16** Instead, these are very subtle differences. But what's really interesting about this is that estrogen makes us attuned to those fine-tuned differences. These are differences that are oftentimes not even noticed by women using hormonal birth control.
- 6.17** Some other work has found, for example, that women who are using hormonal birth control, not only are they not as sensitive to facial cues related to testosterone levels, when women are using hormonal birth control, their ability to discriminate between differences in levels of testosterone is lower than what it is true for naturally cycling women.
- 6.18** Naturally cycling women, particularly when estrogen is high in the cycle, are better able to discriminate between fine-tuned differences in scent-based cues to testosterone levels in ways that women who are on hormonal birth control are not. It dampens their sensitivity to fine-tuned differences in testosterone levels in men, potentially leading them to somewhat different partner choices than what we would find in the case of a naturally cycling woman.
- 6.19** In addition to changing the sensitivity that women have to cues related to testosterone levels when they're using hormonal birth control, researchers have hypothesized that, given that it seems to be decreasing women's sensitivity and interest in qualities related to what we tend to think of as sexiness in men, cues to testosterone levels, scent, etc., that maybe women who are using hormonal birth control, rather than

having their mate choice influenced by these kinds of cues, that instead they may be heavy-loading their mate preferences on other types of qualities.

- 6.20 To introduce you to this idea, I just want to unpack this difference between what's known as direct and indirect mate benefits in the world of evolutionary biology. When an evolutionary biologist is looking at mate preferences of individual organisms within a species, including humans, we can divide up the types of qualities that people are looking for, that an organism is looking for, based on the extent to which it provides the chooser with either direct or indirect fitness benefits.
- 6.21 A direct fitness benefit, when you're choosing a mate, is something that benefits you directly, which is why we call it a direct benefit. So for example, time spent with you, time listening to your problems, buying you dinner, investing resources in you, caring for you, loving you—these are all direct benefits. Resource access, provisioning, protection—these are direct benefits. These are things that you benefit from directly through your choice of mate.
- 6.22 Indirect benefits are those genetic benefits that we're getting when we're choosing a mate with whom we're going to be having children. Ultimately, what we're talking about is good genes. And so whenever somebody is choosing a mate, whether it is something as complex as a human being or something a little bit less complex like a cardinal or a canary, they're tending to manage a preference for both types of qualities.
- 6.23 They're balancing a need for both direct benefits, that is getting investment, but then also indirect benefits, getting qualities related to good genetic quality, getting a partner who has good immune genes, for example, so you'll have healthy offspring. And so, with hormonal birth control, it seems to be decreasing women's attunement to some of these cues related to indirect benefits, because one of the things that we know about testosterone is that it tends to be related to having good immune genes. It's one of these good genetic markers, qualities.
- 6.24 And so researchers have hypothesized that women who are using hormonal birth control, given that the volume seems to be turned down a little bit on their preference for cues related to indirect benefits, these genetic benefits, that when women are choosing partners, that they may be focusing more on these qualities related to direct benefits; that is, that they may be paying more attention to things like their partner's provisioning ability than they are to things like their partner's sexual attractiveness.
- 6.25 And so, in one study, researchers had women evaluating how satisfied they are within their relationship. Some of the women chose their partner when they were on hormonal birth control, and some of the women had chosen their partner when they were naturally cycling. This was a large sample of women. It was over a thousand women.
- 6.26 They asked them about their satisfaction with things like their partner's provisioning ability, their partner's intelligence level, their sexual arousal, sexual adventurousness,

sexual proceptivity, which just refers to the extent to which, if your partner initiates sex, are you wanting to say yes? Sexual attraction, support, how attractive they are. They asked some questions about rejection of their partner, orgasm with their partner, and so on.

- 6.27 What they found was that there were some really interesting differences that were consistent with the idea that when women choose their partners when they're using hormonal birth control, it may turn the volume down on their attunement to cues related to indirect benefits. Instead, women may be putting all of their eggs in the basket of direct benefits, prioritizing cues related to things like provisioning ability, while deprioritizing cues related to things like sexual arousal and sexual attraction. And that's exactly what they found.
- 6.28 So for example, when they looked at partners' relationship satisfaction or looked at women's relationship satisfaction, whether they chose their partner when they were on or off the pill, they found that women who chose their partners when they were using hormonal birth control, that they reported feeling more satisfied than women who chose their partners when they were naturally cycling; they felt more satisfied with their partner's provisioning ability and also their partner's intelligence levels.
- 6.29 Where the really interesting results came in is when we look at the differences, the things that women who chose their partners when they were naturally cycling, the things that they felt most satisfied with in their relationship, tended to be things that were related to sexual arousal, sexual attraction, sexual adventurousness, sexual, sexual, sexual. All of this really makes sense when we consider what birth control pills do.
- 6.30 When you're dampening estrogen release and keeping estrogen levels really low, and estrogen, along with testosterone, help to promote things related to mate choice and attraction, remember, when estrogen is being released in the cycle, it is when your body is optimized for attracting mates and having sex with mates. It's turning the volume down on that, and in so doing, is also turning the volume down on sexual desire and turning the volume down on the extent to which women are prioritizing sexual attraction in the context of partner choice.
- 6.31 What this research suggests is that, if you choose your partner when you're on the pill, it may lead you to prioritize different types of qualities in your mate choice than what you prioritize if you're choosing your partner when you're naturally cycling. And there's other evidence that suggests this very same thing. Research finds, and this is some more recent research that was actually done with a former graduate student of mine, using data that was taken from the natural cycles tracking app.
- 6.32 A cycle-tracking app is one of these phone applications that you can use to keep track of your cycle. And on it, women will also log their sexual behavior. And we were able to collect data from the cycle-tracking app and actually look at attraction and sexual

behavior within relationships among women who are all naturally cycling at the time of the study, but some of whom had chosen their partner when they were using hormonal birth control, and some of whom had chosen their partner when they were naturally cycling.

- 6.33 We sent them out a follow-up survey after they joined the app, asking what type of birth control they were on when they chose their current partner. Then we were able to divide our data into two buckets, looking at the results of women who chose their partners when they were on hormonal birth control, and then looking at data from women who chose their partners when they were naturally cycling.
- 6.34 What we found is that there were striking differences between the two groups of women; and that even though all of these women were currently, at the time of the study and during the time which we were collecting the data, even though all of them were naturally cycling at the time, those who chose their partners when they were on hormonal birth control reported lower attraction to their partners than those who chose their partners when they were naturally cycling.
- 6.35 And again, this just suggests this idea that women, when they're naturally cycling, estrogen is sort of leading them toward things that make them feel sexually aroused and sexually attracted. When they don't get that monthly boost of estrogen, this may be de-emphasizing these qualities in their choice of partners. They report lower attraction to their partners. They also logged less sexual activity across the cycle.
- 6.36 When we looked at the frequency with which women were logging sex across the cycle, the women who chose their partners when they were on hormonal birth control were logging less sex than were women who chose their partners when they were naturally cycling. This was true despite the fact that the whole sample was currently naturally cycling.
- 6.37 Even though everybody in the sample was currently naturally cycling, those who chose their partners when they were on hormonal birth control were logging less sexual activity, suggesting that there might be less sexual attraction within those relationships than what might be observed in relationships where women chose their partners when they were naturally cycling.
- 6.38 This, of course, this idea that something like a medication might influence who we're attracted to and who we choose as a mate is a very provocative idea. It's provocative for a number of reasons because nobody would imagine taking a medication and reading on the warning label "may cause fatigue and also mate choice differences." That's not something that most of us see on medication that we take.
- 6.39 But it's also provocative because so many women are on hormonal birth control during the times in their lives when they're choosing their long-term mates. A lot of college women are on hormonal birth control. They choose a husband, and then they're married to them, and then at some point, they might make the decision to go off of

hormonal birth control. This raises the question of what happens if you choose your partner when you're on the pill or on hormonal birth control, and then you go off of it?

- 6.40 If you're not attracted to the same types of men when you're on it versus off of it, does that mean that your relationship is going to get totally turned upside down? And the answer to this seems to be very complex. And so there's no one answer about what happens if you choose your partner on the pill and then you go off of it.
- 6.41 In one study that was looking at changes in attraction that women experience when they choose their partner on the pill and then go off of it, in one study, research found that about 36% of women reported an increase in attraction to their partner after going off the pill. In a lot of ways, this is consistent with the idea of cycling resuming. It's like, here you are having a hormonal flat line. You're never experiencing the sexual desire increases in response to estrogen rising in the cycle.
- 6.42 And then, all of a sudden, you start cycling again, and it comes back, and you're feeling sexually awake again for the first time in a long time. And then you look at your partner and you're like, "Yes, this is great." And 36% of women experience that feeling, where they feel increased attraction to their partner once they discontinued the pill.
- 6.43 21% of women experience a decrease in attraction. This is where the problems can come in. This is where you get a mismatch between the hormonal state that a woman was in when she chose her partner and then the hormonal state that she's on once she discontinues the pill, and all of a sudden, she's realizing that she's not as attracted to her partner as she used to be when she was on hormonal birth control. So this is something that can happen.
- 6.44 And then about 42% of women reported no change in attraction. So they didn't notice any noticeable differences in how they feel about their partner sexually when they're on compared to off the pill. And so this suggests that if you choose your partner on the pill, it does come with a risk that you're going to experience a change in attraction. And some of the time, the change in attraction is going to be negative.
- 6.45 But this, of course, doesn't guarantee that this is what's going to happen. And many times this isn't what happens, but it is still a potential risk. And what creates this variability? Why is it that some women report feeling more attracted to their partners when they go off the pill, and other women report feeling less attracted to their partners when they go off the pill, and some women notice nothing at all? There's not any consensus on what specifically might be responsible for this heterogeneity in responses, but there are some clues.
- 6.46 One set of clues comes from a really neat and a really nicely done longitudinal study of newlywed couples. In this study, the researchers studied samples of newlyweds, and they followed them over time, measuring their relationship satisfaction, so their marital satisfaction, their sexual satisfaction, their attraction to their partners, and so on, and they measured it over time.

- 6.47** As often happens with newlywed couples who've just become married couples over time, is that, in a lot of these couples, the women started off on hormonal birth control, and then they discontinued the hormonal birth control, oftentimes because they were going to start a family. They were able to look at longitudinal changes in attraction and marital satisfaction in women, depending on whether they chose their partner on or off of hormonal birth control.
- 6.48** What these researchers found was really fascinating. What they found was that, when women discontinued hormonal birth control—so this is women who chose their partners when they were on the pill—when they discontinued hormonal birth control, on average, it led to changes in their attraction to their partners and their relationship satisfaction with their partners. But whether this change was positive or negative depended on how attractive their partner was.
- 6.49** What the researchers found is that women who were partnered to attractive men, when they went off the pill, they reported more sexual attraction to their partner and reported more sexual activity within their relationship. They also reported higher marital satisfaction in their relationships after discontinuing hormonal birth control. For women who were married to unattractive men, the opposite thing happened, and women reported less sexual attraction, less relationship satisfaction, and less marital satisfaction than they did prior to discontinuing hormonal birth control.
- 6.50** And so some of this, again—estrogen, one of the things that it does is it really makes us attuned to quality differences between men and the types of things that make our brain release dopamine, and we get that reward response and things that don't. Estrogen really primes that. It primes our brain to be sensitive to differences between men, leading us toward those that lead to that dopamine release and away from those who don't lead to that dopamine release.
- 6.51** And then when we mask that, and we suppress that with hormonal birth control, and we don't notice these things, and then we choose a partner, if we go off it, it's like the blinders come off. And if you're partnered with somebody that your brain finds attractive, it's like, "Oh, my gosh. Wow. How did I not know that you were here this whole time?" But if you're partnered with somebody that doesn't pluck those same strings, it can have the opposite effect. This can obviously be really devastating for people in relationships.
- 6.52** Some of this, there might be partner-related effects, just depending on who you chose. Is this someone that is genetically compatible and somebody who has some of the qualities that your brain finds sexy? Some of it's luck, where people just so happen to choose somebody that both sets of hormones tend to get along with and like.
- 6.53** Then there's also this idea of the medication and hormone match. One of the things that we're going to learn about in one of our last lectures is how much heterogeneity exists within the area of hormonal birth control. There's close to 100 different

formulations out there on the market, and each one does a little bit different things, pushing different buttons in the body and leading to different types of psychological and behavioral changes.

- 6.54 You take that amount of heterogeneity, and then you add that into the fact that women are so different from one another, that no two women are going to respond the same way to the same hormonal birth control. If I gave everyone in this room the exact same dose of the exact same medication, we would have as many different effects of that medication as there are people in the room, because everybody's body responds to things differently.
- 6.55 We metabolize drugs differently. We have different numbers of hormone receptors on our cells. We have different abilities of our bodies to quickly adjust to hormonal changes. We all differ. For some women, when they take hormonal birth control, it's going to create changes that are more profound than it will in another woman. It's in those women who experience these profound changes that we're going to get the strongest mismatch between how you feel and experience the world when you're on hormonal birth control, compared to when you're off of it.
- 6.56 What this other research showed is that, for a lot of women, they don't really notice a change in attraction, which suggests that they might be the types of people who don't have a profound response to the hormonal changes initiated by hormonal birth control. But for the women who experience an increase or a decrease in attraction, it suggests that they are.
- 6.57 This doesn't necessarily mean that it's going to be for the worse. We see that 36% of women, when they go off of hormonal birth control, are thrilled by the partner that they've chosen and feel even more attracted to them than they did before. But for 21% of women, it can mean bad news, at least according to that study.
- 6.58 Another question that's been raised about the use of hormonal birth control and the way that it affects women is that it might also affect how attractive women are to men. This is another really provocative question. The idea here is that we do know from a great deal of research now that when estrogen is high and rising across the cycle, women look, smell, and move and sound sexier to men than they do in points in the cycle when estrogen is low.
- 6.59 This suggests that if we take a drug like hormonal birth control, which turns the volume down on estrogen and keeps it relatively low and stable, this could potentially make women less attractive to men. Because you're not getting that little sexiness boost mid-cycle that most women get when they're naturally cycling.
- 6.60 And although this isn't something that has been studied explicitly as a research question on its own—there hasn't been research conducted looking at whether women become less attractive to men when they're on hormonal birth control—we do know from some of the research I've already talked to you about that women who are using

hormonal birth control don't experience that little sexiness boost that naturally cycling women get from estrogen.

- 6.61** I'm sure you all can remember when we talked about the study that was done at the gentlemen's club, where the dancers who were on hormonal birth control earned significantly less money than the women who were naturally cycling, particularly when the naturally cycling women were at high fertility, because men find those women more attractive at that time. And that is consistent with the idea that women may be missing out on a little mid-cycle sexiness boost when they're on hormonal birth control, compared to when they're off of it.
- 6.62** There's also been some evidence from studies looking at men's—they call these T-shirt studies, where they have people smelling T-shirts that were worn by people at different points in the cycle, and then worn by women who are using hormonal birth control. And so there's some evidence that suggests that women might not smell as sexy to men when they're on hormonal birth control as they do when they're naturally cycling, particularly in the estrogenic follicular phase of the cycle, near ovulation, when conception is possible.
- 6.63** There's also some evidence from primates, some nonhuman primates, that's really provocative in terms of tapping into this possibility that using hormonal birth control may dampen some of these unconscious cues that females give off that can increase their attractiveness to males. There's been work, for example, in chimpanzees that finds that when you give female chimpanzees hormonal birth control, which they do, by the way, to prevent them from breeding at zoos, that actually prevents the males from going up and attempting to mate with them.
- 6.64** The males have no interest in mating with the females when they're on hormonal birth control. The only times that they want to mate with females is when they're naturally cycling and in estrus, during that period of time when estrogen is high and conception is possible. Outside of that window, the males have no interest in mating with those females.
- 6.65** There's also other work in other forms of primates, other nonhuman primates, that finds that there's other males who will mate with females who are using hormonal birth control, but they won't ejaculate. It's like they won't bother, which is so interesting. It's like, physiologically, there's something going on where there's a switch being made, where they recognize that this is not a fertile female. And so they save the energy of ejaculation because it's a reproductive dead-end in a female who's non-reproductive.
- 6.66** And this is true when those males have sex with females outside of their fertile window, and also when those females are on hormonal birth control. All of this just simply suggests that, when we shut off estrogen, when we prevent ovulation and shut off estrogen release, it may also be turning off other types of subtle cues that increase

our attractiveness to partners, potentially making women who use hormonal birth control unable to capitalize on a free-of-charge, all-natural little boost in sexiness that naturally cycling women get each cycle.

- 6.67** Let me just hasten to add that, again, whenever we're talking about these differences that are initiated by hormonal changes, we're not talking about the difference between night and day. We're talking about night and later that night. A woman's sexiness—it's not like you're going to go on hormonal birth control and all of a sudden turn into a completely different person who nobody is going to want to go out with. Instead, it just means that it may dampen some of the cues that are used to infer attractiveness by preferred mates. And women who are using hormonal birth control, because they're shutting down estrogen, may be missing out on some of these benefits.
- 6.68** So there's a lot of evidence, an increasing amount of evidence, suggesting that the hormonal changes that are initiated by hormonal birth control use can have an impact on women's sexual psychology. This is true both in terms of the act of sex itself and things like sexual desire and sexual function, as we talked about last lecture, and there's also things like sexual attraction and partner choice, which we're talking about here.
- 6.69** But it's not just sex and sex hormones that get impacted when women go on hormonal birth control. In some ways, the sex-related changes are almost obvious. We would expect that. You're changing sex hormones, so of course it's going to change processes related to sex and processes related to attraction. But it turns out that the effects are far wider than just changes in sex and sexuality, and that they also extend to areas related to our mood.
- 6.70** Mood changes on hormonal birth control are very common. They're so common that they're one of the primary reasons why women discontinue using it. They've done studies looking at the rates by which women continue or discontinue hormonal birth control use. What they find is that most women, after they start taking the pill, discontinue it. The rate is something like 50%. It's about half of women, after they start taking it, they quit because they hate it.
- 6.71** The reason that they hate it is because of side effects. The side effects that the women dislike the most are those related to mood. More than half of women who start hormonal birth control discontinue it soon after starting, because of the presence of unwanted side effects. And the number-one side effect that causes women to discontinue using the pill are those related to mood.
- 6.72** There's been a lot of discussion about mood and hormonal birth control use, and you may have even seen some of this in newspaper articles or magazine articles or things that you see on the internet, with some people saying birth control pills and the use of hormonal birth control of all sorts can be related to unpleasant changes in mood and

increase the risk of depression and anxiety, and then other people turning around and saying, "No, that's not true."

- 6.73** And so, what the large research cohorts, when it's done in large populations, where you're able to actually get at some of the nuances related to hormonal birth control use and mood, come in, it suggests that being on hormonal birth control can increase the risk of developing and being diagnosed with depression. And so, I'm first going to walk you through the results of what is one of the most comprehensive and large studies of the relationship between hormonal birth control use and mental health that's ever been conducted.
- 6.74** This was a study that was conducted on data collected from the Danish health registry. Denmark has socialized medicine. Say whatever you will about socialized medicine, one of the benefits of socialized medicine is that you have data on everybody who uses the medical system. Because it's a government-run program, they're able to keep track of everything, and they keep track of every person who lives in Denmark. This allows for the creation of these huge data sets.
- 6.75** In this data set, this is of over a million women living in Denmark and looking at the probability of being diagnosed with depression or being prescribed an antidepressant subsequent to starting hormonal birth control use. What these researchers did is, they were able to secure the data of women ages 15 to 35, and they were able to look at women who were prescribed hormonal birth control or not prescribed women hormonal birth control, and then follow them over time to see whether they were subsequently either diagnosed with depression or prescribed an antidepressant.
- 6.76** They were able to compare women who started a prescription for hormonal birth control with age-matched control women who were not prescribed hormonal birth control. Here's a whole cohort of the entire population of Denmark, over a million women in this study, ages 15 to 35, some of whom start hormonal birth control, some of whom stay off of it, and then they follow them over time.
- 6.77** What they found—this figure is very hard to read, so you're just going to have to squint and trust me, but what this figure shows, the first column is a list of different types of hormonal birth control products. Some of them are oral contraceptives, and you can see that it says oral, and it's got the list, and it's got a bunch of names. These are the names of the progestin that is used in the product, and there's multiple types of progestin, multiple types of this synthetic progesterone that goes into these things.
- 6.78** They break it down by different types of progestin. They break it down into whether it's an oral contraceptive pill or whether it is a non-oral product like the hormonal IUD or the vaginal ring or the hormonal birth control patch. Then they look: What is the odds ratio? What is the probability of being—in this first column, this is looking at being prescribed an antidepressant subsequent to starting hormonal birth control use. What you can see is that, for every product that they measured, the ratio is higher than 1.

- 6.79** So 1 is the reference group. This is just women who are not using hormonal birth control. Everything over 1 suggests that there's an increased risk of being prescribed an antidepressant, based on being on that type of hormonal birth control. What you can see, for example, if you're on the hormonal birth control patch, the probability of being prescribed an antidepressant is 3.1. That means that it's three times higher than what it is for a woman who's naturally cycling.
- 6.80** For women who are on the vaginal ring, it's 2.9 times higher. For women who are using a progestin-only pill, it's 3 times higher. They found that there's a significantly higher risk of being subsequently prescribed an antidepressant, which is consistent with a depression diagnosis, and they also find that there's a significantly greater risk of being diagnosed with depression. What they found was that the risk factor seemed to differ across products. In particular, they found that the lowest risk product is the combined hormonal contraceptive pill.
- 6.81** We can see that the probability of being diagnosed with depression is almost twice as high in women who are using the hormonal contraceptive pill compared to women who are naturally cycling. But when you look at products like, for example, the hormonal birth control patch, it's almost three times higher. There's differences depending on the product that's being used. But across the board, there is an increased risk of being either diagnosed with depression after starting a hormonal birth control or of starting an antidepressant after starting a hormonal birth control.
- 6.82** Another thing that they found in the study was that the effect was significantly stronger and higher in teens. This is something that's been replicated in a number of different studies. When you look at this figure, what it's showing you is, it's showing you the increased probability of being diagnosed with depression, based on the type of product that you're on.
- 6.83** For example, the yellow circles are the combined oral contraceptive pill, which, as I said, for all forms of hormonal birth control, is the one that has the lowest risk profile of an increased risk of depression. And you can see it there on the bottom. And then we also have the hormonal birth control patch, the vaginal ring, the progestin-only contraceptive pill, and the hormonal intrauterine device, the hormonal IUD, which is the blue line.
- 6.84** On the x-axis, what you can see are the different age ranges. In the first one, it's ages 15 to 19. The second one is 20 to 24. This is followed by 25 to 29, and lastly, 30 to 34. What you can see is that, as women get older, the probability of being diagnosed with depression after starting hormonal birth control decreases significantly. The risk of developing depression from hormonal birth control is asymmetrically shouldered by teenagers.
- 6.85** When you look at the difference between that 15- to 19-year-old bucket, you can look at the odds ratios. In some cases, such as with the vaginal ring, for example, the

probability of being diagnosed with depression is three times higher in these teens. It's very significant. Most people aren't told about this. Teenagers are being put on hormonal birth control for acne, and nobody shows this to their parents. Nobody shows this to the girls themselves.

- 6.86 But when the brain is developing, and you're still in the midst of that pubertal transition, where your brain is developing and transitioning from the child version of itself into the adult version of itself, it is primed to be sensitive to sex hormones. And then they're being flat-lined and then replaced with the synthetic. And that's what we get. We get this really high increased risk of developing depression in adolescence and early adulthood.
- 6.87 You can see that the 20- to 24-year-olds still have an elevated risk relative to what's observed in the 25- to 34-year-olds. This also, again, tells us that this heightened risk that we see of developing mental health-related illnesses isn't shared equally in the population; and that, for older women, women who are young adults, 25 and older, the risks aren't as great. This doesn't need to be as much of a concern for this population, but it does for the younger women.
- 6.88 This pattern is repeated in other large cohort studies. This was one that was just recently published a couple of years ago. This was using biobank data from the United Kingdom, where they're looking at over 250,000 women. This time, they're looking at women ages 12 to 30. Again, we're including this adolescent group, these really young women. What you need to know from this figure is that they found a very similar pattern to what was observed in the big Danish cohort study.
- 6.89 That was that being on hormonal birth control increases the probability of being diagnosed with depression, and those are the results that you see at the top. Then it also increases the risk of scoring poorly on what's known as the mental health questionnaire. The mental health questionnaire is not a diagnostic tool, but it does evaluate the state of your mental health, and people who score higher on this have poorer mental health than people who score lower.
- 6.90 What this research found is that being on hormonal birth control is associated with a higher risk of depression and a higher risk of scoring poorly on the mental health questionnaire. The risk is asymmetrically higher for teenagers than it is for adults—again, very consistent with what they found in the original study. This study also found that ever being on hormonal birth control is related to an elevated risk of developing depression and scoring poorly on the mental health questionnaire, even after you go off of it.
- 6.91 This also suggests that there may be a lasting effect, and this lasting effect seems to be particularly strong for the adolescent population—again, these teenagers who are oftentimes being prescribed hormonal birth control for really minor issues that aren't nearly as important as something like fertility regulation.

- 6.92** Even more devastatingly, we see similar patterns for suicide risk. This is another study that came out of Denmark using their health registries, and this was a study looking at more than 500,000 women. They were able to follow these women and then look at the risk of a suicide attempt or a successful suicide. What they found was that using hormonal birth control was related to having a two times greater risk of attempting suicide. It doubles your risk of attempted suicide. It triples your risk of completed suicide.
- 6.93** Again, like in the other studies, the results found that the highest relative risk was for adolescents. Again, these 15- to 19-year-old women are the ones who have the greatest risk. If you look at the second figure on this slide, which is the one in the middle, you can look at the hazard ratio based on a woman's age, and you can see these really young women, the 15- to 19-year-olds, they're the ones who have double the risk factor of suicide, whereas when we look at the older women, it is lower.
- 6.94** The bottom figure shows us another really important finding that they discovered from this study, and they discovered that the risk is the greatest, and it peaks about two months after beginning hormonal birth control. What this means is that there might be a window in which women are particularly vulnerable to the negative mental health side effects that can come up with the use of hormonal birth control. Again, this is stuff that is usually not communicated to women or to their mothers or mothers and their daughters.
- 6.95** This is a time of intense vulnerability for women who are using hormonal birth control. Peaking two months after use, you can see there, the very bottom figure, it has months of use as the x-axis and then the hazard ratio. You can see that, at about two months of use, the probability of suicide is two and a half, and then it drops down after they've been on it for seven years of use or more, it begins to decrease, but it really seems to hit a peak early on in use, which is also, again, useful information that oftentimes isn't communicated, and because of that, can create issues.
- 6.96** Another study that's worth mentioning when it comes to women's mental health and the use of hormonal birth control is the fact that when you study women who are naturally cycling and study women who are using hormonal birth control, and you follow them for any period of time, either in a relaxed state or in a response to a stressful situation, you have elevated levels of subjective stress, and you have lower levels of positive mood in the pill-takers compared to the natural cyclers.
- 6.97** This was one study that was done looking at differences in how women who use and do not use hormonal birth control respond to a stressful situation that's created in the laboratory. What the researchers found is that, when they looked at the differences between women who are using hormonal birth control and women who are naturally cycling, is that they are different across the board.

- 6.98** When they measure women's, for example, the first figure that you're looking at on the left is showing women's subjective stress levels. You can see that, before the stressor, which is that red bar that says TSST, that stands for Trier Social Stress Test, and that's actually a stress-causing paradigm that's used in research laboratories to see how people respond to stress. It makes people give an extemporaneous speech in front of people and video cameras, and then it makes them do backward math.
- 6.99** They have to count backward in steps of 13 from 1,132 going backward. If they mess up, they have to start over again. All this is being recorded and people are frowning at them, and it's this really terrible setup. And it creates a lot of stress. And so, we do this as a laboratory setup, this Trier Social Stress Test, because it reliably makes people feel terrible and stresses them out. Then it allows us to see how they respond to stress.
- 6.100** As we were doing this study, this was a study that came out of my lab with one of my graduate students as the lead author. We followed the women who were both using hormonal birth control and then naturally cycling, to see how they were responding to stressful situations. It was really interesting to observe that not only were there differences in how women felt in response to stress, but there were differences in how women felt without stress.
- 6.101** The women who were using hormonal birth control reported feeling significantly more stressed out prior to the stressful paradigm, during the stressful paradigm, and then after the stressful paradigm. You can see that their recovery was actually a little bit slower. You can see that the steepness of their slope is less steep than that of the natural cyclers, who were able to more quickly recover from stress in terms of their subjective experiences.
- 6.102** Similar patterns were found when we looked at the positivity of women's moods. Again, regardless of whether we were stressing them out or not stressing them out, the women who were on hormonal birth control had lower mood. That was true in the presence or in the absence of a stressor.
- 6.103** All of this is consistent with the idea that hormonal birth control can cause changes, psychological changes, in the way that our bodies regulate stress, are able to manage stress and regulate mood and manage mood, and that these things can have really, oftentimes, really difficult outcomes for women in terms of their mental health, and sometimes really devastating outcomes in terms of some women wishing to end their own lives, and sometimes even successfully doing so, in response to the psychological changes initiated by the birth control pill.
- 6.104** And so, when we pick up next time, we're going to be talking about what it is mechanistically that's going on in the brain that's responsible for these differences. Why is it that we see that women who are using hormonal birth control experience an increased risk of depression and an increased risk of anxiety and less resilience to stress

and less resilience to an absence of stress, for that matter? And we'll talk about also the ways that changing women changes the world. So, thank you.

- 6.105** From a mental health professional standpoint, is there any standard practice for evaluating someone for depression in relation to whether or not they are on hormonal birth control? Do they consider those questions invasive, or do they go ahead and prescribe them an antidepressant, without gauging whether or not a different intervention might be more appropriate, such as maybe tapering off hormonal birth control?
- 6.106** Yeah. The usual way that it's dealt with is, women are prescribed an antidepressant. That's usually what happens. And this isn't even something that's common knowledge amongst most health care practitioners. Most health care practitioners, when I show them these research results, they're shocked. They have no idea that this exists, and so they don't even think to ask the question.
- 6.107** But I think that informed care would require asking those questions, and then also asking the question of whether or not being on hormonal birth control is necessary, because a lot of these women don't know this. Their moms don't know this. As soon as they learn that and they think about how they're feeling, they're willing to try something else. If it's for their acne, they can do something different.
- 6.108** But yeah, currently, the way that it's usually dealt with is usually at the same doctor's office that prescribed the hormonal birth control. Then they'll go and report that they're feeling low mood, and then the doctor will prescribe them an antidepressant. Even the doctors don't always put two and two together and think, "Well, maybe a better solution would be to try something else for your acne, or try to figure out something else for making your menstrual cycles more bearable."
- 6.109** It's to tag along to that question, as far as true informed consent. So in a perfect world, you're in an office, and you have a 14, let's say 17-year-old girl there with her mother, and the prescriber has 30 seconds or less to give true informed consent about depression, anxiety, low mood, and suicide risk. What would you say?
- 6.110** I would say, research shows that there's an increased risk of this, this, and this happening from the use of hormonal birth control. This doesn't happen to all women, but it does happen to some women. And this is something that you should be looking out for. Because the fact is, even if women are going to be on it, they need somebody looking out for them. I think that, if you let their mother or their partner, whoever it is, know, "I'm starting this drug that's associated with an increased risk of this," does that mean that all—no.
- 6.111** There's a lot of girls who go on this, even as teenagers, and they feel fine. But we need to know that. We need to have people looking out for these girls. Because the thing about depression is that your brain doesn't think, "I've got depression." It doesn't think, "I've got depression, and it's probably from the medication I'm taking, and none of this

is real." Instead, you think, "Why do my friends hate me? Everything is terrible. My job, I just don't find joy in the work I do anymore. Gosh, my life is just really bad."

6.112 Your brain can't think about itself as having depression. Your brain just thinks that the world is terrible. It's seeing the world through the lens of depression. You need to have other people looking out for you, because they're going to be a better gauge of whether or not you're having mood changes, oftentimes in a person themselves, because they might say, "I'm not having mood changes. Just, everybody's terrible, and my job sucks." And so, having other people looking out for you, I think, is really key.

6.113 I would say, communicate the risk, and then also talk to people who love the people who are being prescribed about the fact that this is a risk, so that they know what to look out for. Because I think that there are so many tragedies that could have potentially been prevented just by virtue of information, and having people know that this is a risk and that this is something to be looking out for.

Lecture 7: Birth Control & Mental Health

At a glance — Birth control and mental health.

- 7.1 When we left off last, we were talking about research that shows a relationship between hormonal birth control use and the risk of being diagnosed with depression, particularly among teenagers, and a number of studies have replicated this effect. However, it is important to put all of this in context. Just because you go on hormonal birth control doesn't necessarily mean that you're going to develop depression.
- 7.2 Some studies, for example, find no relationship between hormonal birth control use and the risk for depression. This is particularly true with combined oral contraceptive pills. As noted in an earlier lecture, when you look at the risk profile of different forms of hormonal birth control, the one with the lowest mental health-related side effects are the combined oral contraceptive pill, particularly when taken in adulthood.
- 7.3 Some studies have found that there's no relationship between these things, especially when they're looking specifically at the pill, but they continue to find a relationship when they look at progestin-only products, looking at the hormonal IUD, the vaginal ring, and the hormonal implant. If you're looking at the risk of hormonal intrauterine devices, or IUDs, research does indicate that having a lower-dose IUD has a lower depression risk than those with a higher dose in the IUD. So again, showing that there's differences depending on the product.
- 7.4 It's also important to note that women who have severe PMS or PMDD, which again is short for premenstrual dysphoric disorder, and this is a very severe form of premenstrual mood change, where women experience extreme depression, which is oftentimes accompanied by suicidal ideation, and these women oftentimes will experience improved mood when they're using hormonal birth control.
- 7.5 To show you the results of some research looking at this, this was a very recent meta-analysis done, looking at mood-related effects of hormonal contraceptives on women who have severe PMS and premenstrual dysphoric disorder. What they find is that some of the products make the mood-related symptoms worse, and those are the ones that you can see shaded in blue. Some of the products make the mood changes worse, but there are some products, particularly those that contain the fourth-generation drospirenone, that seem to make symptoms better.
- 7.6 The reason that hormonal birth control can be palliative to women who have premenstrual dysphoric disorder is that a lot of women who have this disorder have an extreme sensitivity to hormonal changes. One of the things that we've been learning about in this class is the fact that for a natural menstrual cycle, for somebody who's naturally cycling, you get a lot of dynamic hormonal changes across the cycle. You get a big rise and fall in estrogen in the first half of the cycle, and a big rise and fall in progesterone in the second half of the cycle.

- 7.7** If you're somebody who's very sensitive to those hormonal changes, and your body isn't able to make the necessary adjustments to be able to ride the wave of those hormonal changes without it feeling disruptive, it can lead to emotional challenges and depression in the case of PMDD and hormonal birth control by flat-lining hormones. This, of course, is going to prevent all of that variability in hormone levels that can contribute to PMDD symptoms.
- 7.8** So even though, for a healthy woman who doesn't have a history of mental health problems, using hormonal birth control potentially can lead to an increased risk of depression, for women who suffer from premenstrual dysphoric disorder and also who have severe PMS, some of them find the use palliative. And again, it's important to note that when you're taking hormonal birth control for one of these syndromes or any other syndrome, whether it's PCOS or endometriosis, this isn't a cure for the condition. It's a way to manage the symptoms.
- 7.9** This is simply going to make things better for you, without addressing the underlying cause. With things like premenstrual dysphoric disorder, the research is still ongoing into what the cause is. We're staying tuned on that one, but for some of the other issues that hormonal birth control is used as a treatment for, there are alternative treatments that will address the issue instead of just simply providing relief.
- 7.10** Now we're going to talk a little bit about why it is that the pill— again, my shorthand for hormonal birth control—why it is that it's related to changes in mood. There are a number of different mechanisms at play, and so we're going to walk our way through these different mechanisms so we can better understand what's responsible for the increased risk.
- 7.11** The first thing that's responsible for the changes in risk profile of hormonal birth control use and depression are changes in neurotransmission that occur in response to taking hormonal contraceptives. The first of these has to do with the neurotransmitter serotonin. You're probably familiar with serotonin. It's one of our feel-good neurotransmitters. It's something that helps stabilize our mood, and it improves our mood and makes us feel good.
- 7.12** Serotonin is one of the things that our brain gives us when we're doing something that feels good. If we're doing something that, historically, would have promoted survival and reproductive success, one of the ways that our brain rewards us is by releasing serotonin. It makes us feel good, and more of it is generally better than less of it. Estrogen, our wonderful sex hormone that we tend to get a lot of right near ovulation, tends to increase our sensitivity to serotonin.
- 7.13** It upregulates serotonin function. It increases synthesis of serotonin. It increases the expression of serotonin receptors on neurons, meaning that there's more receptors available to grab onto all of that feel-good serotonin and bring it in, bring it into the neuron. Hormonal birth control, of course, blunts this. By turning off women's sex

hormones and blunting levels of estrogen, women are getting less serotonergic function in the brain. They're getting less exposure to serotonin, and this is something that can lead to a lowered mood.

- 7.14** Another contributing factor that can lead to flattened mood among women who are hormonal birth control users is that you get reduced amygdala activity and reduced functional connectivity between the amygdala and the rest of the brain among women who are hormonal birth control users, compared to natural cyclers. And although you might think to yourself, "Reduced amygdala activity, isn't that a good thing?"
- 7.15** The amygdala, of course, is the fear center of the brain, and you might think that if you turn the volume down on amygdala connectivity, that this might make you feel better, but poison is all about the dose. You can have too much amygdala activity, and that's going to make you feel anxious. But if you have too little amygdala activity, this can actually make you feel worse. This can flatten affect and lead to anhedonia, which is characteristic of depressed affect and depressed mood. Another potential mechanism contributing to hormonal birth control users having an increased risk of depression is this decrease in connectivity between the amygdala and the rest of the brain, leading to flat affect.
- 7.16** The last of these changes—and this is by far the one that's been studied the most—is changes in neurotransmission as a result of the fact that the progestins in birth control are not the same as our endogenous-produced progesterone. Progesterone, as produced by the body and as contained in forms of hormone therapy that include micronized progesterone, when those are being broken down in the body, they release a compound called allopregnanolone.
- 7.17** Allopregnanolone is a very potent modulator of the GABA-A receptors in our brain. So, first, I'm just going to tell you about the GABA-A receptor, and I'm going to tell you about GABA. GABA is the most frequently used form of inhibitory neurotransmission that goes on in the brain. And the role of GABA is to decrease the brain's reactivity to stimuli from the environment. It essentially turns down the volume on neurotransmission, making our brain less reactive to things.
- 7.18** This actually plays a really important role in terms of stabilizing our mood. GABA, when you think about what it feels like when GABA receptors are being stimulated, and one of the examples I gave in an earlier lecture is, when your GABA receptors are stimulated, this is usually the way you feel when you're getting a massage, if you're doing yoga, if you're meditating.
- 7.19** You're in your pajama pants and sitting by the fire, that's activating GABA receptors. It's that calming neurotransmission that goes on in your brain, slowing your brain down, calming you down, and stabilizing your mood. GABA receptors are also what gets activated when you drink alcohol. When you have a drink, and you feel yourself

beginning to relax and calm down, that's the feeling of your GABA receptors being activated.

- 7.20** One of the most potent forms of GABA receptor activity in women comes from allopregnanolone, this really lovely neurosteroid that gets released when progesterone is being broken down in the body. Women who use hormonal birth control don't release any progesterone because they're shutting down their cycles, and ovulation is how women's bodies make most of their hormones.
- 7.21** This means that women who are on hormonal birth control, because they're taking a progestin, which doesn't release allopregnanolone when it's being metabolized, and they're not producing any of their own endogenous progesterone, their levels of allopregnanolone are significantly lower than those of naturally cycling women. And this is believed to be one of the largest contributors to the increased risk of depression that's observed in women who use hormonal birth control.
- 7.22** Because there's a lot of research that indicates that lower levels of allopregnanolone are related to an increased risk of mood disorders, including anxiety disorder and depression. One of the contributors is this reduction in this important neurosteroid, again contributing to the heightened risk of depression among hormonal birth control users.
- 7.23** Other differences that we see in hormonal birth control users that can contribute to the increased depression risk have to do with the HPA axis. This is something I've talked a little bit about in this course already, but as a reminder: your HPA axis is your hypothalamic-pituitary-adrenal axis, which is why we call it the HPA axis, because nobody wants to say that. The HPA axis is essentially the pathway of communication between your brain, your pituitary gland, which is your brain's medium between itself and the endocrine organs, and then your adrenal glands.
- 7.24** Your adrenal glands are little blobs that sit on top of your kidneys that release the stress hormone cortisol. So your HPA axis is your brain's stress axis. That's the easiest way to think about it. What research finds is that, when most people are experiencing a stressful event— imagine yourself having to give a speech or you're sitting in traffic, and it's really bad, and you want to get home—when you're feeling stressed out, your body will release cortisol.
- 7.25** That's your brain telling your adrenal glands, "There's stress going on, release some cortisol." Even though we tend to think about cortisol as a bad guy because it is related to stress, it's actually a good guy in the context of acute stress. The reason for this is that cortisol prepares our body to be able to learn from stress, and it also prepares our body to be able to escape from situations that potentially are dangerous.
- 7.26** What cortisol does, for example, is it dumps fat and sugar into our bloodstream, which allows us to be able to make a quick getaway if something terrible is happening or something's chasing us, because, of course, historically, these are some of the contexts

that would have led to the release of cortisol, was having some sort of a physical danger. Having all that fat and sugar in your bloodstream is also going to allow your brain to get access to the glucose that it needs to be able to make quick decisions.

- 7.27** It also helps to prompt the birth of new neurons in the hippocampus, so that way you remember things that are going on in the context of stress, so that way you're able to better cope with similar stressful situations in the future. Cortisol, even though we tend to think of it as a bad guy, in the context of acute stress, it's a good guy. Because of this, when people experience something stressful, their bodies will start to release cortisol to allow their bodies to adjust to and cope with the stressful situation. Women who use hormonal birth control are found not to release cortisol in response to stressful situations the way that normal, healthy women and men do.
- 7.28** This is some research that's been done for more than 30 years now, where researchers find that if you put women in a stressful situation, and some of the women are on hormonal birth control, and some of the women are naturally cycling, the women who are naturally cycling, if they're experiencing something stressful, they'll dump cortisol into their bloodstream, as most people do when they're responding to a stressful situation. But then women who are using hormonal birth control don't.
- 7.29** And although this might seem like a get-out-of-jail free card to stress, "OK, no cortisol; that sounds great," it's actually not great. Remember, cortisol is part of what our body is using to cope with and learn from stress, so not having that change is something that's maladaptive. In fact, this is a type of pattern that we tend to see in people who've suffered from things like PTSD or people who've experienced childhood trauma.
- 7.30** People who've experienced chronic stress in their life have this same type of blunted cortisol response to stressful situations that we tend to see in women who use hormonal birth control. This, of course, can lead to compromises in coping. It can lead to reduced coping from stress. There's some evidence suggesting that women who are using hormonal birth control tend to feel more overwhelmed from stress and less able to cope with stress compared to naturally cycling women.
- 7.31** It can also compromise the anti-inflammatory signaling that needs to happen in response to a stressful situation. One of the things that we've talked about is the fact that, when we're experiencing stress, our body, in addition to releasing stress hormones, also increases inflammatory activity. The body does this because, historically, the types of situations that would prompt the need to release stress hormones were also related to an increased risk of injury. For example, the threat of a bear attack or a threat of a lion attack or a threat of a rival community coming to take your resources and land—these are types of contexts in which the risk of injury are potentially very high.
- 7.32** So, our body, we've inherited this tendency to release inflammatory activity in addition to stress hormones when we're in stressful contexts. The issue, if you aren't releasing

cortisol at the same time that you're releasing inflammation, is that this can lead to a compromised downregulation of the inflammatory response that cortisol is responsible for.

- 7.33** Because one of the things that cortisol does, in addition to doing the dumping of the fat and sugar into the bloodstream and birthing the new neurons in the hippocampus, is it also helps to downregulate inflammation. Lacking that response, in response to stress, can lead to exaggerated inflammatory activity, which, of course, can contribute to diseases of aging, the development of autoimmune disorders, and so on.
- 7.34** Why is it that women experience blunted cortisol in response to stress? This is something that researchers don't actually have a good answer to, but we do have some clues. The clues are all pointing in the direction of the progestins in hormonal birth control because they are, again, not a perfect match for progesterone receptors. Remember, these are being synthesized primarily from testosterone, or occasionally, they are synthesized from a diuretic called spironolactone.
- 7.35** They're an imperfect match to progesterone receptors, but they're able to get picked up by progesterone receptors to enough of a degree that they suppress ovulation. But they also have non-perfect or imperfect binding specificity, meaning that they'll bind to other receptors that they don't belong to. One of the receptors that they are found to bind to occasionally are glucocorticoid receptors, and glucocorticoid receptors are receptors in cells that pick up cortisol.
- 7.36** If your body is getting constant cortisol stimulation, which would happen if you're taking a daily progestin that's binding to glucocorticoid receptors, this is going to make your brain and body think that it is in a state of chronic stress. When your body is in a state of chronic stress, where there's nonstop glucocorticoid signaling, the body will just shut the stress response down. It does this because having constant glucocorticoid stimulation is no good for the body.
- 7.37** Because one of the things that our stress hormones do is they redistribute resources that are being used for everyday bodily activities—things like cell repair and immune response and growth and respiration and digestion— it takes those energetic resources and directs them toward dealing with the stressor. And your body won't let that go on long term because then there's no resources for cell repair, immune function, growth, digestion, and so forth.
- 7.38** And so when the body is experiencing what it believes to be chronic stress, it starts to release what are known as glucocorticoid binding globulins. And these bind to the cortisol, and make it inactive. It does this to prevent the body from being totally overwhelmed with stress hormones. This is a pattern that we see in people who have PTSD, who've experienced chronic stress. Their bodies will start releasing high levels of glucocorticoid binding globulin, and they have very low levels of cortisol in response to stress.

- 7.39** This is also what we see in people who've experienced childhood trauma, and this is what we see in women who've used hormonal birth control. This suggests that their bodies may be responding to the progestins in a way that feels like chronic stress. Eventually, their body shuts the stress response down, and then the response to this is that they're no longer able to cope with stress the way that they normally would be if they were not using hormonal birth control.
- 7.40** Another factor that's believed to contribute to the greater risk of developing depression among hormonal birth control users is exaggerated inflammatory activity. I've already hinted at this on the last slide when we were talking about what one of the jobs of cortisol is, and that is to help regulate inflammatory activity in response to a stimulant, whether it's an immunological stimulant or a stress stimulant. We know that inflammation is also higher on average in hormonal birth control users compared to non-users, which is consistent with this idea.
- 7.41** The thing about inflammation is that, in addition to being related to diseases of aging and aging in general, inflammation is also a potentially negative mood killer. It is a potent depressant and anxiogenic, meaning it increases anxiety. The reason that inflammation leads to depression is actually kind of interesting. I'm going to have a little sidebar here. That is because inflammation is essentially telling your body there's something going on with the immune system.
- 7.42** The immune system needs resources to be able to fight whatever this is that's either about to happen or is happening. And so, the body will essentially turn down the volume on mood, as a way to keep you wanting to stay home and in bed and not doing anything and seeing anybody, because it's trying to conserve resources for the immune response. When we experience inflammation, our mood gets the volume turned down on it. Chronic inflammation, in addition to being related to diseases of aging, is also related to lowering of mood.
- 7.43** This is also believed to be a contributing factor among hormonal birth control users who have a greater risk of developing mood disorders, because there is cross-sectional and cohort data, so looking at this relationship in different types of research methods, finding that that combined oral contraceptive pill use is related to higher levels of CRP, and that's C-reactive protein. And this is a nonspecific marker of inflammatory activity in the body. We also find elevated levels of CRP in hormonal IUD users, which, again, suggests that you may get increased inflammatory activity in that context.
- 7.44** Some experimental work that came out of my research lab that was led by one of my former graduate students also found, you get a greater pro-inflammatory response to stress in women who use hormonal birth control compared to naturally cycling women as well, which, again, is consistent with this idea that hormonal birth control may promote exaggerated inflammatory activity or a runaway inflammatory response, in response to stress, and that this may be a contributing factor to both the risk of worsened mood, but also potentially a risk of autoimmune disorder.

- 7.45 Who is most at risk? We do know that the relationship between hormonal birth control use and lowered mood or depressed mood is something that not everybody is equally susceptible to. What are some of the things that you need to look out for if you're somebody who's on hormonal birth control or somebody that you love is on hormonal birth control?
- 7.46 The first thing that you need to consider is age. One of the things that we learned about in our last lecture—and I've reminded you about this morning—is the fact that adolescents consistently show the highest risk of depression and suicide in response to hormonal birth control use. This is the most vulnerable group, and these are the ones that you need to be watching over most carefully.
- 7.47 How do I define an adolescent? In the literature, it's generally defined from age 12 to 19. Within this age range, you tend to see the greatest risk of developing depression. You generally see the risk begin to lower when women are in their 20s. After 25, the risk becomes fairly low, especially for the combined oral contraceptive pill.
- 7.48 Another contributing factor that can be used to predict a person's probability of developing mood disorders in response to hormonal birth control use has to do with their history of sensitivity to the pill or their sensitivity to hormonal changes. If you have prior pill-related mood changes—let's say that you were on the pill in your teens and now you're in your 20s and you're considering whether you want to go on it again—if you had mood changes the first time, you're at a much greater risk of developing mood disorders the second time. The same is true if you have relatives who've had issues related to the pill, and they experienced depressed mood, you're going to be at a greater risk of also developing depressed mood on hormonal birth control.
- 7.49 If you have sensitivity to hormonal changes, like PMDD-like sensitivity, although some women who have PMDD find taking hormonal birth control is incredibly palliative and makes them feel a lot better, if you are sensitive to hormonal changes in ways that lead you to have those types of symptoms, you should just take extra care in observing yourself if you go on a prescription, because it's one of these things where it either goes all right or all wrong, where some women feel better, and some women, it makes them a whole lot worse, and they're already experiencing a lot of suffering psychologically, and so it can be a dangerous set-up. This is another group of women that needs to be watched carefully.
- 7.50 If you have a family or personal history of mood disorders, this is also something that is going to increase your probability of experiencing mental health side effects with hormonal birth control. You should know your family history and know your personal history. Again, if you are somebody who has this history, make sure that people are looking out for you because the fact is, the brain doesn't always recognize when it's depressed. Other people oftentimes can recognize it in us before we can recognize it in ourselves.

- 7.51** Because when we feel bad, oftentimes we externalize those bad feelings. We blame it on issues with our relationships or issues with our jobs, rather than recognizing the fact that our brain is just not seeing the world accurately and is seeing it in a way that's associated with having depression. If you fall into any of these different buckets where there's an increased risk of developing a mood-related disorder, I really encourage you to have somebody keeping an eye on you, while you're also keeping an eye on yourself.
- 7.52** The type of product that you use also has an impact on the probability of developing depression. In particular, what research finds is that these progestin-only products tend to be worse when it comes to mood than products containing ethinyl estradiol. In particular, the research seems to find that the birth control shot is the worst. When you look at the different risk profiles of the different products with respect to mood and unpleasant mood changes, what the research finds is that the hormonal birth control shot has the worst mood profile.
- 7.53** So if you're somebody especially who has a history of mood disorders, this probably is not the right one for you. The progestin-only pill, which is the one that's now available over the counter, also doesn't have a very favorable mood profile, which concerns me that this is out there, without people keeping an eye on the people who are using it.
- 7.54** And the reason that the products that have ethinyl estradiol tend to make women feel better is, estrogen feels good to women. It generally feels good. It's an unusual circumstance to have a body that has incredibly low levels of estrogen. Because remember, when women are on hormonal birth control and not ovulating, they're not producing any estrogen of their own. Then if they're taking a product that doesn't have any estrogen in it, that's a really biologically unusual experience for a woman's brain and body to be in.
- 7.55** And so women oftentimes feel different and not like themselves when they're on these progestin-only products. And so if you do have mood-related issues or you're concerned about mood-related concerns, then I would recommend one of the combined products that contains both ethinyl estradiol and the progestin. And in fact, the reason that ethinyl estradiol is even included in most forms of hormonal birth control isn't because of a contraceptive-related effect, but rather because it makes women feel better. These products tend to have a lower risk profile than do the progestin-only products when it comes to mood.
- 7.56** The only issue is, the products that contain ethinyl estradiol have a greater cardiac set of side effects. Women, for example, who have a history of migraine with aura or have a history of other types of migraines or have a history of any heart-related disease, are not recommended to take any products containing ethinyl estradiol because it is thrombotic, and it can have cardiovascular risk factors. There is a risk-benefit ratio that you have to determine in a conversation with your physician.

- 7.57** The risk of the IUD, as noted, is dose-dependent. There's different IUDs that are out there: there's hormonal IUDs and non-hormonal IUDs. The non-hormonal IUDs have a very low risk profile in terms of the probability of mood disorders because they don't contain hormones. When you're using the non-hormonal IUD, you continue to ovulate, and so you continue to produce sex hormones, so you continue to feel pretty much like yourself.
- 7.58** The hormonal IUDs come in different flavors. They come in the higher hormone levels, all the way down to very low-dose hormone levels. The ones with the lower doses are related to a lower risk of mental health-related disorders or the development of a greater risk for depression, and the ones with the higher doses of hormones are related to a greater risk of depression.
- 7.59** Lastly, the impact of hormonal birth control and the probability of developing mood-related disorders is going to depend on everything ranging from the formulation of hormones that you're using, the dose of the hormones that you're using, and the route of administration. So whether it's a vaginal ring, whether it's a patch, whether it's an arm implant or a shot or a pill, the research finds, when you look at the risk profiles in terms of the mental health-related side effects, it seems like different products have different effects—even the exact same product, if it's delivered in different ways.
- 7.60** So, for example, they've done research looking at the differences in depression risk between women who are using an IUD that uses a specific type of progestin, and then looking at a birth control pill that uses the exact same progestin, and the risk profile is very different. And so, even the way that it gets into the body seems to have an impact on how we respond to it. And so, keeping an eye on all of these things and making note of what works and what doesn't work is really important.
- 7.61** Now, I want to switch a little bit. We've been talking about the fact that using hormonal birth control is related to an increased risk of developing depression for some women. And now I want to talk a little bit about some of the downstream consequences of using hormonal birth control. This is actually from some research that's being done in my lab right now with some of my graduate students, and we're looking at the relationship between hormonal birth control use and alcohol craving use and use intentions, in response to alcohol cues.
- 7.62** We predicted that we would expect to see that women who use hormonal birth control would be at a greater risk of developing these types of psychological and behavioral profiles that increase the risk of developing alcohol use disorders because, when you're taking hormonal birth control, you're not getting to experience endogenous progesterone. Remember, progesterone, when it gets broken down, leads to the release of allopregnanolone, which has that calming influence on the brain by stimulating GABA receptors.

- 7.63 And so, we predicted that women who are using hormonal birth control, because they're not getting that GABA receptor stimulation naturally, may be seeking outside sources to get GABA receptor stimulation. And alcohol, that's how it works. Alcohol works by stimulating GABA receptors in the brain, and that's what makes people feel relaxed. And so we've done a series of studies now looking at the relationship between hormonal birth control use and alcohol craving and use intentions, and then people's responses to alcohol cues.
- 7.64 So if you show them scenes from a bar, the degree to which that activates alcohol-related cravings, what we find is that, in hormonal birth control users, we get a greater risk profile for these psychological and behavioral changes that are related to the development of alcohol use disorders. It seems that hormonal birth control use does, in fact, potentially lead to an increased risk of alcohol use disorders.
- 7.65 What are some of the other consequences, some of the other downstream consequences, of hormonal birth control use? This one is really important, and this is one that is not ever discussed. I want to make sure that we talk about it here, because it's something that we need to think about. We talked about the fact that adolescents tend to be at a greater risk of developing mood-related disorders when it comes to hormonal birth control use. But it also may be the case that teens may experience long-term developmental changes to their brain as a result of hormonal birth control use.
- 7.66 One of the things that we talked about on one of our very first lectures, so during our very first time together, we talked about the difference between activational and organizational effects of hormones. Activational effects of hormones are what I would experience if I were to take a birth control pill right now, that I would have these effects of the hormones in me today, and then tomorrow they would be gone, as my body metabolized the hormones and they left. Those are activational effects. It's the effects that you get of the hormone right now.
- 7.67 Organizational effects are the effects of the hormone on the long-term organization of the brain and the rest of the body. Sex hormones have both types of effects. And during the pubertal transition, when the brain is transitioning from its child version of itself into the grown-up version of itself, one of the primary drivers of that post-pubertal brain transformation is your sex hormones.
- 7.68 During this time, when your brain is still developing—and our brains are still developing until we're about 25, but most of the heavy lifting is done by the time we're 20— during this time, our sex hormones, in addition to regulating our cycles and everything else, are also playing a role in brain development. When we put adolescent girls on hormonal birth control during these years while their brain is still developing, how could that not affect brain development? It would be virtually impossible for it not to affect brain development.

- 7.69** How many studies do you think have been done looking at the developmental consequences of teenage birth control use? Yeah, almost zero. It's almost zero. These research studies are just starting to be done. What these research studies—and I can count them on one hand. That's how few there are. Nobody's even thought to ask the question. We're running these experiments on children, giving them birth control for things like acne and for cramps, without knowing what this might be doing to their brains, which I think is tragic.
- 7.70** The human studies suggest that using birth control, using hormonal birth control in adolescents, does seem to be related to an elevated increase in the risk of developing major depressive disorder throughout your lifetime after you go off of it. I think that most women and most mothers of teens, if they heard this information, would try to figure something else out for their teen's acne; and yet, this information isn't being communicated.
- 7.71** There's also some new rodent work that's come out, where you're able to do experimental rodent work, where you look at the developmental consequences of hormonal birth control use during adolescence. And what this rodent work finds is that, when they give teenage rodents hormonal birth control, that it leads to structural changes in the brain that last throughout adulthood after the drug is removed. Also, it leads to alterations in fear extinction. This is something that's found, specifically if rodents are given progestin-only products.
- 7.72** One of the reasons for this, likely, is the fact that, again, when our brain is transforming from its teenage version of itself to the grown-up version of itself, one of the primary things that it's listening to as it's reorganizing is estrogen. If you give a teenager, teenage rat or a teenage human, a progestin-only product, there's no estrogen. They're not producing any estrogen of their own because they're not ovulating. The brain is going to be different. I mean, there's just no way around it. The brain is waiting for the instructions from estrogen, and they're never arriving.
- 7.73** What this work in the rodents finds is that you get alterations in fear extinction. This is something that can be really difficult. Fear extinction plays a really important role in preventing us from getting PTSD. It plays a role in helping us get over PTSD. It's the ability to not be afraid of everything that's ever caused a problem for us. We tend to find that, like I said, in these rodent paradigms, where you get difficulties with fear extinction in response to hormonal birth control use in the teenage years.
- 7.74** Other consequences of hormonal birth control use that are worth mentioning is talking about the increased risk of breast cancer when hormonal birth control is being used as a form of menopausal hormone therapy. MHT means menopausal hormone therapy, also known as hormone replacement therapy. There was a big—headlines were raised back in 2002. The Women's Health Initiative study did a big study looking at hormone therapy, and what they found was that, when you use hormone therapy—and they use

this term very broadly—when you use hormone therapy, this is related to an increased risk of breast cancer.

- 7.75** So all of the headlines are saying hormone therapy is linked to an increased risk of breast cancer. The thing that they didn't tell you in the headlines is that, in the Women's Health Initiative study, the type of hormone therapy that was being used wasn't biologically identical hormones. It was estrogen plus medroxyprogesterone acetate, MPA, which is a progestin. One thing that we've been learning about in this class is that progesterone and progestin are not the same thing. They do completely different things in the body.
- 7.76** Later research revealed that the type of progestogen used in hormone therapy matters dramatically when it comes to the impact on breast cancer risk. In particular, what the research finds when you break it all down—so first, we'll talk about the Women's Health Initiative study, that 2002 study that scared women away from hormone therapy. That study was looking at estrogen plus MPA, again that progestin. What they found is that women who were taking estrogen plus progestin, that they had an elevated risk of cancer. They had a 24% higher rate of developing breast cancer than did the control group.
- 7.77** What they found for women who were on estrogen alone, so no progestin, they actually found a lower breast cancer rate. The breast cancer rate was 23% lower than baseline. Using estrogen only related to a lower cancer risk, using estrogen plus progestin, 24% higher rate—very different outcomes. Since that time, there've been other studies that have come out. There's one that's called the E3N study. The E and the N stand for French words that I'm not going to try to pronounce because I'll embarrass myself.
- 7.78** But in these studies, they actually looked at the difference between taking estrogen with micronized progesterone, again, that biologically identical, real-deal progesterone, like the stuff our body makes, and then comparing that to women taking estrogen plus progestins. What they found was that, when women were taking estrogen plus micronized progesterone, again, no increased risk of developing breast cancer. In fact, they found that there was actually a 10% lower rate of developing breast cancer in women taking estrogen plus progesterone.
- 7.79** But when they looked again, at estrogen plus different synthetic progestins, and they assigned women to different types of progestins to see if there's something special—is it just this one progestin that's doing women dirty in terms of breast cancer risk, or is it all of them?—what they found was that, when women were taking estrogen plus a synthetic progestin, this was related to a significantly higher increased risk of developing breast cancer, anywhere from 16 to 69% higher rate of breast cancer in this group of women.

- 7.80** Progestins and progesterone are not the same thing. There's been a lot of women feeling very afraid of hormone therapy because of these headlines that say, "Hormone therapy linked to breast cancer risk." You have to look more closely to understand that when we look at the effects of actual biologically identical hormones on breast cancer risk, it's nonexistent.
- 7.81** There were actually some headlines that came out over the summer about this, saying, "Young women on hormone therapy at a greater risk for breast cancer." You look at the research study, and it shows the exact same patterns that these other studies are showing. The women who were on estrogen-only had no increased cancer risk. The women who were on estrogen plus progestin had a higher risk.
- 7.82** The thing that makes this particularly frustrating is that, when you actually look at the research literature, many times the researchers who are studying this stuff will interchangeably use the words progestin and progesterone. You actually have to look and see what product they're talking about to figure out whether they're talking about progesterone or a progestin. Many times, these things are being totally used interchangeably and confused, adding to the amount of confusion that so many women have about hormone therapy.
- 7.83** True hormone replacement therapy, meaning hormone therapy for menopause that's using biologically identical hormones, is not the same as hormonal birth control. This is an important point because hormonal birth control and menopausal hormone therapy, in some ways, are actually opposites of one another. Because hormonal birth control, you're taking a woman who's producing hormones and you're shutting them down. You're shutting them down using low doses of synthetics that lack the benefits of actual hormones.
- 7.84** You don't get that allopregnanolone with that GABA release in your brain. You don't get the benefits of having that nice surge in estrogen that has all of those really nice neuroprotective effects and also has a lot of positive effects on mood. With hormonal birth control, you're shutting down hormone production. With hormone replacement, when you're doing that with biologically identical hormones, you're actually supplementing the hormones in women who are producing less of their own.
- 7.85** In a lot of ways, it's the opposite ends of the spectrum. One is turning down the volume on hormones or shutting them down altogether. The other one is trying to help women get a little bit of a boost once their body [stops] naturally producing as much sex hormone of their own. So, despite being imperfect, nonetheless, many women are going to be continuing to use hormonal birth control.
- 7.86** One of the things that you have to appreciate about hormonal birth control is that it really has opened doors for women by being able to regulate their fertility. When a woman is able to regulate their fertility, rather than becoming reckless and irresponsible in the ways that some people might think or argue, generally what

happens is, women actually become more responsible and start taking control of their future because they actually feel like they do have control over what happens next.

- 7.87** Women are able to do things like finish getting an advanced degree and build a large career because they can do all of this without fear of an unexpected pregnancy, leading them to get benched. Because of this, and because we don't have a lot of options, unfortunately, to this day, this is 2025, and we still don't have a lot of good options for fertility regulation for either men or for women.
- 7.88** There's a lot of women who feel like they have their back up against a wall with hormonal birth control, where they feel like they know it's imperfect, they know it's going to shut down their hormones, they know it has all of these sets of risk profiles and can change everything ranging from their sexual desire to their mood, but nonetheless, they need some way to protect themselves from pregnancy because they have career plans.
- 7.89** Next time, we're going to be talking about what some of the things are that people should be paying attention to if they're choosing to be on hormonal birth control. What are some of the factors that should be taken into consideration when making those decisions? How to be able to make those decisions? Then we're also going to be talking about, in closing, talking about the way that changing women has changed the world. We'll pick up talking about those things next time.
- 7.90** You had mentioned alternatives to treating PMDD besides birth control. Can you go into any specifics or advisements you could give on that?
- 7.91** If you want to go the medication route, the answers that are usually given are hormonal birth control and then SSRIs. Just a really quick interesting note about SSRIs in the context of PMDD is that, unlike in the case of depression, when you have to take SSRIs every day and you have to take it for several months before you start feeling positive effects, with PMDD, you can take it only when symptomatic, and you get the results right away. The reason for that is that the mechanisms of action are different in the case of depression and the case of PMDD.
- 7.92** In PMDD, the reason that it works right away is, by keeping serotonin in the synaptic cleft, it actually slows the breakdown of allopregnanolone, which prevents women from having as jarring of a hormonal transition. That tends to lead to a more stabilized mood. That's if you want a medication-based answer. Non-medication-based answers, cognitive behavioral therapy has been found to be really effective for women with PMDD, because a lot of times what happens with these women is their really sensitive threat detectors in their brain are going crazy.
- 7.93** They're feeling terrible and they're feeling anxious, and they're feeling sad. They lash out at people that they love, and then they start to think that people hate them. There's a lot of little thought loops that they get stuck into. Cognitive behavioral therapy does a really good job of helping women identify what those thought loops are

when they're starting to get into them and come up with counter-narratives that they can introduce psychologically to be able to talk themselves out of feeling the way that they're feeling. And so, cognitive behavioral therapy has been found to be fairly effective.

- 7.94** We're doing a study right now where we're looking at vagus nerve stimulation, and I'm super stoked about this project, because there's just no way that it's not going to work. I'm completely convinced that, I would bet my house on this, because it's been found, when you stimulate the vagus nerve, it's been found to be really palliative in even untreatable depression. But it also even affects things like endometriosis. It's been found to be effective with endometriosis and slowing endometrial growth.
- 7.95** Anything where the body is misbehaving, it seems to do a really good job of helping to calm it down. It helps with epilepsy. I could go on and on and on, and so I'm really excited about that work. I'm hopeful that we're going to find good results that will offer women relief. But there's actually a product out there by a company called OHM Body, that's O-H-M Body. They're using vagus nerve stimulation that you can do at home for women's health-related issues.
- 7.96** We're also going to be doing a study looking at it in the context of perimenopause and see whether it also calms symptoms that way, because that's a nice thing that you can do at home for 2 hours every day while you're doing the dishes or doing something else that can help to alleviate symptoms. So that's another option. Then it's just the pillars of good physical health—exercise, having loving relationships with people that you love, making sure that you're getting enough sleep and so on and so forth.
- 7.97** In the context of cortisol and birth control or hormonal birth control, let's say, dampening or turning it down to some extent, when we put on top of that for women maybe personality traits such as agreeableness and societal pressures and norms such as fear of offending people, and then now we're adding the trifecta of literally changing biology, what do you feel like the real-world consequences could be for young women right now?
- 7.98** Oh, my gosh. I mean, I think that we're seeing it everywhere. I was telling my daughter this. I have an 18-year-old daughter. I call it young women's syndrome. I think young women's syndrome is this blend of overachieving, because in feeling like you need to please everyone, and perfectionism, and I think that all of these things, and in some ways, birth control, in addition to all the biological things that it does, it also sets up this context where women are expected to be both women and men.
- 7.99** There's so much pressure on women. These young girls, growing up expecting that they're supposed to be doing everything, and they're supposed to be doing it all the time, and then they're in this crazy biological context that is not typical, it's really hard. I see so many young girls have—and I say young girls and young women—have such

mental health struggles as a result of this toxic blend of overachieving and the need to please other people.

- 7.100** And the societal expectations that, now that women can regulate their fertility, everybody needs to be doing this and getting these advanced degrees and building these big careers and doing all this, but while also looking gorgeous and also not letting the kids suffer. I mean, it's exhausting even talking about it, let alone having to live that. And so, yeah, I think it's contributing to the decline of mental health in young women.
- 7.101** In regards to hormonal birth control, it might increase our rates of depression, possibly even anxiety, OCD, mental health issues. So would you actually consider hormonal birth control to potentially be a gateway, let's say, prescription drug into more psychotropic medications, in particular SSRIs for women?
- 7.102** Oh, yeah, 100%, yes. So anytime that you start getting comfortable with the idea of medicating your body—and there's a lot of research supporting this, that when people take regular medications, that additional regular medications oftentimes follow. So I do think that there's this comfort level that happens. On top of that, because a lot of women, when they take hormonal birth control, it causes these side effects, then they're comfortable taking a medication, and then they're having these side effects for which there's another medication-based solution.
- 7.103** I do think that, yes, 100%. Like I said, there's research showing that, once you sort of break that barrier of being comfortable taking daily medications, the number of medications increases. It just does. People get comfortable with the idea, and then that's their solution.

Lecture 8: Birth Control & Choice

At a glance — Birth control and choice — what to consider when deciding.

- 8.1** Welcome back. We're going to start today talking about some of the things you should be looking for if you are going to be using hormonal birth control, just so that way, you can best find something that's going to work for you. The first thing that you need to know if you're choosing a hormonal birth control is that the progestins that are in hormonal birth control are different. There's a lot of different progestins that are currently in use.
- 8.2** The way that these progestins are divided up is based on what's known as a generation. It just describes when the progestin came onto the market. The first-generation progestins were the first ones out on the market, followed by the second generation, and so on. There's four generations of progestins. We're going to walk our way through these four different generations and talk about the similarities and differences between them.
- 8.3** That way, if you're making a decision about choosing a hormonal birth control, you have a little bit more of an idea of what it is that you're choosing. The first generation of progestins, these were the first ones on the market. Just to make sure you understand that, even though we talk about a first-, second-, third-, and fourth-generation progestin, they are all currently in use. So these are all currently still in use.
- 8.4** The first-generation progestins are mildly androgenic. These are synthesized from testosterone. Again, with the progestin, they're molecularly manipulating the molecule. So that way, it will stimulate progesterone receptors, but oftentimes, they also will stimulate androgen receptors or testosterone receptors. The first-generation progestins are mildly androgenic, meaning that they do stimulate some testosterone receptors, which can lead to some masculinizing type of side effects like facial hair growth and acne.
- 8.5** Second-generation progestins are the most androgenic of all four generations of progestins, so they're the ones that have the most androgenic-related side effects. These ones are also synthesized from testosterone, and so they do stimulate progesterone receptors, but as noted, they can also stimulate testosterone receptors, leading to things, again, like sometimes hair growth, sometimes acne. Water retention tends to be higher with these categories of progestins.
- 8.6** Third-generation progestins are also synthesized from testosterone, but they're not androgenic like the first- and second-generation progestins are. So they don't have as many of those types of side effects related to acne or water retention or hair growth in unwanted places.

- 8.7** The fourth-generation progestins are the only ones that are synthesized not from testosterone, but instead from a diuretic called spironolactone. And these ones actually have anti-androgenic effects. And this means that, in addition to providing and stimulating the progesterone receptors in a way that inhibits ovulation, they also block some testosterone receptors. And this can actually have a really positive impact on things like water retention. So it can have an anti-water retention effect, so it has a diuretic effect. And it also has an anti-acne effect. This is one of the progestins that's most frequently used for women who experience acne and are looking for a treatment for that.
- 8.8** These different types of progestins, these different generations of progestins, because they're synthesized in slightly different ways, some of them are synthesized from testosterone, some of them are made from spironolactone, the specific formulations and the specific ways that they manipulate the molecules to be able to stimulate the progesterone receptors is a little bit different from one another. This means that they're going to have different types of effects in the body.
- 8.9** So if you're somebody who's going on hormonal birth control, the first thing that you want to make note of is: What is the type of progestin in your birth control? Making note of this is important because, if you take a form of hormonal birth control and you don't like the way that it's making you feel, the first thing that I would do when you're troubleshooting is troubleshoot the progestin.
- 8.10** If you take something and let's say that you're taking a hormonal birth control that has a fourth-generation progestin, and it's making you feel crazy, and you want to try something different because you would like to stay on hormonal birth control, the first thing I would do is try a different generation of progestin. I wouldn't stick within the same generation because, even though there's differences among the different progestins within a generation of progestin, they're more similar to one another than they are different. And so, the first thing I would do is try a different generation of progestin. That's the first thing I would troubleshoot.
- 8.11** The second thing I would troubleshoot is the dosages. Some of the progestins are going to be paired with 20 milligrams of ethinyl estradiol or 30 milligrams of ethinyl estradiol. There's different levels of estradiol that's in the forms of hormonal birth control. Also make note of that. You can have conversations with your doctor about whether more or less estrogen makes sense in the context of the experiences that you're having with the pill, and then you can be able to make informed choices about what's going to feel best for you.
- 8.12** The second thing to note, if you're making decisions about hormonal birth control, is that when you look at the different side effect risk profiles—for example, when we talked about things like the risk for depression or the risk for sexual dysfunction, in addition to differing across different categories of progestins, there's also differences depending on delivery method.

- 8.13** One of the things that we talked about that's a little bit surprising is that you can have the exact same progestin being delivered in different ways and have completely different side-effect profiles. What this means is that when you're considering how you're feeling on hormonal birth control and whether you want to try something different than what you're currently on, another thing that you should consider is how it's entering your body.
- 8.14** Are you taking this as a pill? Is this an implant? Is this an injection? Is it a patch? Is it a ring? Or is it an IUD? Because each one of these has different side-effect-related profiles. And so, if you're on, for example, a contraceptive pill and you don't like the way that it makes you feel, you can try the same type of progestin in a different delivery method, and you may have a totally different experience. And so, it seems that the way that we take the drug also has an impact on how we're feeling. This is something else to consider as you're making your decisions.
- 8.15** Now, a lot of people have heard the story, usually from their doctors, unfortunately, that the hormonal IUD is locally acting. I have heard from so many women who say, "I think that everything that you're talking about with the effects of hormonal birth control on women's psychology and the way that women experience the world is really interesting. But I'm on the hormonal IUD, and those hormones only act locally, so I don't have to worry about that." That's just not true. There's no such thing as a locally acting hormone.
- 8.16** Even at the end of this class, you know this. We learned about how hormones go through the body. When hormones get into the body, they go into the bloodstream, and then they go everywhere that blood travels. This means that they can't stay local, because hormones move. They move throughout the body. Now, the hormonal IUD, this being said, the hormones in the hormonal IUD do go everywhere, just like hormones always do. They get into the bloodstream, they go everywhere that blood travels, and that means that they affect even the brain.
- 8.17** Because of this, what research finds, when you look at women who are using hormonal IUD, especially when they're first on it, most of them are not ovulating. The reason they're not ovulating is because those progestins that they're being told are acting locally are getting into the bloodstream, reaching the brain, and telling the brain not to stimulate the ovaries. It's suppressing ovulation, so it's getting to the brain.
- 8.18** Now, what's interesting about the hormonal IUD is that the levels of hormone that are in it are very low. And so, when you have the hormonal IUD, when women are first on it, about 80% of women do not ovulate when they're on it during the first year—meaning that, for 80% of women, the amount of hormone that's reaching the brain is sufficient to shut down ovulation. But for about 20% of women, it's not. Ovulation doesn't shut down.

- 8.19** So it still protects from pregnancy, just because anytime you have something in the cervix and uterine area, it won't lead to implantation. You won't get implantation of an embryo because your body won't allow it, because there's an obstruction, and the body doesn't want to get pregnant if it thinks that there's an obstruction in the uterus, at least that's the current wisdom about how it is that it works.
- 8.20** So for some women, they continue to ovulate when they're on the hormonal IUD. For these women, they're going to feel pretty normal because they're still producing hormones. And so women's experiences on the IUD tend to be incredibly varied. This is all depending on whether or not they're ovulating or not. If you're on the hormonal IUD and not ovulating, because the levels of hormone in it are so low, a lot of women will feel really crazy, just meaning that they feel in a way that's biologically not natural for themselves.
- 8.21** Because it's not natural for women to have really low levels of hormone. That's just not, biologically, usually the way that things are for us. When women are on this and aren't ovulating and aren't getting a lot of hormone from exogenous sources, it feels a little crazy. But what happens with the IUD that's also sort of interesting, but makes it really messy to understand, is that when women are on it for a long period of time, generally, they'll start ovulating, because the levels of hormone are low.
- 8.22** And at some point, the brain picks up on the idea that this does not actually mean pregnancy, and so ovulation will resume. What research finds with the hormonal IUD is that during the first year when women are on it, about 80% are not ovulating. But if you look five years later, women who are on it for five years or more, about 80% of women are ovulating.
- 8.23** What this means is that you're going to have some women who feel really good on the IUD the whole time that they're on it because they never stop ovulating. Then you're going to have some women who feel absolutely terrible on the hormonal IUD the entire time that they're on it because they never ovulate. And so, this is a very idiosyncratic form of hormonal birth control, just meaning that, depending on what a woman's body does in response to taking it, that's going to impact her side-effect profile.
- 8.24** When women ask me, "All of these things that you're talking about, does this apply to me? I'm on the hormonal IUD," my question in response to that is, "Are you ovulating?" Because if you're not ovulating, then the answer is yes. You're at the same risk as a woman who's on a hormonal birth control pill of having any of these effects that we talked about. But for a woman who is ovulating, the answer is no. You probably are not at as great of a risk of developing these different types of effects that we talked about, because many of them are a result of what happens when you shut off ovulation.
- 8.25** If you want to troubleshoot your birth control, you're going to also want to troubleshoot the delivery method. What we know from research, when you look at all of the different types of side effects that women experience, or the different types of

psychological and behavioral outcomes that tend to shift in response to pill use, the one that tends to have the lowest risk profile across all categories of risk, except for the physical risks of thrombosis, as noted, because estrogen does have cardiovascular risks associated with it.

- 8.26** But if we forget about the heart for a minute, if we forget about the cardiovascular system for a minute, combined hormonal contraceptive pills have the lowest risk profile. They're the ones that have the lowest risk of— they're the ones that have the lowest risk of developing depression. They're the ones that have the lowest risk of developing sexual dysfunction and so forth.
- 8.27** The highest risk profile generally goes to the progestin-only products, and these are the ones that are safest for the cardiovascular system. So again, it's always a trade-off. Progestin-only products, especially in teenagers, tend to have a greater risk of developing mental health disorders. There's a greater risk of weight gain, especially with the birth control shot. There's a greater risk of having sexual dysfunction and so forth.
- 8.28** As noted, the IUD is highly variable, simply because it depends on whether or not you're ovulating or not, the extent to which you're going to be having these side effects. Many women report really loving the IUD after they've been on it for a period of time, generally after they start ovulating. But as noted, teenagers in particular, when they're given the hormonal IUD and they're not ovulating and they're not getting any hormone, that's one of the ones that has the highest risk profiles in teens.
- 8.29** But then in adults, it has one of the lower risk profiles, and so again, all of this—you have to take all of this contextual information into play when you're making decisions about choosing a hormonal contraceptive that makes sense for you.
- 8.30** How do you know which one is best for you? How do you know whether something is making you feel better or making you feel worse? I always recommend to people that they become an expert in themselves. No one is going to be more of an expert in you than you—not your doctor, not your therapist. You need to be an expert in you.
- 8.31** One of the things that I recommend, if somebody is considering starting hormonal birth control or switching from one type to the other, is to begin before they start by doing what I call objective journaling. I love objective journaling because you're essentially treating yourself as a research participant in your own experiment. I call this "me-search". It's me-search. You can use objective journaling to evaluate and rate yourself on a 1 to 7 scale, with 1 to be extremely negative and 7 to be extremely positive, and rate the things that are important to you.
- 8.32** For example, things like sexual desire, mood, energy level, your self-esteem, like how good do you feel about yourself? And evaluate yourself every day over a period of time. Then you get to act like a scientist and you can actually create averages. You can run a mean. During the month where I was on this form of hormonal birth control, my self-

esteem was, on average, a 4, and my sexual desire was, on average, a 5. You're able to create an average to see what your levels are, and then you can compare when you change.

- 8.33** If you change and you start hormonal birth control, or you try a new one, and all of a sudden you see, "Oh, on average, my sexual desire is now a 6, and my self-esteem is a 7," you'll know that things are getting better. Or if all of a sudden, on average, your sexual desire is a 2, and your self-esteem is a 3, you know that things are getting worse. That's why I really like objective journaling, because it does give you the opportunity to create an average, and you can actually compare to how things were in the past and then how things are going to be in the future, to make decisions about whether or not something is working for you.
- 8.34** It's really important that you learn to become an expert in yourself, because sometimes when we're taking a medication in particular, we'll have experiences that other people do not have. I have this figure up here, showing a relationship between two variables. Let's say it's hormonal birth control use and mood, and everybody else is clustered around this area where the more of, let's say it's the higher level of estrogen in the dose, the better the mood, and you're way off in outer space somewhere, that happens.
- 8.35** We are outliers on a lot of different things. Most of us are average in a lot of different things, and then we also have areas in our lives in which we are an outlier. Always keep track of yourself because, even though something might not characterize most people's experience with a hormone, it may influence your experience with the hormone. Knowing that is important, because the only data point that matters—you know, I've been talking about all of this science and all of this research, and research finds this and research finds that, but the only data point that matters in your life is yours.
- 8.36** Keeping track of that is the most important thing and really knowing how you're responding, so that way you can be able to make adjustments to allow you to feel the way that you want to feel. Just because research says no to a side effect, and when I say that, I mean, just because research says, "No, we do not find a relationship between hormonal birth control use and whatever it is that you think that you're experiencing," that doesn't make it implausible.
- 8.37** The fact of the matter is, no two women will respond to the same birth control method in the same way. And so idiosyncratic effects happen all the time. And so listen to your body and learn to believe it. And not only is it—sometimes people experience things that just don't happen to everybody. We have these idiosyncratic effects.
- 8.38** But also, you have to understand that the way that researchers generally study the impact of hormonal birth control on any type of an outcome is done in an incredibly simplistic way, where they're just looking at the difference between users and non-

users, without considering all the myriad mitigating factors that are likely to interact with birth control use to lead to differences in outcomes.

- 8.39** To give you an example of this, I'm going to show you this figure. This is a study that was done. It was a really nicely done study, in that it was an experiment. They randomly assigned women to take either hormonal birth control or a placebo. Then they looked at changes in weight over time, because they were interested in answering the question: Does hormonal birth control use lead to weight gain?
- 8.40** Because a lot of women report that they gain weight on hormonal birth control. Researchers say, based on most of the science, they're like, "There doesn't seem to be a relationship." So these researchers wanted to do this really carefully controlled experiment, which they did. They randomly assigned women to take birth control or a placebo, and they looked at changes in weight over time. And what they found was no differences between hormonal birth control users and natural cyclers.
- 8.41** And so the researchers concluded, "No, hormonal birth control does not cause weight gain." But other researchers subsequently went to their data and said, "Wait a minute, wait a minute, wait a minute. We can't be treating all women equally. We might find that a woman's pre-existing body weight interacts with her use of hormonal birth control to influence her risk of weight gain." And so, they re-analyzed the data, looking at the women's starting weight, and then looked at the differences.
- 8.42** What they found was that there are differences and there are changes in weight that can happen when you use hormonal birth control; and that, for heavier women, when they go on hormonal birth control, they actually lose weight. For women who are lighter, so women who weigh less, they end up gaining weight. These things are masked when you lump all women together.
- 8.43** And so, a lot of times, the data are being analyzed in the very simplistic way that this was analyzed the first time, where you just look at users compared to non-users and don't check for mitigating or interacting factors like a person's starting weight, for example; or if we're talking about sexual side effects, a person's starting levels of sexual desire, or if we're looking at partner attraction, what a person's baseline of attraction is.
- 8.44** There's all of these different factors that can potentially interact with the use of hormonal birth control to impact the outcomes that oftentimes aren't studied. What this means is that, just because science so far doesn't say that this side effect definitely happens, it could be happening to you. If you perceive it, start to track it, and then see whether or not it might make sense for you to make a change with what you're using, so that way you can feel a different way.
- 8.45** The hormonal birth control pill and hormonal contraception in general changes women. We've been spending the last several hours together, talking about all of the different ways that hormonal birth control does this—the way that it changes, can change our moods, and can change who we're attracted to, and change our energy levels, etc.

- 8.46** But in addition to changing women, hormonal birth control changes the world, because when we change women, women live in an interconnected world that's surrounded by their friends and surrounded by their families and surrounded by their romantic partners and surrounded by their colleagues. So when we have something that makes a change in women, because women live in an interconnected web of other people, it also means that we change the world. I want to end our lecture series talking about some of the ways that hormonal birth control has changed the world and what it might mean moving forward.
- 8.47** One of the things that we know that hormonal birth control does is that it changes the costs of sex. Women, when they're taking hormonal birth control and they're able to be in a sexual relationship and they no longer have to fear the possibility of unanticipated pregnancy, one of the things that I've been talking about is that this allows women to be able to dream bigger and achieve more than anybody ever would have thought possible from women before the advent of safe and effective contraception came on the market.
- 8.48** Just to show you a snapshot of some of the ways that hormonal birth control has changed the world, I want you to look at this drawing. What this is illustrating is the proportion of female applicants to medical and law school right prior to hormonal birth control being made legally available to single women. So it was only 1 in 10. One in 10 applicants to law and medical school. And law and medical school are things that require you to stay in school until you're almost 30 years old.
- 8.49** Now, flash forward just 20 years later, and then the proportion of female applicants to medical and law school became 4 out of 10. Just this idea, just by virtue of opening up the possibility that you might be able to go into an advanced degree program and not have to worry about pregnancy, has likely contributed to the growing number of women seeking advanced education.
- 8.50** If we look at this now, the proportion is 5.6 out of 10, so more than half of applicants to law and medical school are now women. Again, showing some of the ways that having access to a technology that allows us to regulate our fertility allows women to be able to make plans and be able to go out and achieve goals in ways that weren't possible before.
- 8.51** Another way that hormonal contraception and access, again, to something that allows women to be able to regulate their fertility, is that women no longer need to prioritize good-dad qualities or commitment to consent to sex. One of the things that we learned about in our very first lecture together was that choosiness acts as a function of the need to invest. This is something that's true across species of reproducing organisms or across species of sexually reproducing organisms.
- 8.52** We find that the more investment that a sex has to make in terms of the potential need to invest in offspring, the choosier that sex is in terms of being willing to consent to sex,

and then the choosier they are about the qualities possessed by those people that they're going to mate with or those organisms that they're going to mate with. With hormonal birth control, it removes that cost.

- 8.53** There's no longer that need to have to invest, just in order—if you're going to have sex, the probability of going to need to exert paternal care or parental care is much lower in the world of hormonal birth control than it is in the absence of it. Because of this, we've seen major shifts in sexual behavior and sexual attitudes over time that have coincided to the lowered cost of sex for women. Research finds that hormonal contraceptives, since the time that they arrived on the market, it's dramatically reduced the costs of casual sex.
- 8.54** When you ask women about the costs of casual sex, they acknowledge the fact that it comes at a much lower cost than it used to before there was access to contraception. When you compare women who are using hormonal birth control to women who are not using hormonal birth control, in terms of their perceptions of the costs of sex, they're very different from one another, with the naturally cycling women recognizing a larger cost because there's a larger probability for pregnancy in that context than there is in the context of using hormonal birth control.
- 8.55** We've also seen that, after the time that hormonal birth control became available on the market and available to single women, this has also led to a greater acceptance of casual sex. And it's also been one of the contributing factors to the rise of hookup culture among young adults. This is where people are forming these short-term casual sexual relationships with one another without any commitment or expectations of commitment. And all of these things, all of these shifts, have been impacted in part by the availability of safe, effective hormonal contraception.
- 8.56** Some research suggests that simply by virtue of being on hormonal birth control and having a lower cost related to sexual activity, this changes women's sexual attitudes. These figures here are looking at the differences between men, naturally cycling women, and women using hormonal birth control, and looking at their sociosexual attitudes. These different labels, it says SOI, and it's A, B, D, and then total. This stands for the sociosexuality inventory. This is a scale that evaluates how sexually unrestricted people are in terms of their attitudes, their behaviors, and their desires.
- 8.57** What you find when you look at the differences between men, women who are naturally cycling, and women using hormonal birth control is that women who are using hormonal birth control become more similar to men when they're using hormonal birth control in terms of having positive attitudes toward casual sex. They tend to be more likely to have casual sex. They tend to have more desire for casual sex. And overall, they have more sexual permissiveness relative to women who are naturally cycling.

- 8.58** And this, again, is illustrating that when we change the costs of sex, that it changes sexual behavior. That's one of the big themes from evolutionary biology, is just that costs predict choosiness. When we lower the cost, it lowers restrictedness, and it also lowers choosiness. This slide here is looking at the results of a study comparing men, women who are naturally cycling, and women using hormonal birth control on their minimum standards for partner choice across different types of dating contexts.
- 8.59** I showed you all a slide that looked very much like this, that was comparing men and women. This was a slide that was showing us that men become increasingly choosy about their partners as commitment increases, and that women become increasingly choosy as commitment increases, but that men become pretty non-choosy in the context of short-term casual sex, where men have low standards in a low-commitment context. Again, this is just going to illustrate that the need to invest predicts choosiness.
- 8.60** The more you might need to invest in a relationship or an offspring resulting from that relationship, the more choosy you're going to be in partner choice. And so in the context of short-term sex, because for a naturally cycling woman, the need to invest is still potentially high because she could end up with a pregnancy, but for a man, it's relatively low because he does not have to worry about a pregnancy, we generally find that men's choosiness decreases in the context of short-term casual sex and that it does not for women.
- 8.61** When you compare women who are naturally cycling to birth control-using women and then compare to men, you can see that the women who use hormonal birth control, and they're the ones shown in the dashed lines, they become less choosy in the context of short-term casual sex, just like men do. When we lower the cost of casual sex, it lowers people's standards for their partners that they have as casual sexual relationship partners.
- 8.62** Given that it's pretty clear from when you look at the data— both the data I showed you and then other data that have been collected in other types of context— that having access to hormonal birth control and having access to safe and effective birth control that allows a person to have sex without the fear of pregnancy, that this does increase women's sexual permissiveness. It increases women's willingness to engage in casual sex, and it lowers their standards in terms of the partners that they're willing to consent to sex with, researchers applying these ideas to a sexual economics perspective have predicted that by lowering the cost of sex for men, when women become more sexually permissive and become less choosy about the partners that they're going to have sex with, it's been hypothesized that this may be something that demotivates men, leading them to achieve less than they have before in history.
- 8.63** There's this great quote by Aristotle Onassis, who said, "If women didn't exist, all the money in the world would have no meaning." Just this idea meaning that one of the most powerful sources of motivation for men to achieve that's ever been identified is trying to get sexual access to women. And when women are willing to grant sexual

access to men who haven't really achieved anything, this may actually have the result of demotivating men's achievement motivation.

- 8.64** When we look at the achievement gap, this is consistent with the idea that men's achievement motivation seems to have declined over time. When we look at the changes in educational attainment, for example, after the advent of the birth control pill being legally available to single women, what you see from the darker-colored lines on this figure is that women are beginning to achieve more.
- 8.65** This y-axis is showing the share of fall enrollment and degree-granting post-secondary institutions by sex of student. 1970, right when the pill was becoming legally available to single women, through 2026, which is just a projection. What you can see is that once women were able to understand that if they start college, they'll be able to finish it because they don't have to worry about an unplanned pregnancy, you can see women's rates of enrollment in colleges go through the roof.
- 8.66** Women have done a lot better, but at the same time, men are doing a lot worse. It's not just that women are doing better, it's that men are achieving less. It's been hypothesized that one potential contributing factor—and there are a lot of contributing factors to these trends, so this is by no means exclusive—but that one contributing factor may be that women are willing to consent to sex with men, and they're exhibiting less choosiness when they're doing so.
- 8.67** This may be a demotivating factor because men are still able to get sexual access without having achieved anything, and this may potentially lead to some of the changes that we're seeing in achievement motivation in men. There was one experiment that was done looking at this possibility, where men and women were randomly assigned to read one of two newspaper articles. They would read a newspaper article that they said was recently published in the New York Times.
- 8.68** And the newspaper article was either saying that members of their opposite sex were getting choosier about their romantic partners, and that dating and sex is getting harder, or they read an article saying that members of the opposite are becoming less choosy about their romantic partners and that dating and sex is getting a lot easier. Then they had men and women report on their career aspirations and grit. The grit scale is something that looks at your willingness to continue to persevere in the face of setbacks.
- 8.69** What they found was that, for men, but not for women, they reported lower career aspirations and grit in response to the article, after they read the article that said that dating is easier and that women are becoming less choosy. This is consistent with the idea that men's achievement psychology may be tuned in to what is necessary in order to be able to gain access to partners.
- 8.70** Roy Baumeister had this really great quote, where he said, "We have found no evidence to contradict the basic principle that men will do whatever is required in

order to obtain sex, and perhaps not a great deal more." Maybe that's true. And so, it could be that, by virtue of the birth control pill, lowering the cost of sex for women, which then allows women to have casual sex and become less choosy about their partners, this may be a demotivating factor. It may remove a very powerful source of motivation that, historically, has been around to help promote achievement motivation in men.

- 8.71** This isn't to say that this is women's responsibility, because it's absolutely not women's responsibility to be making sure that men are motivated enough to go out and achieve. There are clearly other factors at play in the lack of achievement motivation and the decline in achievement motivation that we've seen in men in the last 30 years. But the changing costs of sex do change mating dynamics. Those changes in mating dynamics can change the face of the world.
- 8.72** Changing the cost of sex has also created societal shifts. We've seen since the advent of hormonal birth control that the age of marriage has steadily increased since the late 1950s. In the late 1950s, things started to decline. Then once the birth control pill became legally available to single women, which started at the end of the 1960s, early 1970s, you can see that the age of first marriage really sharply increases. The age of first marriage, the median age of first marriage, in 1960 for women was 20 years old back in 1960, and it's currently 28 years old.
- 8.73** This is a big shift, and things are continuing to get pushed further and further away, where the first age of marriage is getting later. This is also creating a marriage squeeze, in particular for women. The reason that is particularly happening for women is that women, when they get married later, bump up against a wall with their fertility. Women generally want to get married earlier than men, because women have their fertility to be concerned about, and if they want to start a family, they need to get married at an early enough age that will allow them to have children.
- 8.74** What happens is, you have women within a certain age, let's say 25 to 30, who are looking to settle down and get married. Men don't want to do that. They're able to get sexual access. They're hanging out in the dating market. Now we also have dating apps, which makes it really easy for people to find partners nonstop. Men oftentimes want to hang out in the dating market longer than women want to hang out in the dating market, before they're willing to enter the marriage market.
- 8.75** The marriage market, men generally enter that much later than do women, because they don't have fertility concerns. What happens is, then the marriage market becomes populated primarily by women, because there's so many fewer men who are in it, because men enter so much later. That's created a marriage squeeze, where especially educated women, because women are hypergamous and generally want to mate with somebody who has at least as much educational attainment and achievement motivation as they have themselves, this puts these women in a squeeze where they're having a more difficult time finding marriage partners.

- 8.76** And not surprisingly, in response to this, we're also seeing changes in marriage rates. We're seeing a greater number of people never getting married. In particular, women are leaving the marriage market if they're not able to find what they want. There's a growth in the number of single women that we've had since the 1950s and even in the last 20 years. This here is just looking at changes in marriage rates, looking at the rise of never-married adults. In this one, it is not sex-differentiated.
- 8.77** In 1960, only 9% of adults that were 25 and older were never married. Right now, well, in 2020, it was 35%. That's a huge increase. We've seen the world change. And again, hormonal birth control has a lot to do with this, because it used to be, back in the 1960s, before you had safe, reliable, and effective contraception, if you were in a sexual relationship with somebody, chances are somebody was going to get pregnant at some point, usually the woman.
- 8.78** There's a chance of pregnancy that occurs, and that generally leads couples to get married. They're going to settle down and they're going to care for this child. Now that we don't have that as a concern, oftentimes what people are doing is, if they're not worried about having to care for children, they don't get married. And you also, as noted, you get these asymmetries in the sex ratio in the marriage market being overwhelmed by women compared to men, who then will leave the marriage market if they're not successful. And we have a lower number of married people than we ever have before.
- 8.79** Another change that has happened in response to the advent of the hormonal birth control pill are changes in fertility. People are now able to voluntarily make the decision about whether they want to have children or not, and when they want to have children or not. Because of that, people are generally postponing pregnancy and having children until later on in their life. Generally, they want to acquire more education. Then as a result of that, they wait to have children until later. They choose to have fewer children, so that way they can invest heavily in those children.
- 8.80** We can see that the birth rate has dropped really dramatically. This is a figure showing the number of births per 1,000 women between the ages of 15 to 44. You can see that, in the 1960s, it peaked at about 120, and currently, it's at about 56.1. You can see the biggest decrease and then plateau right around 1970 when hormonal birth control became legally available to single women.
- 8.81** This is also availability of the hormonal birth control pill and the ability to be able to regulate our fertility is also changing the evolutionary trajectory of humans. This is such an interesting thing to think about, because by virtue of introducing something that allows us to make decisions about when and whether to have children, it means it's going to put evolutionary pressure on the desire to have children.
- 8.82** For most of history, if you were a person who enjoyed sex and you were in a relationship, you were going to be a parent, whether you liked it or not. The desire to

have children was totally immaterial from the decision. It was just about, if you like to have sex and you're in a relationship, you're going to end up with kids. And so, wanting children was never a requirement for becoming a parent.

- 8.83** Now, it has to be. People can make the decision, "I don't want to have children," and then they don't have children. But this is going to put direct selection pressure on the desire to have children. We should expect to see that, in the next thousand years, that there will be a greater proportion of people who actively want to have children than we do currently. Right now, we're going to be seeing evolutionary change going on, because people who don't want to have children are not having children.
- 8.84** But those genes related to not wanting to have children aren't going to get inherited. Over time, we're going to see changes in gene frequencies that go on in response to people's decision-making, which is really fascinating. Currently, there are a number of people who actively don't want to have children. Over generations, those people are going to not have children. Those genes related to not wanting to have children are eventually going to disappear, because now it's going to create direct evolutionary pressure on the desire to have children.
- 8.85** This is the first time in history that this has been true, because it used to be that it didn't matter whether you wanted to have children or not. You were just going to have them anyway. Then it's really interesting to sit and think about all of the other things that it might be changing. How is it that the use of hormonal birth control, using something that allows women to remove the cost of sex from sex, in what other ways is that changing the world? It's an interesting thought experiment.
- 8.86** When we change women, we change the world. This is true both for better, like in the case of women's ability to be able to regulate their fertility allows them to achieve more; it's also sometimes for the worse. We do know that, for some women, the use of hormonal birth control can be related to psychological side effects that can be nothing short of devastating, such as what we get with the increased risk of suicidal thought and suicidal behavior. It changes us for better and for worse.
- 8.87** To close out our class, I want to have a little reminder of all of the things that we learned. We learned that men and women have two distinct evolutionary paths that stem from the differences in our reproductive biology. Just the size of our sex cells leads us off into two different evolutionary paths, where men and women have been confronted with different adaptive challenges that they've had to solve in order to successfully survive and reproduce. This has led to psychological and behavioral differences.
- 8.88** We talked about the fact that ignoring the fact that these differences exist is bad for all of us. It does nobody any favors to pretend that biological females and biological males are identical to one another because they're not. It's bad for us medically, it's bad for us legally, it's bad for us socially.

- 8.89** We talked about the fact that our sex hormones are part of what makes us us. They're part of the signaling machinery that our brain uses to create the experience of being the people that we are. This means that when our hormones change, we change. This doesn't mean that we need to get rid of the birth control pill. This doesn't mean that the birth control pill is bad or that it's dangerous, because it's none of those things. It simply means that we need to be educating women and men about the trade-offs that it entails when you take hormonal contraception.
- 8.90** We also learned that when we change women, this also means that we're going to be changing the world. With that, I say thank you, and I hope that you learned a lot. If you want to learn more about any of these topics, you can check out my books: *This Is Your Brain on Birth Control* or *The Period Brain*. You can also find a lot of free resources on my website at sarahehill.com. And you can follow me on social media to see things there, @sarahehillphd.
- 8.91** I was wondering if there were any proven environmental factors that also affect hormone balances in men and women that can affect fertility in a similar way to how intentional consumption of hormonal birth control can.
- 8.92** Yeah, it's a really great question. Certainly, in our current environment, we are exposed to a lot of endocrine-disrupting compounds, and those do have the effect of changing our hormone levels. Just to give some examples of those things, like phthalates and BPA and other things that are in plastics, those things can act as xenoestrogens, and they can have estrogenic effects on both male and female bodies, but not like positive estrogenic effects, negative ones.
- 8.93** Other factors, one of the biggest factors that's not really discussed is excess body weight, especially in men. Body fat acts as an endocrine organ, and it leads to the production of estrogen. It also releases this compound known as aromatase, and aromatase converts testosterone into estrogen. We've seen that men's testosterone levels over time have declined dramatically.
- 8.94** Two of the big factors responsible for that decline are excess body fat or excess adiposity on men that lowers testosterone levels by aromatizing a lot of their testosterone into estrogens. Then also exposure to these endocrine-disrupting compounds, which also tend to have a feminizing effect on men. Another big source of these endocrine-disrupting compounds is receipts. If you handle receipts, wash your hands. That's my message there.
- 8.95** It seems that objective journaling is the best proactive step a female should take in determining what birth control to be on. But what is your recommendation to a female who is chronically ill, as it relates to what birth control to be on? Because objective journaling for a relatively healthy female is a lot more attainable than for a chronically ill female, whose symptoms are so compounded and multifaceted that journaling to find tangible results seems impossible.

- 8.96** The thing I like about doing the journaling is that you can really do it in a way that's specific to you and your concerns. For example, if you have a woman who does have a chronic illness, and let's say that the symptoms are X, Y, and Z, I would recommend measuring X, Y, and Z. In a naturally cycling woman, you can measure it over the cycle, because there are cycle-based changes and symptoms of chronic diseases.
- 8.97** Then, if you're talking about birth control, going on or off it or switching kinds, you can measure X, Y, and Z and how they respond to the different formulations of the hormonal birth control. But definitely, whenever you have an added layer of complexity, it always does make everything a little bit more challenging. But what's nice about creating an objective journal is that you get to pick what you're measuring. And so, if there are specific symptoms that are most concerning in the context of the disease, you can choose that and then have those be the things that you follow over time to see whether they get better or get worse.
- 8.98** OK, I've got a heavy for you.
- 8.99** All right, let's hear it.
- 8.100** I love the concept or the terminology of the marriage squeeze. With women putting off marriage and even having children to later in life for, let's say, college, career or advanced degree, but then in the end of that, when you ask women, "What is the most important thing in your life?", most of them will say, "My spouse, or motherhood, or my children." Because if you look at children, maybe over advanced degree, they'll say, as a mom, "I have grown the most, learned the most, learned more about life, love, relationships, patience, spirituality, and even myself through motherhood."
- 8.101** We're taking birth control to push all this off and pushing us into, I don't know what you said, the environmental squeeze later in life when women are going to want it, and it might not be available as much. That was super fascinating. So if we're pushing ourselves into that spot, what would be your recommendation for women to not get left behind and be, like, "Wait, wait. What should I have done?"
- 8.102** Yeah, gosh. I think it's so woman-specific. I really think it's really woman-specific. And I think that the recommendation I can make to all women is to recognize the limitations on their fertility. I do think that, when you have a lot of celebrities who are having babies at 45, it makes it seem like this is a really common occurrence and that it's just going to be super easy to get pregnant anytime you decide to do so. Most women don't realize that their fertility begins to decline a lot earlier than they think it does.
- 8.103** I would say, really being honest with yourself about fertility and also understanding there's a lot of people who make money making women believe that they can push their fertility into perpetuity with IVF and freezing their eggs, and that doesn't always work. Also, understanding that, because I think that sometimes women are given unrealistic pictures of how easy it might be to be able to get pregnant by just freezing their eggs and waiting until later.

- 8.104** Then also, for women who are considering that, it's just understanding that there's never an ideal time, and people figure it out. I had my first child, and I was writing my dissertation and getting ready to defend my PhD. It was not a great time, but I pulled it off, and most people can. I think that we get so much pressure put on us as young women to achieve and achieve and achieve that it can feel like a character flaw to want to slow down and have children.
- 8.105** I also think rewriting that narrative about what it means to be successful and being willing to listen to what you want yourself, instead of just listening to what the cultural paradigm is trying to shove down your throat, is also really useful. When we talk about the way that birth control pill might be changing the evolutionary trajectory of humans, I'm sure that the fertility clock on women is going to increase.
- 8.106** I don't think it's going to increase to be 75 years old because it's never—biologically, it's just not going to happen. But because women who aren't able to reproduce later in life, and so many women are not having babies in their 20s, I do think that the average age of menopause will start to very slowly creep up, just because women who are able to reproduce later in life will be passing on genes, and women who aren't will not be. And so I think we'll start to see that creep up, just the way that we've seen birth weight creep up with cesarean sections.
- 8.107** What would you say would be the top two or three misconceptions? You've shared a lot with us. And what would be the best way to educate and address these?
- 8.108** I think the one I hear the most is that the IUD acts locally. That one I hear all the time, like, "Oh, no, don't worry about it. It acts locally," and that's not true. With that one, it's about understanding how hormones work.
- 8.109** Number two is that hormones are something that happens to you, instead of a part of who you are, because sometimes people get really resistant to the idea that taking a hormonal contraceptive can have the effect of having changes to their personality and their desires and their motivational states. But really understanding that your hormones are part of the machinery that your brain uses to make you you. I think that educating yourself on the role of hormones and what they do in the body, I think, is incredibly important.
- 8.110** Then the third thing about hormonal birth control in particular is that, if it didn't do X, Y, Z to me, then it doesn't do X, Y, Z to anybody. I'll go and talk to people about, "Oh, hormonal birth control has these different risks." We spent some time in this class, talking about the ways that it might impact mental health and the ways that it might impact sexual desire and partner choice. Sometimes I'll encounter women who say, "Well, that didn't happen to me. This doesn't happen." It's like, those are two separate statements, and one is true, and the other one's not.
- 8.111** Just because something doesn't happen to you doesn't mean that that doesn't happen to some women and that it's not something that's worth exploring. I think, ultimately,

what it boils down to is understanding the role of hormones in the body, because once you understand that, it's clear to understand how it's impossible for the birth control pill not to change women. You can't, because it changes your hormones, it changes you.

- 8.112** Just really educating yourself on the role of hormones in the body, and then the range of potential effects that are possible with hormonal birth control use. That way you can better keep track of yourself, if you're somebody who's using it, or keep track of the ones that you love, if they're going to be using it.
- 8.113** Do you feel like, because it does affect relationship and who we choose, do you think it could be a treatment option to, say, if someone's in a non—let's say, a dating relationship—to say, maybe try going off birth control and seeing if this is really the person you would choose? In all seriousness, do you feel like that could be a treatment option?
- 8.114** Yes, as long as the person has another plan for contraception in place. If we take that out of the picture, then yes. Because I think that people sometimes, again, when we have an effect of a medication, a lot of times we don't recognize it. To give you an anecdote, for my own life: I spent more than a decade in my life on birth control, and I have no regrets on that. But I am a different person than I thought I was.
- 8.115** For example, I used to think that I was a little anxious and depressive. I thought that I was somebody who was susceptible to having really low moods. I had this view of myself that I carried around for a long time, and it took me being off of it for more than a decade to realize that wasn't me. That was the pill. It's only after I've been off of it for so long and realized that, no, I kind of roll with the punches. I do OK. My mood is pretty stable, actually. But I thought that I was a different person than I was, because I was on it.
- 8.116** And I didn't think there was anything wrong. If you would have asked me, "Are you depressed?", I would have said, "No, I'm not depressed." But it's because I didn't know anything other than that. It's like, I went on it when I was about 18, and so you're still getting to know who you are. Then that developed a sense of myself that included this mood stuff that's not me, and it was my birth control, and I had no idea.
- 8.117** So if you have people who are not feeling right, or they're experiencing relationship trouble and other things, I think that you can learn a lot by going off of hormonal birth control, and then spending some time off of it, and getting used to your body again. Most people, when they experience, aren't going to have something where they stop taking it and the next day they wake up and like, "Oh, my god, what was I doing? Who am I, and what was this?"
- 8.118** But instead, it'll be a couple of months later, and they'll just be driving and say, "I feel really good." That was my experience. I was off it for about three or four months, and I was like, "I feel really good. I feel alive. Isn't it great to feel alive?" It's usually subtle. It

sneaks up on you. But I definitely do think that there's— especially if you're feeling like life doesn't feel right, and there's not a high cost of going off of it, I would totally try going off of it.